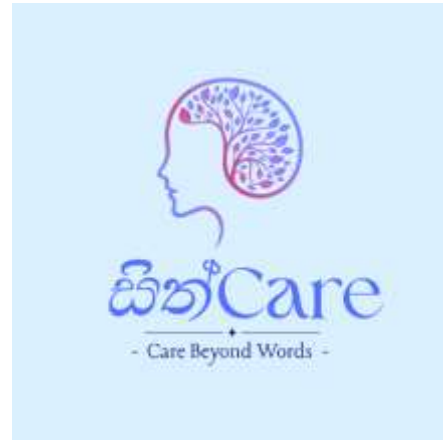




SENG 31242 – System Design Project

System Design Specification

Document

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1. Introduction

1.1. Purpose

The purpose of this Software Design Specification (SDS) document is to describe the detailed system design and technical architecture of the SithCare Mental Health Care Platform. This document translates the functional and non-functional requirements specified in the Software Requirements Specification (SRS) into a structured technical design that guides the development, implementation, testing, and deployment phases of the project.

The SDS defines:

- The architectural design of the system
- System components and interactions
- Database and deployment structures
- User interface design considerations
- Security mechanisms
- Technology stack selections
- and infrastructure configurations.

This document serves as the primary technical reference for developers, testers, supervisors, and stakeholders throughout the system development lifecycle.

1.2. Scope

SithCare is a web-based mental health care platform designed to improve accessibility, privacy, and efficiency of mental health support services in Sri Lanka. The platform provides secure digital healthcare services for patients, mental health specialists, clinic administrators, volunteers, moderators, and system administrators.

The system includes functionalities such as:

- User authentication and role-based access control
- Appointment booking and queue management
- Telemedicine consultations
- Emergency mental health assistance
- Anonymous community support
- AI-assisted specialist recommendations
- Clinic administration and scheduling
- and payment processing



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The SDS focuses on the internal technical design and implementation structure required to realize these functionalities using the MERN technology stack and a layered MVC-based architecture.

1.3. References

The following references were used during the preparation of this document:

1. SithCare Software Requirements Specification (SRS) Document
2. Project Guidelines and Assessment Criteria
3. IEEE 1016 – Software Design Description Standard
4. MongoDB Documentation
5. React.js Documentation
6. Node.js and Express.js Documentation
7. Tailwind CSS Documentation
8. JWT Authentication Standards
9. PayHere API Documentation
10. Jitsi Meet API Documentation

1.4. Document Overview

This Software Design Specification document is organized into several major sections describing the overall system architecture, design components, deployment strategies, and implementation details of the SithCare platform.

The document includes:

- System overview and relationship to the SRS
- Architectural and deployment design
- Technology stack justifications
- Component and package diagrams
- Database and class design
- User interface wireframes and prototypes
- Sequence diagrams
- Security and infrastructure considerations

The SDS acts as the technical blueprint for implementing the SithCare system according to the requirements identified during the analysis phase.



2. System Overview

SithCare is a centralized digital mental health care platform developed to connect patients, mental health specialists, clinics, volunteers, and administrators through a secure and scalable web-based system.

The platform addresses several challenges currently faced in mental healthcare services, including:

- Limited accessibility to mental health specialists,
- Long waiting times and inefficient appointment management,
- Lack of anonymous support systems,
- Emergency response delays,
- and social stigma associated with seeking mental health assistance.

The system enables users to securely access mental health services through multiple integrated modules such as appointment scheduling, telemedicine consultations, AI-assisted specialist recommendations, emergency support systems, anonymous community interactions, and peer support services.

2.1. Relationship to the SRS

This SDS document is directly derived from the Software Requirements Specification (SRS) document prepared during the requirements engineering phase.

The SRS defines:

- Functional requirements
- Non-functional requirements
- User roles
- Use cases
- System constraints
- and business objectives

The SDS transforms those requirements into:

- Detailed architectural structures
- System components
- Database schemas
- Interface designs
- Deployment configurations
- and implementation-level technical decisions



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All design decisions described within this document are aligned with the requirements, constraints, and quality attributes defined in the SRS document.

2.2. High-Level System Architecture

SithCare follows a Layered MVC Architecture enhanced with modular service-oriented design principles.

The system consists of the following major architectural layers:

Layer	Responsibility
 Presentation Layer	User interfaces and frontend interactions
 Application/Controller Layer	API routing, validation, and RBAC
 Business Logic Layer	Core healthcare workflows and services
 Data Access Layer	Database communication and persistence

Figure 1 : Major Architecture Layers

2.3. Technology Overview

SithCare is implemented using the MERN technology stack:

- MongoDB Atlas - Cloud database
- Express.js - Backend API framework
- React.js - Frontend SPA development
- Node.js - Backend runtime environment

Additional technologies include:

- Tailwind CSS
- JWT Authentication



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- GitHub Actions
- PayHere API
- Twilio SMS API
- Jitsi Meet API
- and Google Maps API

2.4. Infrastructure Overview

The system uses cloud-based infrastructure for scalability and availability.

Infrastructure Component	Platform
Frontend Hosting	Vercel / Netlify
Backend Hosting	Render / Railway
Database Hosting	MongoDB Atlas
Version Control	GitHub
CI/CD Pipeline	GitHub Actions

Figure 2 : Infrastructure Components

The deployment workflow includes automated testing, linting, pull request validation, and continuous deployment pipelines to improve maintainability and development efficiency.



3. Architectural Design

3.1. Introduction

SithCare is a web-based mental health care platform developed to improve accessibility, privacy, and efficiency of mental health services in Sri Lanka. The platform provides appointment booking, telemedicine consultations, emergency mental health assistance, anonymous community consultation, AI-assisted specialist recommendation, volunteer listener support, and clinic administration services.

The system is designed as a scalable and modular healthcare platform capable of supporting multiple stakeholders including patients, mental health specialists, clinic administrators, volunteers, moderators, and system administrators.

This document describes the architectural design, component structure, deployment strategy, scalability considerations, and system-level technical decisions adopted for the SithCare platform.

3.2. Architectural Pattern Selected

SithCare will employ a Client-Server Architecture utilizing a Layered (N-Tier) model on the backend, tightly integrated with the Model-View-Controller (MVC) design pattern. The application is built using the MERN stack (MongoDB, Express.js, React.js, Node.js).

The layers are defined as follows:

- **Presentation Layer (Client):** A React-based Single Page Application (SPA) handling UI, routing, and state management for Patients, Specialists, and Admins.
- **Application/Controller Layer (Server):** Express.js REST APIs that intercept HTTP requests, validate payloads, and manage Role-Based Access Control (RBAC) via JSON Web Tokens (JWT).
- **Business Logic Layer (Server):** Node.js services managing core functionalities such as appointment scheduling, queue management, AI triage (SithBot), and emergency routing.
- **Data Access Layer (Database):** Mongoose Object Data Modeling (ODM) wrapping a NoSQL MongoDB database.

The architecture is designed to:

- Separate system responsibilities clearly,
- Simplify maintenance and debugging,
- Improve scalability,



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- Support secure healthcare workflows,
- Enable future migration toward microservice-based deployment.
- Instead of implementing a fully distributed microservices architecture during the academic project phase, the platform uses a modular service-oriented structure where each major functionality is isolated into independent service modules.
- This approach provides:
 - Enterprise-level scalability planning,
 - Easier future service decomposition,
 - Better maintainability,
 - Reduced development complexity.

3.3. Justification Referencing Quality Attributes

The chosen architecture directly supports the Non-Functional Requirements (NFRs) defined in the SRS:

3.3.1. Scalability (NFR-SC-01 to 03, NFR-P-03)

Node.js utilizes a non-blocking, event-driven I/O model which makes it exceptionally well-suited to handle the target metric of 500 concurrent users and 5x traffic spikes (NFR-SC-03). The stateless REST API design allows for horizontal scaling. MongoDB's flexible schema easily accommodates the schema-per-tenant design required for managing multiple clinic branches (NFR-SC-02).

3.3.2. Maintainability (NFR-M-01 to NFR-M-05)

The MVC and Layered approach enforces a strict separation of concerns. Frontend UI modifications will not impact backend business logic. This structure aligns perfectly with the development team's expertise and allows for parallel development. Using a single language (JavaScript) across the entire stack reduces context switching and improves code maintainability.

3.3.3. Security & Privacy (NFR-S-01 to NFR-S-08)

A dedicated Controller layer acts as a gatekeeper, verifying JWTs and enforcing RBAC (Patient, Doctor, Admin) before any business logic is executed. MongoDB allows for field-level encryption, aiding in PDPA compliance (NFR-S-02), and the isolated backend layer prevents direct database exposure to the internet.



3.3.4. Reliability & Performance (NFR-R-01, NFR-P-01)

React's virtual DOM ensures fast UI rendering, meeting the sub-3-second page load time requirement (NFR-P-01). The separation of external services (PayHere, Jitsi) from the core backend logic ensures that failures in third-party integrations do not crash the primary application, satisfying the 99.5% uptime goal.

3.3.5. Usability & User Experience (NFR-U-01 to NFR-U-05)

The React-based frontend combined with Tailwind CSS provides a clean, responsive, and intuitive user experience. Simplified navigation flows, queue tracking, emergency shortcuts, and low-complexity interfaces reduce cognitive load for mental health patients and improve overall usability.

3.4. High-Level System Components

The SithCare platform consists of the following major components:

1. Patient Module

Handles:

- Registration and authentication
- Therapist searching
- Appointment booking
- Queue tracking
- Emergency support
- Community interaction

2. Mental Health Specialist Module

Handles:

- Appointment management
- Schedule management
- Telemedicine consultations
- Patient communication



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2. Clinic Administration Module

Handles:

- Clinic profile management
- Doctor allocation
- Queue management
- Appointment monitoring

4. SithCare Admin Module

Handles:

- Tracking of all the 3 portals
- Actions related to users, clinics, doctors
- Risk controlling and maintaining the progress of the application

5. Community Support Module

Handles:

- Anonymous consultation
- Recovery stories
- Peer support
- Volunteer listener sessions

6. AI Assistance Module (SithBot)

Handles:

- Symptom-based triage
- Specialist recommendations
- Navigation assistance

3.5. Component / Package Diagram Specifications

Below is the descriptive specification for the Component Diagram. (Please generate the visual diagram in Draw.io or UML tool of choice using these exact components):

- Frontend Components (React): AuthModule, Patient-Dashboard, Doctor-Portal, Clinic-AdminPanel, SithCare-AdminPanel, SithBotWidget, VideoConsultationRoom.



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- Backend API Packages (Node/Express): AuthController, AppointmentService, QueueManager, NotificationService, PaymentGatewayAdapter (PayHere), EmergencyHandler.
- External Interfaces: PayHere Checkout API, Jitsi Meet SDK, Twilio SMS API, Nodemailer Email Service, Google Maps API

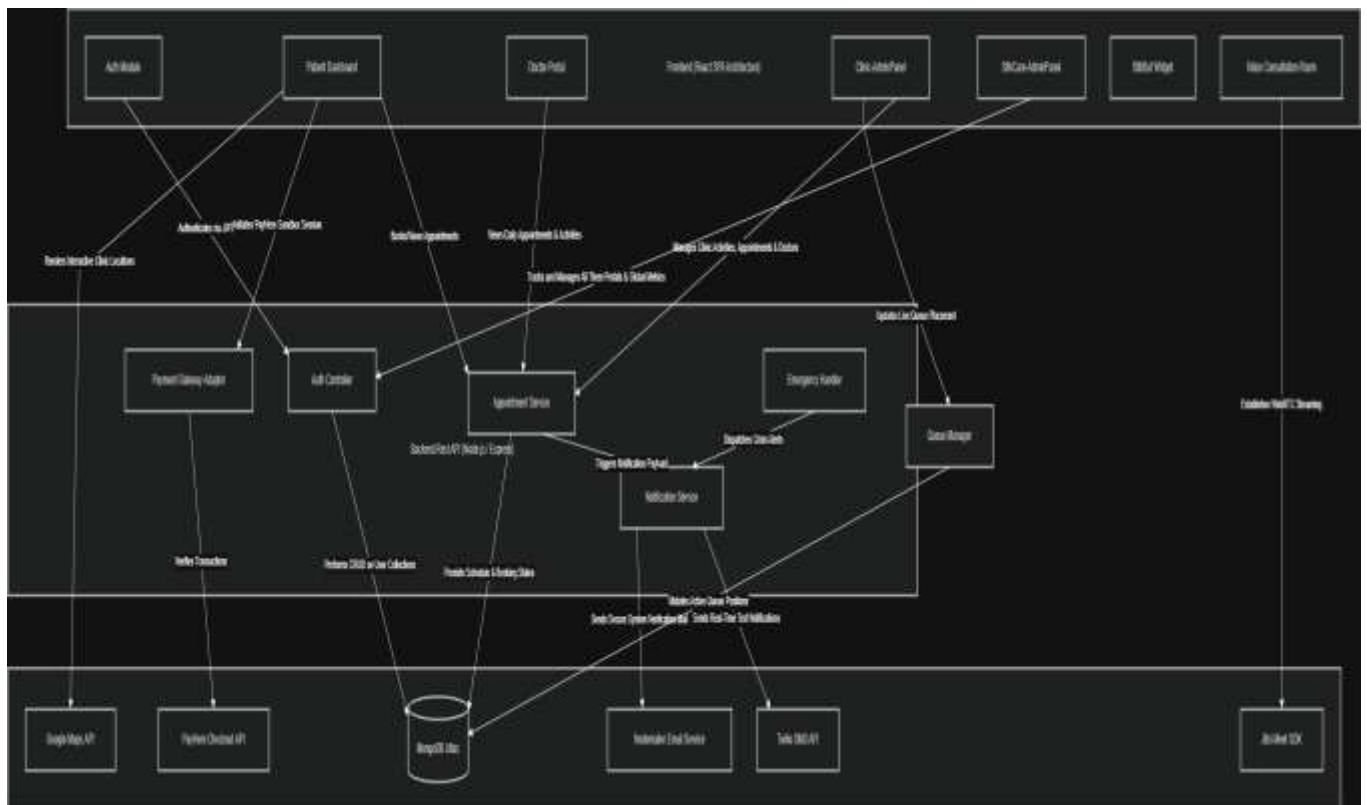


Figure 3 : Component Diagram



Mermaid.js Diagram source code

flowchart TB

subgraph Frontend ["Frontend (React SPA Architecture)"]

AM[Auth Module]

PD[Patient Dashboard]

DP[Doctor-Portal]

CAP[Clinic-AdminPanel]

SAP[SithCare-AdminPanel]

SB[SithBot Widget]

VC[Video Consultation Room]

end

subgraph Backend ["Backend Rest API (Node.js / Express)"]

AC[Auth Controller]

AS[Appointment Service]

QM[Queue Manager]

NS[Notification Service]

PGA[Payment Gateway Adapter]

EH[Emergency Handler]

end

subgraph External ["External Interfaces"]

PH[PayHere Checkout API]

JM[Jitsi Meet SDK]

TW[Twilio SMS API]

NM[Nodemailer Email Service]

GM[Google Maps API]

DB[(MongoDB Atlas)]

end



System Design Specification Document – SithCare Channeling App

%% Frontend Interactions to Backend Services

AM -->|Authenticates via JWT| AC

PD -->|Books/Views Appointments| AS

PD -->|Initiates PayHere Sandbox Session| PGA

DP -->|Views Daily Appointments & Activities| AS

CAP -->|Manages Clinic Activities, Appointments & Doctors| AS

CAP -->|Updates Live Queue Placement| QM

SAP -->|Tracks and Manages All Three Portals & Global Metrics| AC

%% Direct Front-End Script Integrations

VC -->|Establishes WebRTC Streaming| JM

PD -->|Renders Interactive Clinic Locations| GM

%% Backend Service to External Systems

PGA -->|Verifies Transactions| PH

EH -->|Dispatches Crisis Alerts| NS

AS -->|Triggers Notification Payload| NS

NS -->|Sends Real-Time Text Notifications| TW

NS -->|Sends Secure System Verification Mail| NM

%% Backend to NoSQL Persistence Layer

AC -->|Performs CRUD on User Collections| DB

AS -->|Persists Schedule & Booking States| DB

QM -->|Mutates Active Queue Positions| DB



3.6. Deployment Diagram Specifications

The deployment topology utilizes free-tier cloud services optimized for MERN stack applications, adhering to the project's zero-cost infrastructure constraint.

- Client Devices: Web Browsers (Chrome, Firefox, Safari) accessing the SPA via HTTPS (TLS 1.3).
- Frontend Hosting (e.g., Vercel / Netlify): Hosts static React assets and provides a global CDN for fast initial load times.
- Backend Web Server (e.g., Render / Railway): Node.js instance running the Express API, exposed via a reverse proxy handling SSL termination.
- Database Cluster (MongoDB Atlas Free Tier): Cloud-hosted MongoDB cluster handling all data persistence.

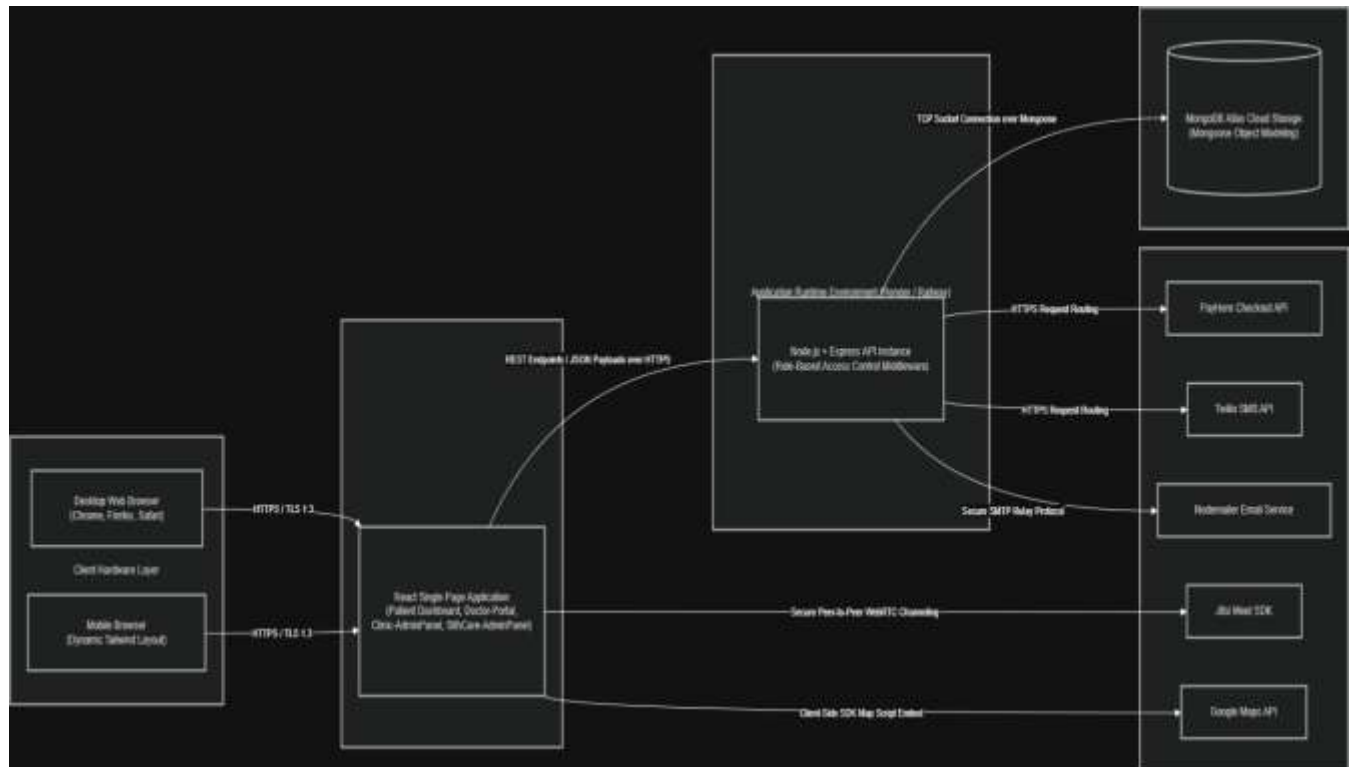


Figure 4 : DeploymentDiagram



Mermaid.js Diagram source code

flowchart LR

subgraph Clients ["Client Hardware Layer"]

Browser["Desktop Web Browser\n(Chrome, Firefox, Safari)"]

MobileBrowser["Mobile Browser\n(Dynamic Tailwind Layout)"]

end

subgraph CDN ["Static Distribution Layer (Vercel / Netlify)"]

SPA["React Single Page Application\n(Patient Dashboard, Doctor-Portal,\nClinic-AdminPanel, SithCare-AdminPanel)"]

end

subgraph AppServer ["Application Runtime Environment (Render / Railway)"]

NodeAPI["Node.js + Express API Instance\n(Role-Based Access Control\nMiddleware)"]

end

subgraph DBServer ["Database Cluster Layer"]

MongoCluster["MongoDB Atlas Cloud Storage\n(Mongoose Object Modeling)"]

End

subgraph External ["Third-Party External Interfaces"]

PH["PayHere Checkout API"]

JM["Jitsi Meet SDK"]

TW["Twilio SMS API"]

NM["Nodemailer Email Service"]



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GM["Google Maps API"]

end

%% Execution Network Vectors

Browser -->|HTTPS / TLS 1.3| SPA

MobileBrowser -->|HTTPS / TLS 1.3| SPA

SPA -->|REST Endpoints / JSON Payloads over HTTPS| NodeAPI

SPA -->|Secure Peer-to-Peer WebRTC Channeling| JM

SPA -->|Client-Side SDK Map Script Embed| GM

NodeAPI -->|TCP Socket Connection over Mongoose| MongoCluster

NodeAPI -->|HTTPS Request Routing| PH

NodeAPI -->|HTTPS Request Routing| TW

NodeAPI -->|Secure SMTP Relay Protocol|

3.7. Core Technology Stack: MERN

The platform will be built using the MERN stack (MongoDB, Express.js, React.js, Node.js). This full-stack JavaScript environment allows the team to maintain high development velocity using a unified language ecosystem, fulfilling the 8-week MVP design and development timeline.

3.7.1. Frontend Technologies

1. React + Vite

- Purpose: Frontend web application development.

- Justification:

- Component-based architecture improves maintainability.



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- Fast rendering and reusable UI components.
- Efficient state management.
- Suitable for scalable single-page applications.
- Large community and ecosystem support.

2. Tailwind CSS

- Purpose: Responsive UI styling.

- Justification:

- Rapid UI development.
- Utility-first responsive design.
- Consistent design system.
- Simplifies glass-morphism UI implementation.
- Lightweight production builds.

3.7.2. Backend Technologies

1. Node.js + Express.js

- Purpose: REST API backend development.

- Justification:

- Lightweight and fast.
- Event-driven asynchronous architecture.
- Suitable for real-time features.
- Large middleware ecosystem.
- Easy integration with frontend applications.

3.7.3. Database Technology

1. MongoDB Atlas

- Purpose: Cloud-hosted NoSQL database system.

- Justification:



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- Flexible document-based schema
- Suitable for dynamic healthcare records
- High scalability
- Cloud-native deployment support
- Faster iterative development

2. Mongoose ODM

- Purpose: Object Data Modeling (ODM) library for MongoDB.

- Justification:

- Simplifies MongoDB database operations
- Provides schema validation
- Improves query handling
- Supports middleware and model abstraction
- Improves maintainability of backend services

3.7.4. Authentication and Security Technologies

1. JWT (JSON Web Tokens)

- Purpose: Session authentication and authorization.

- Justification:

- Stateless authentication
- Secure API authorization
- Scalable session handling
- Suitable for role-based access control

2. bcrypt

- Purpose: Password hashing and encryption.

- Justification:

- Secure password storage
- Industry-standard password protection
- Reduces credential exposure risks

3. TLS 1.3

- Purpose: Secure client-server communication.

- Justification:



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- Encrypts data transmission
- Prevents interception attacks
- Secures REST API communication

3.7.5. Third-Party Service Selection

1. PayHere Gateway

- Purpose: Online payment processing.

- Justification:

- Local Sri Lankan payment support
- Easy integration for MERN applications
- Supports multiple payment methods

2. Twilio SMS API

- Purpose: SMS notifications and alerts.

- Justification:

- Reliable SMS delivery
- Appointment reminders
- Emergency notifications

3. Nodemailer

- Purpose:

- Justification:

- Lightweight email integration
- Appointment confirmations
- OTP verification emails
- Password recovery support

4. Jitsi Meet API

- Purpose: Telemedicine video consultation support.

- Justification:

- Open-source video conferencing
- Lightweight integration



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- Minimal infrastructure overhead
- Suitable for MVP telemedicine implementation

5. Google Maps API

- Purpose: Location and emergency mapping services.

- Justification

- Accurate clinic location mapping
- Consent-based emergency location sharing
- Easy frontend integration

3.7.6. Infrastructure and CI/CD

1. Frontend Hosting

- Selected Platforms:

- Vercel
- Netlify

- Justification:

- Robust free-tier support
- Automatic SSL provisioning using TLS 1.3
- Seamless GitHub repository integration
- Continuous deployment pipelines
- Fast global CDN delivery
- Simplified React SPA deployment

2. Backend Hosting

- Selected Platforms:

- Render
- Railway

- Justification:

- Free-tier Node.js hosting support
- Simplified cloud deployment
- Easy GitHub integration
- Automatic deployment workflows
- Environment variable management



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- Scalable backend hosting capabilities

3. Version Control

- Selected Tools:

- GitHub
- GitHub Actions

- Purpose:

- Source code management
- Branch protection
- Pull request workflows
- Sprint issue tracking
- Team collaboration

- Justification:

- Code consistency
- Deployment reliability
- Team collaboration
- Faster issue detection
- Maintainability

3.8. Architectural Style Justification

SithCare adopts a Layered MVC Architecture with modular service-oriented design principles.

The architecture consists of:

1. Presentation Layer
2. Application / Controller Layer
3. Business Logic Layer
4. Data Access Layer

This architectural style was selected due to the following reasons:



Architectural Benefit	Justification
Maintainability	Independent layers simplify debugging and updates
Scalability	Modular services support future system expansion
Security	Centralized authentication and validation
Flexibility	External APIs can be independently managed
Testability	Layers can be independently tested
Extensibility	Easier future migration toward microservices

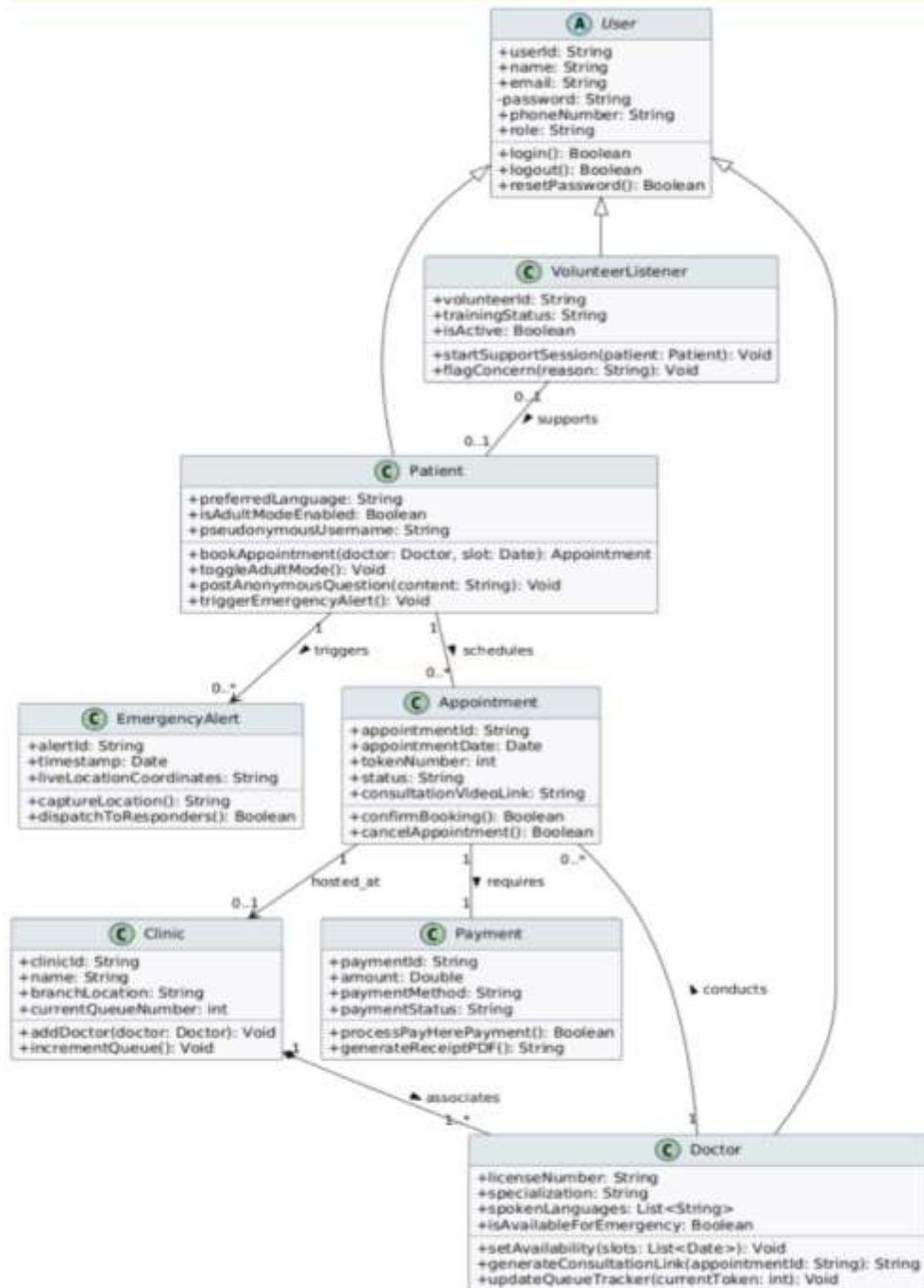
3.9. Conclusion

The selected MERN technology stack and architectural decisions provide a scalable, secure, maintainable, and practical foundation for the SithCare platform. The architecture balances academic feasibility with enterprise-level software engineering principles while supporting future system evolution and healthcare scalability



4. Detailed Design

4.1. Class Diagram





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A separate detailed document of the class diagram was created and uploaded to the Documents Repository of the Github Organization. Click the following links to access the document

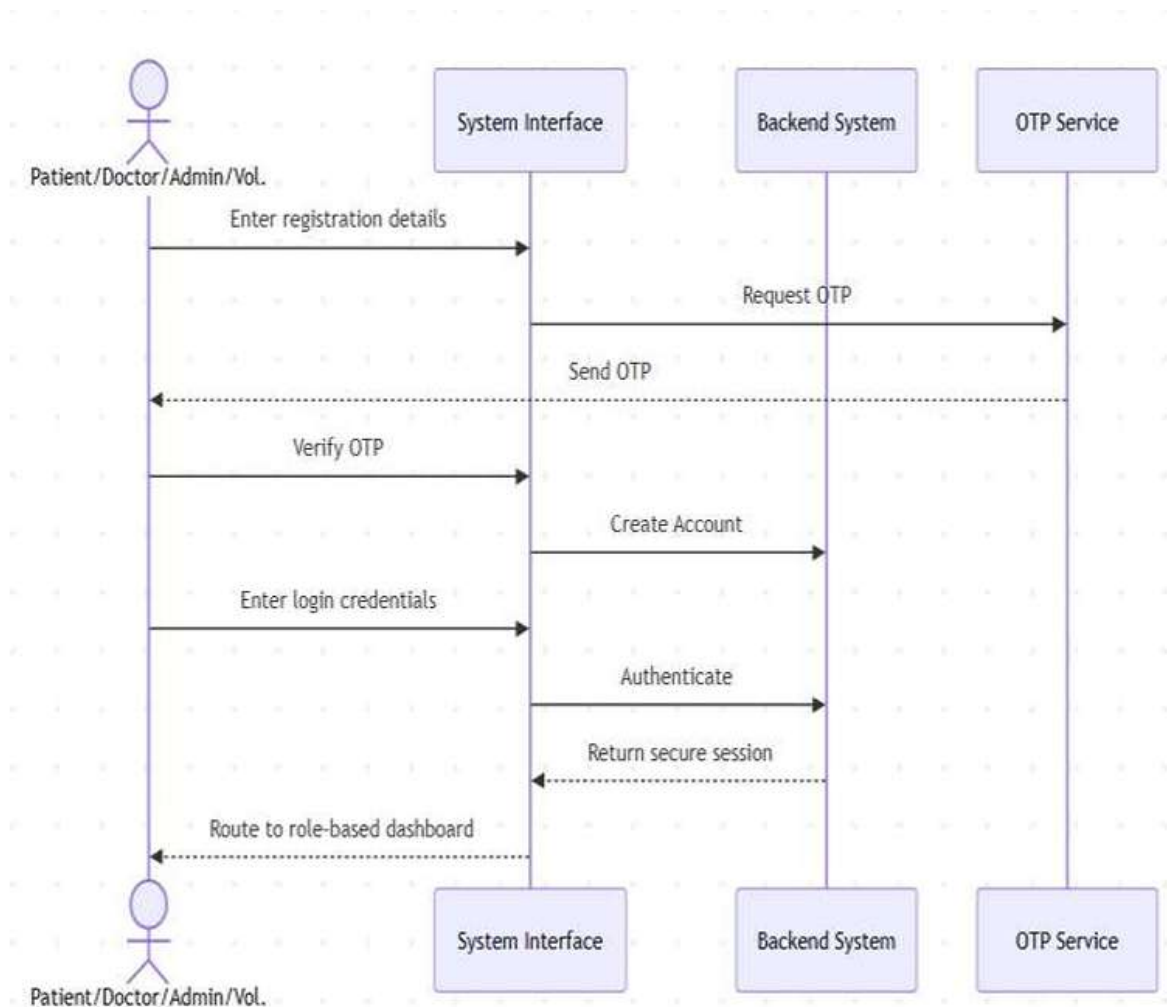
Link1 - [Class Diagram in Github](#)

Link2 - [Class Diagram in Google Drive](#)

4.2. Sequence Diagrams

Following sequence diagrams have been drawn for the major use cases of the application

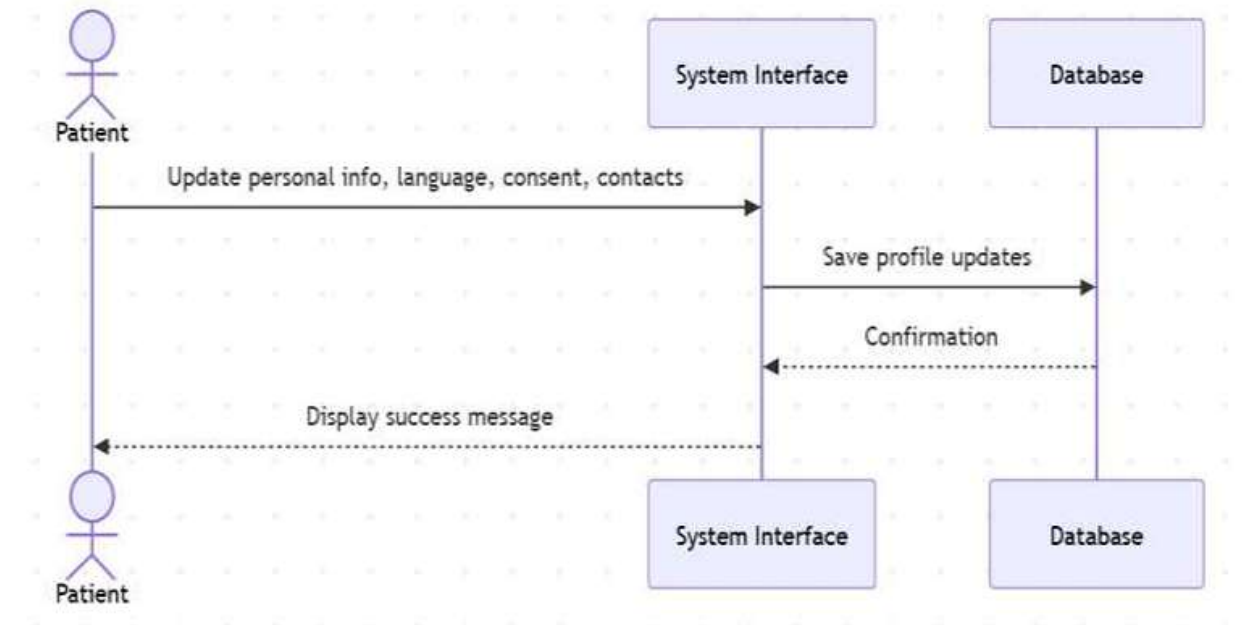
1. UC-01 Name Register and Login



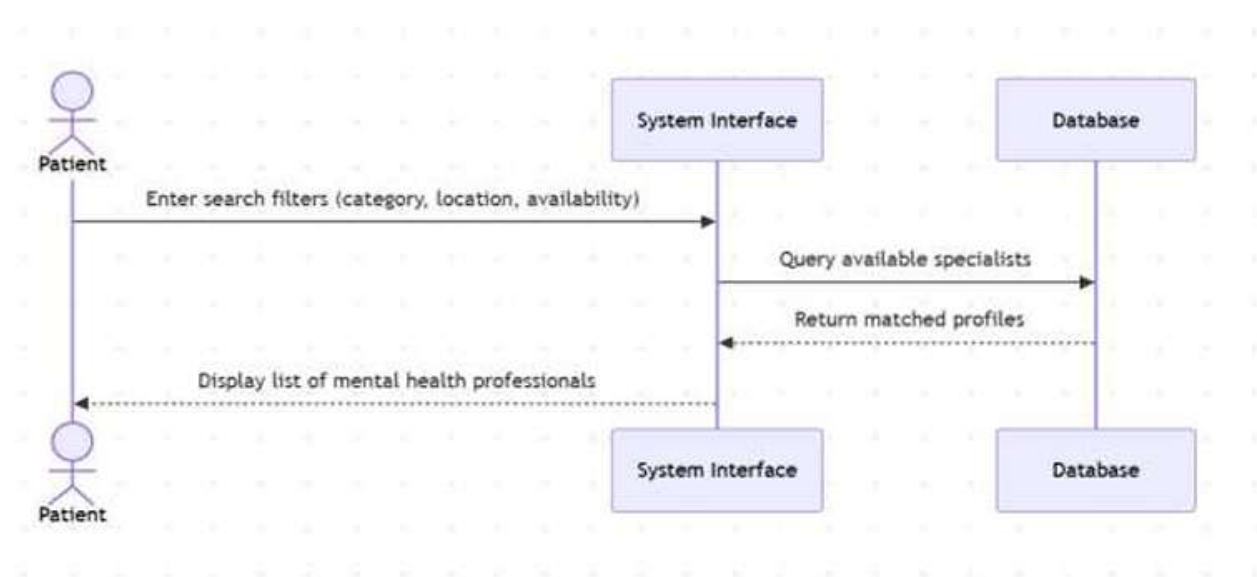


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2. UC-02 Manage User Profile and Consent



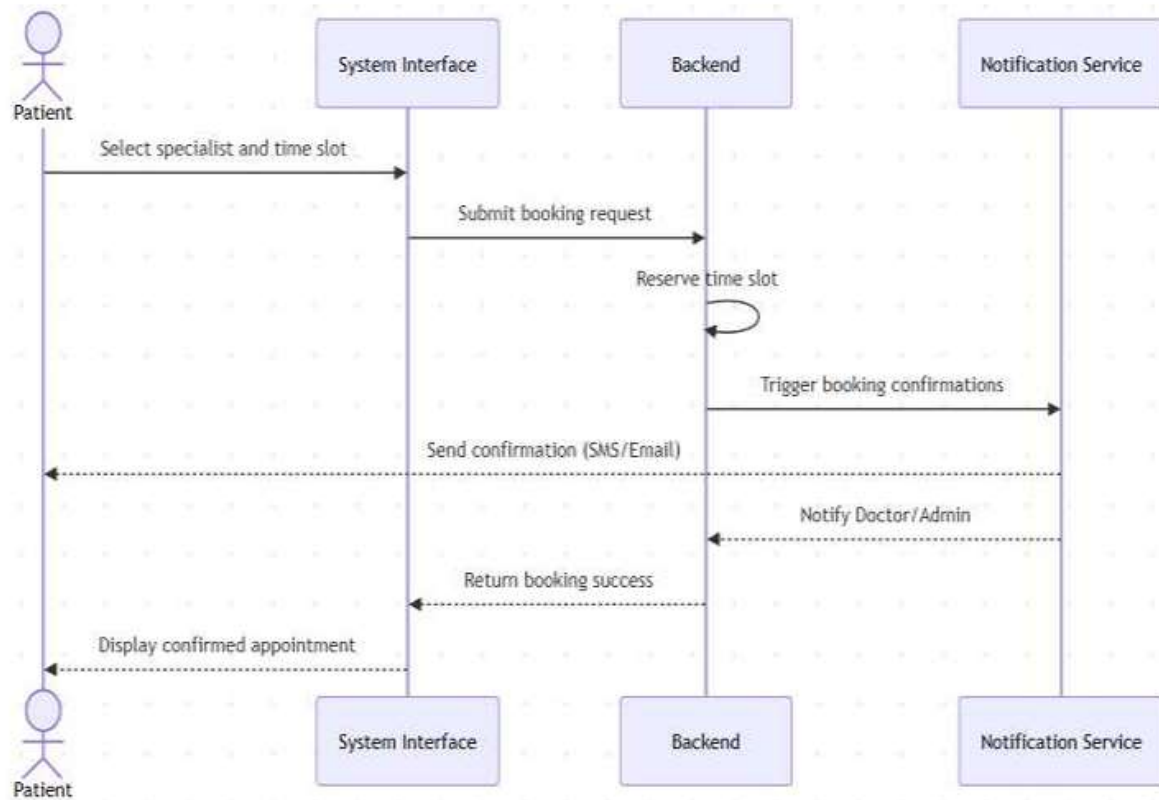
3. UC-03 Search Mental Health Specialist



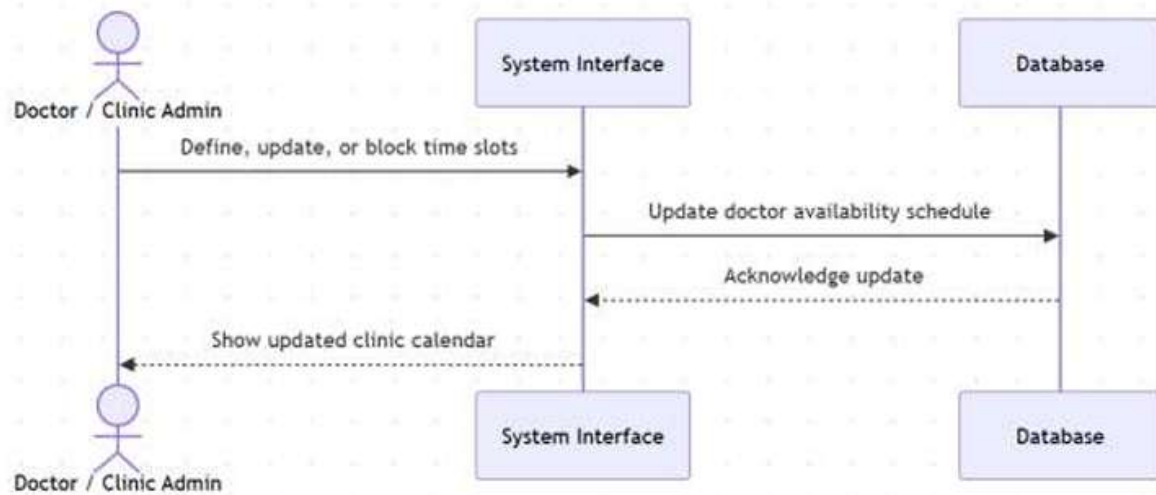


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4. UC-04 Book Appointment



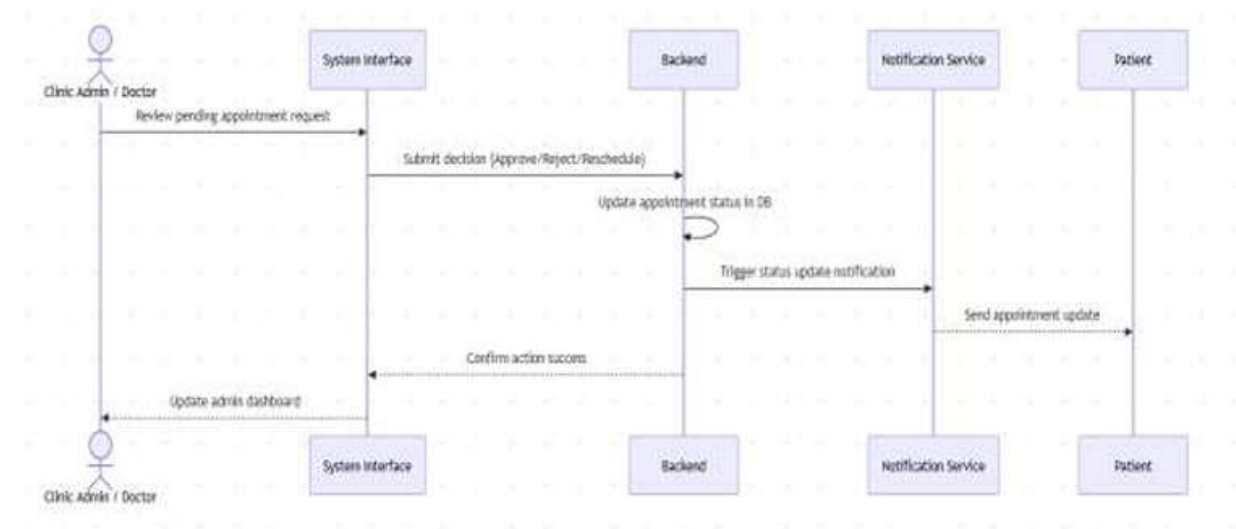
5. UC-05 Manage Doctor Availability



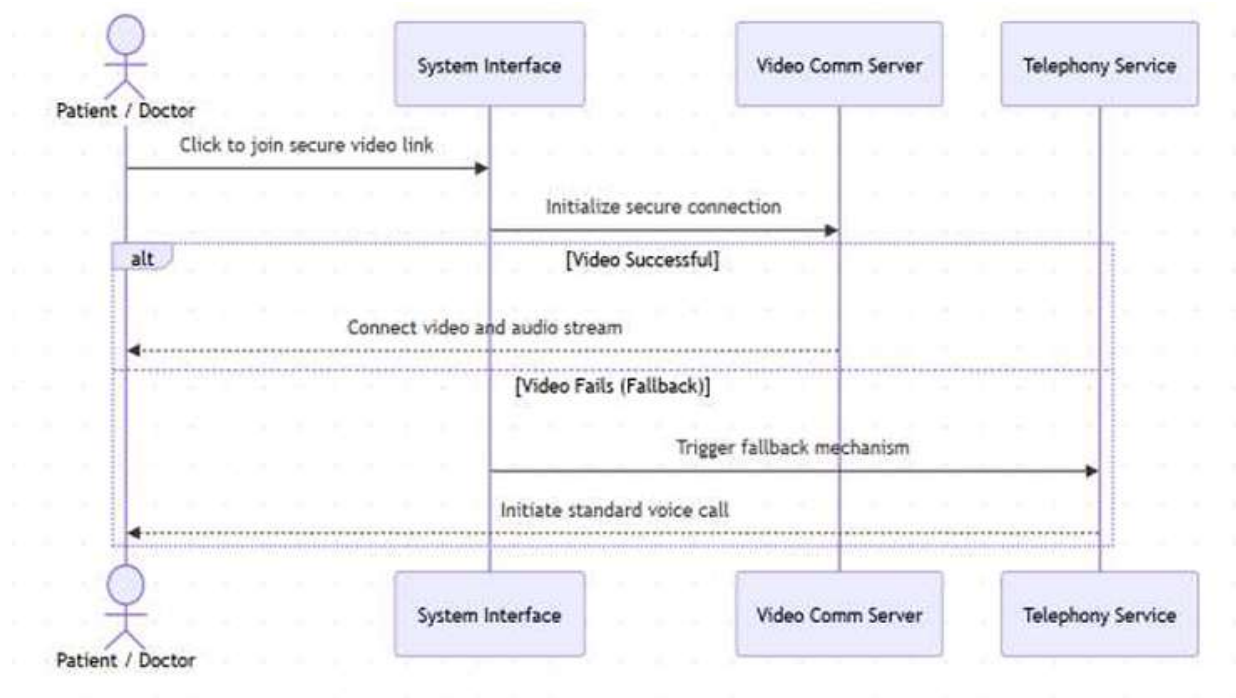


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6. UC-06 Approve / Reschedule Appointment



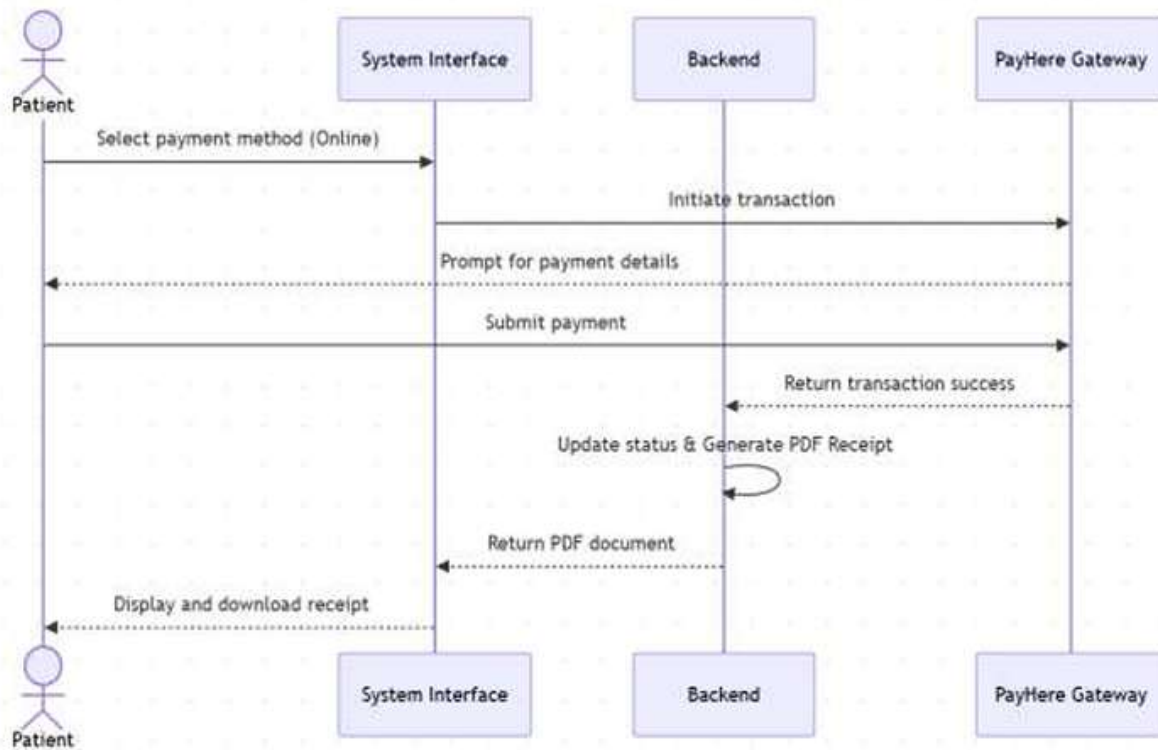
7. UC-07 Join Telemedicine Consultation



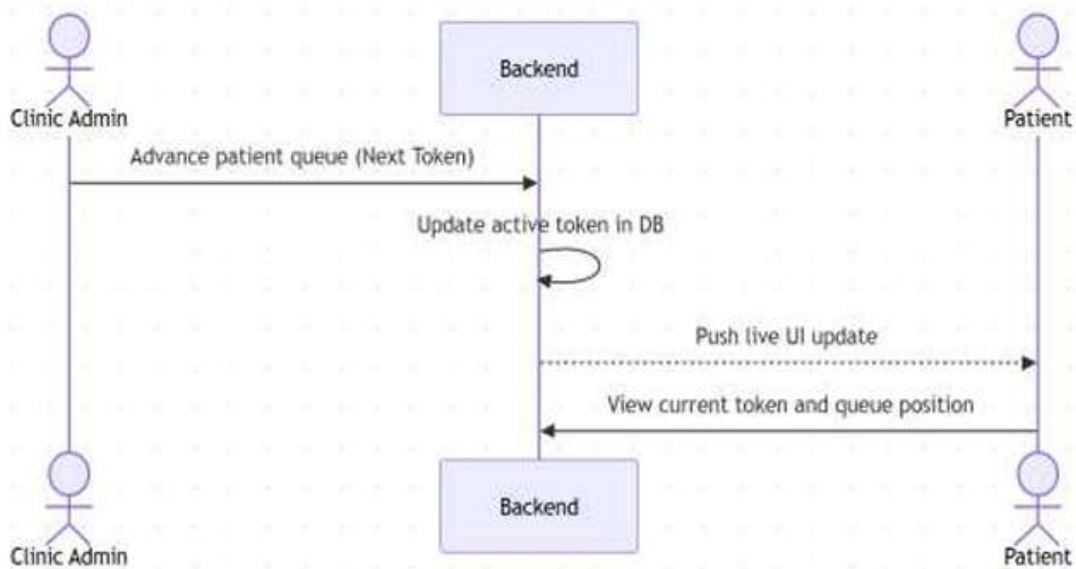


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8. UC-08 Process Payment and Generate Receipt



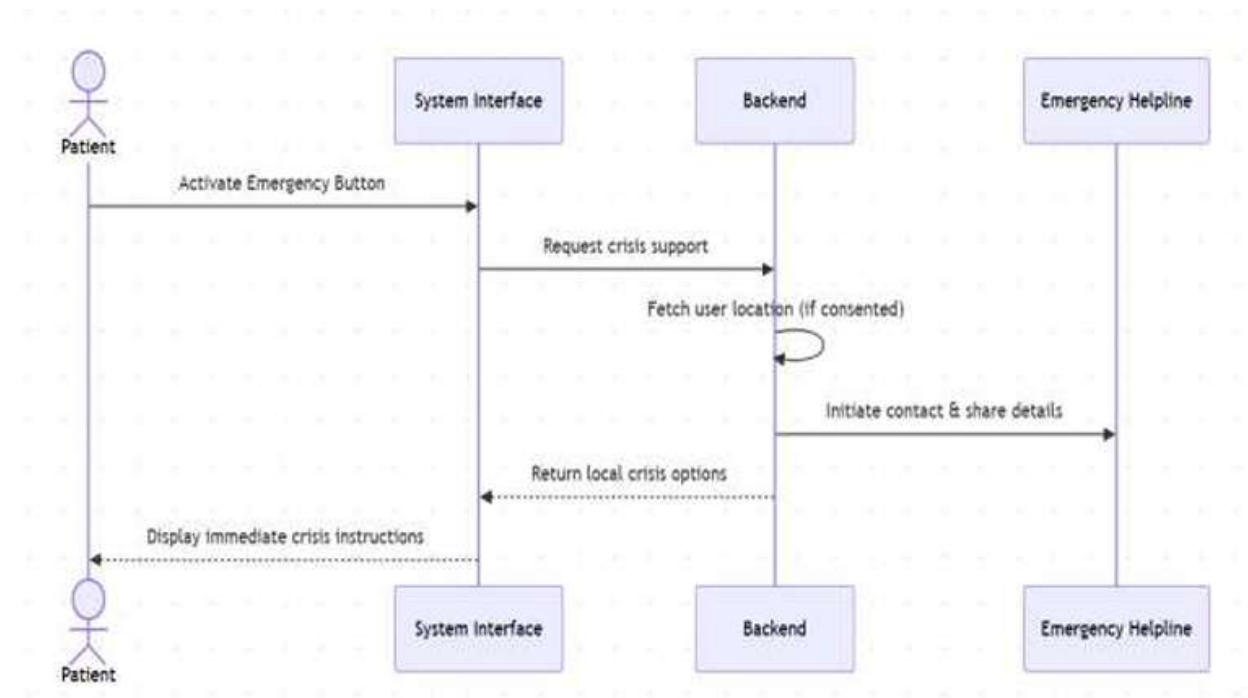
9. UC-09 Track Live Patient Queue



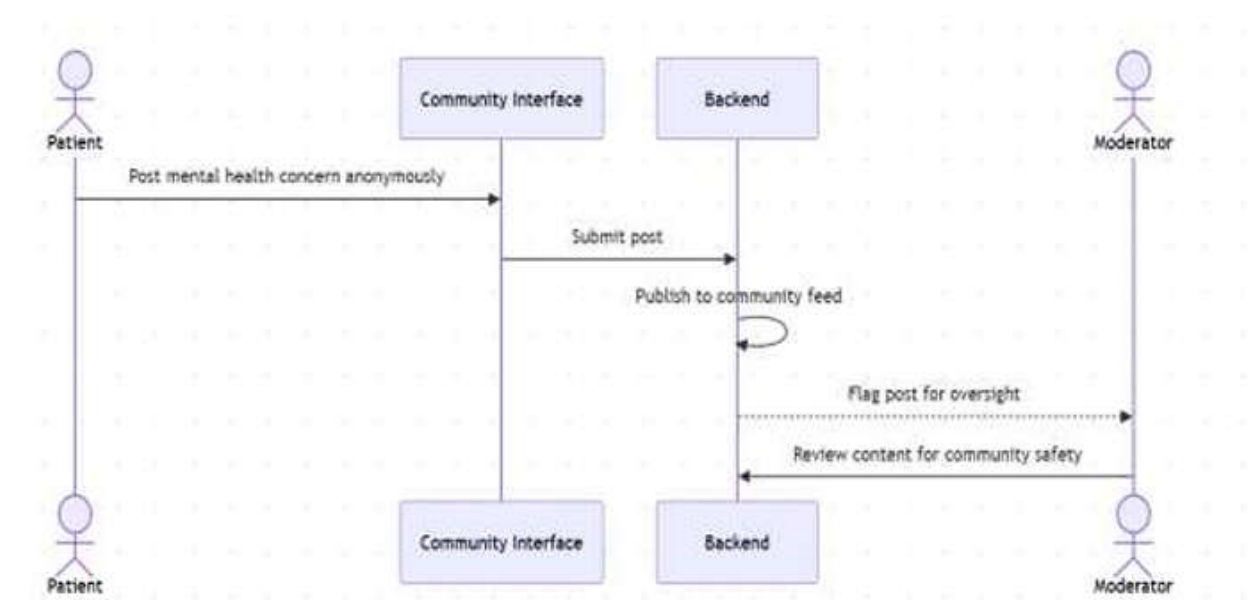


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10. UC-10 Request Emergency Support

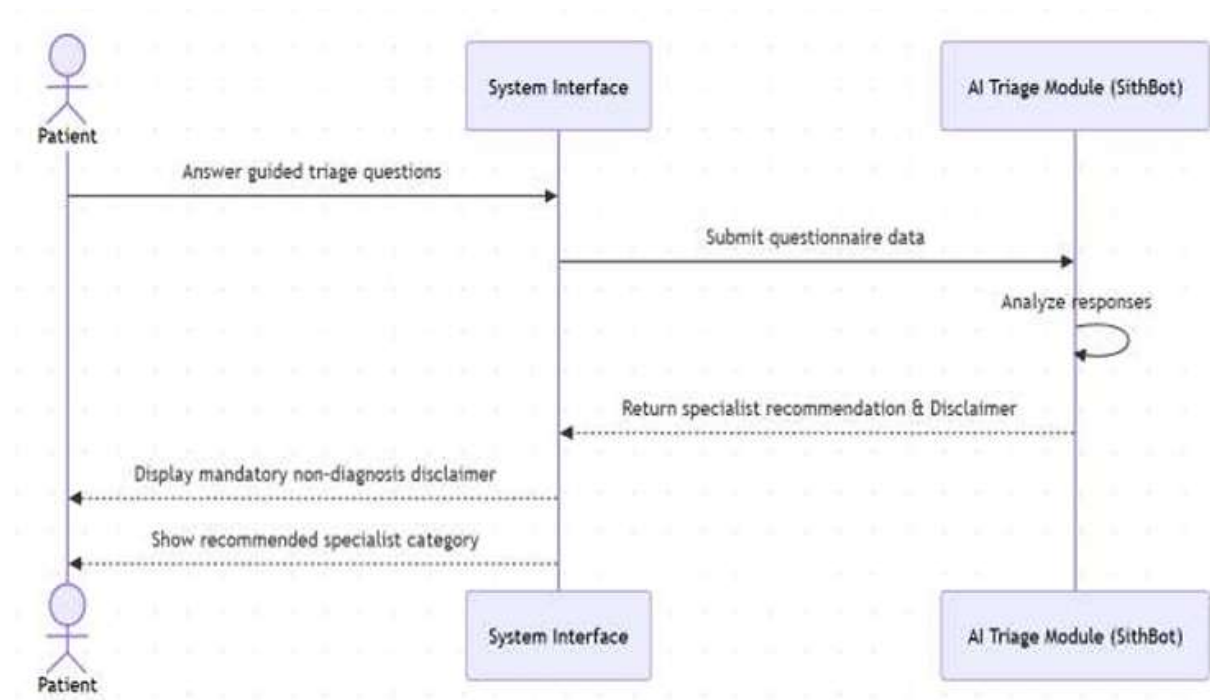


11. UC-11 Use Anonymous Community Consultation





12. UC-12 Complete AI Triage Questionnaire (SithBot)



A separate detailed document of the sequence diagram was created and uploaded to the Documents Repository of the Github Organization. Click the following links to access the document

Link1 - [Sequence Diagram in Github](#)

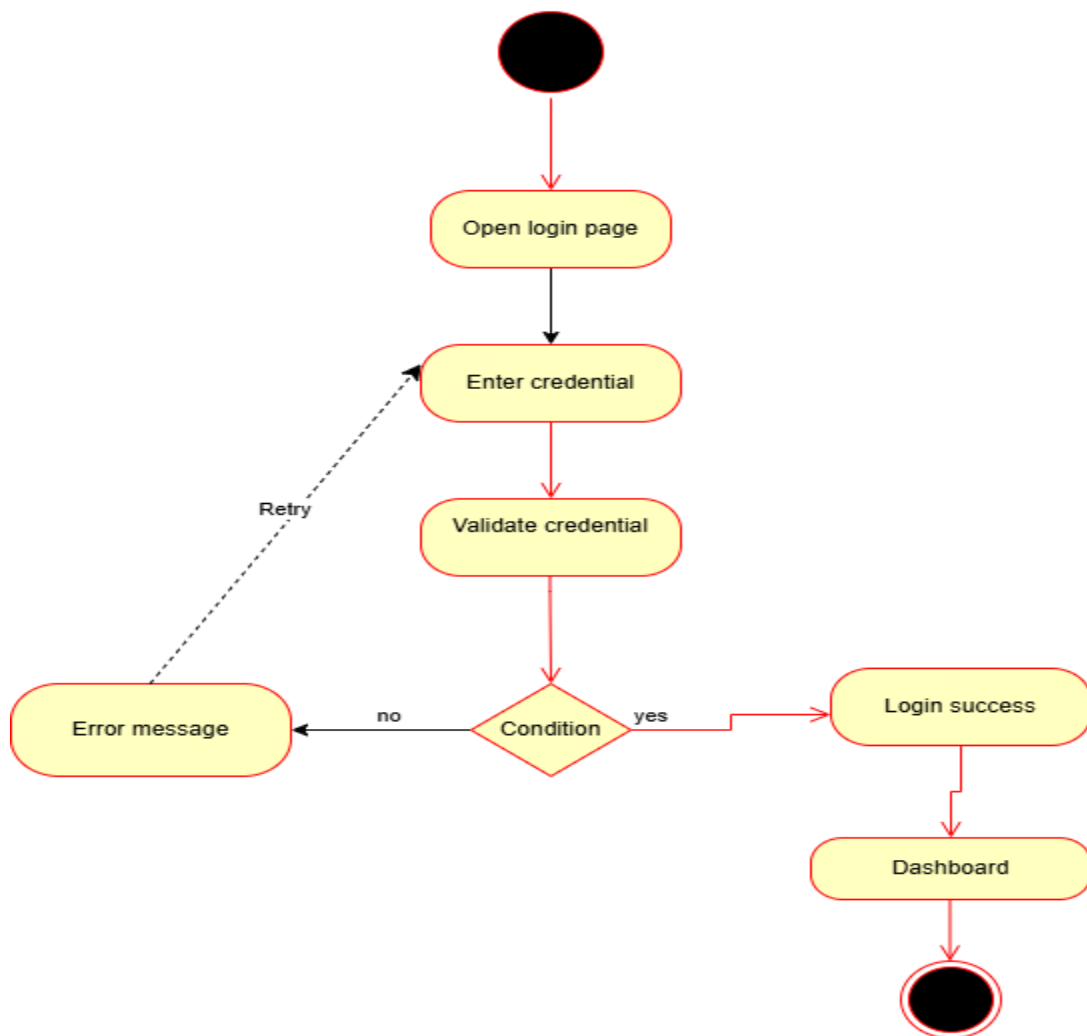
Link2 - [Sequence Diagram in Google Drive](#)



4.3. Activity Diagrams

Below are 5 key activity diagrams drawn to depict the behavior of the SithCare application.

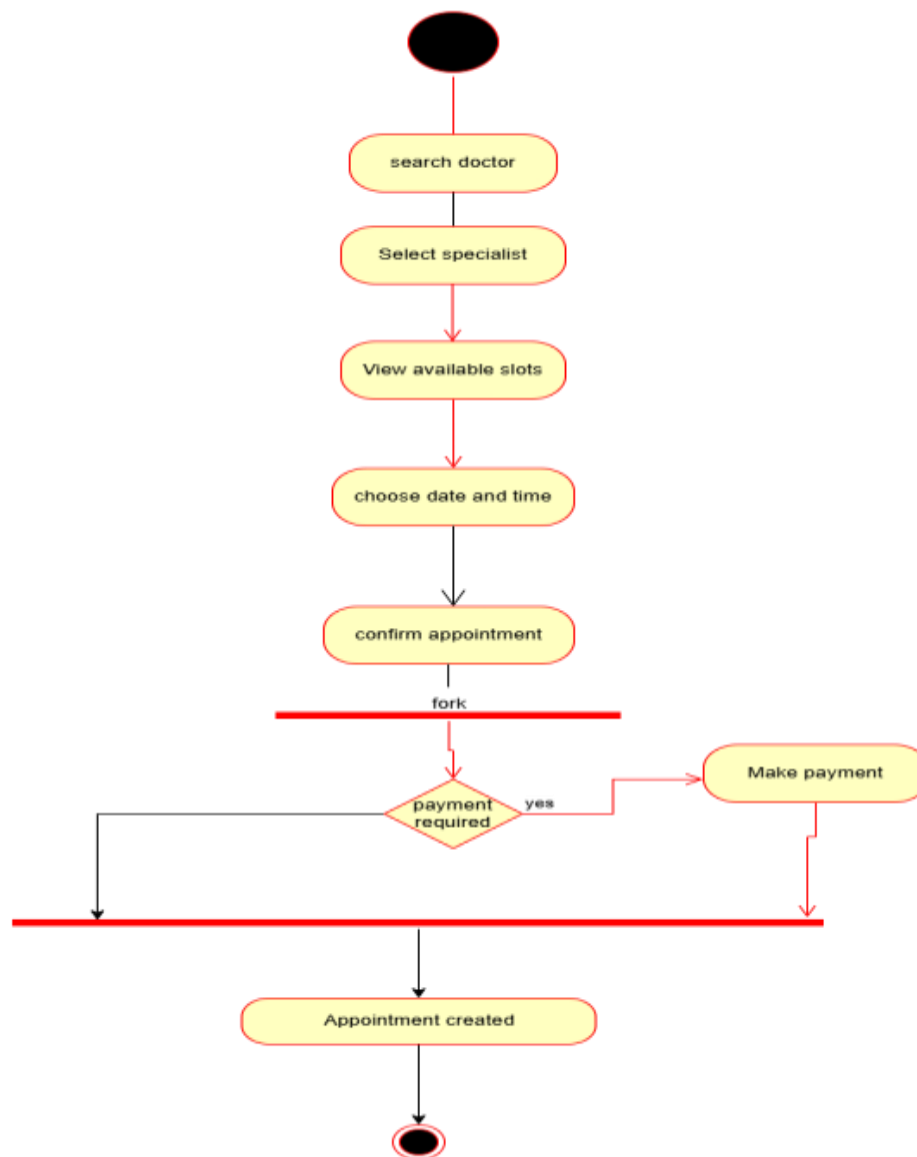
1. User Authentication



Source code used to draw diagram : [Source Code](#)



2. Booking an appointment

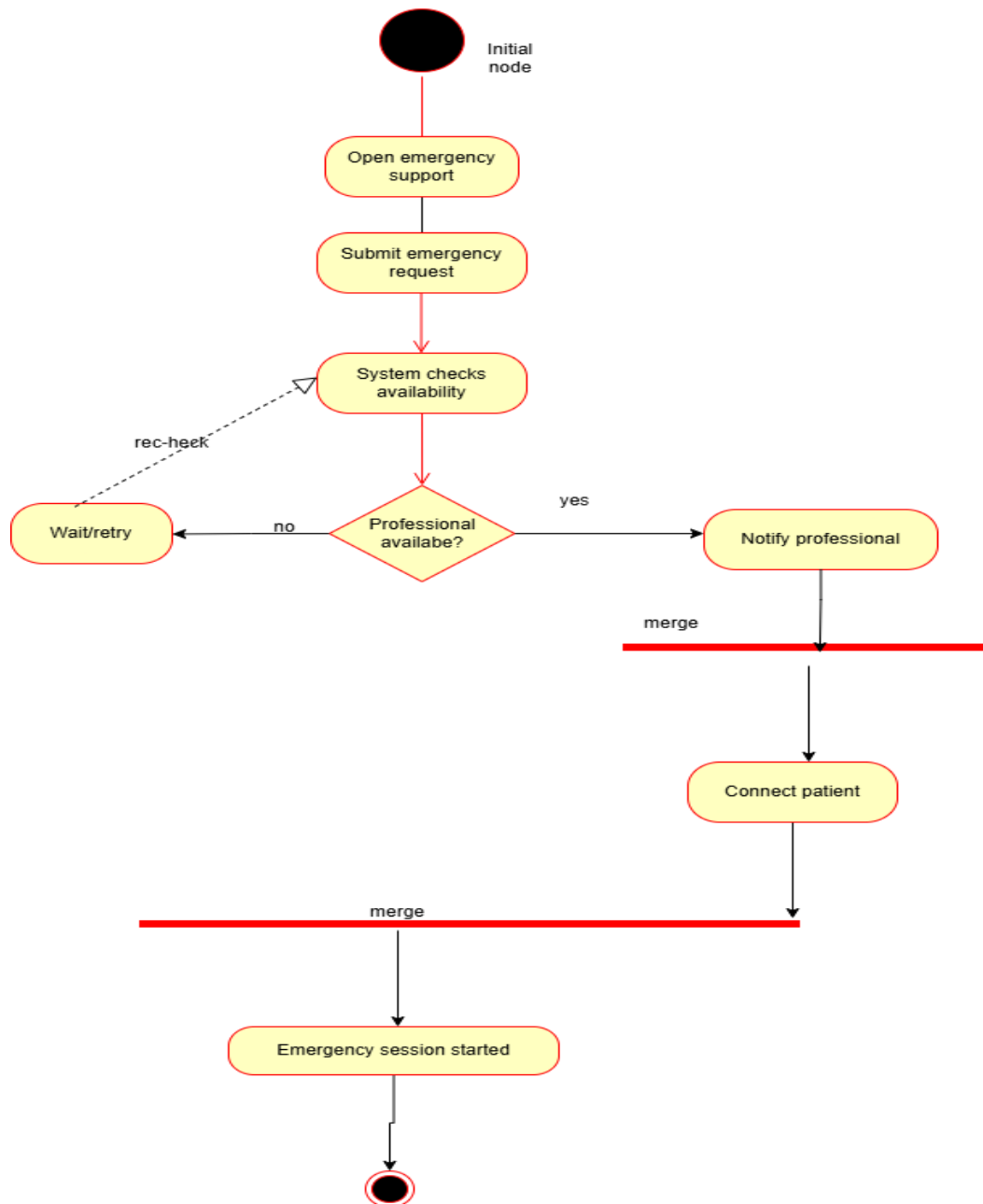


Source code used to draw diagram : [Source Code](#)



System Design Specification Document – SithCare Channeling App

3. Emergency-support

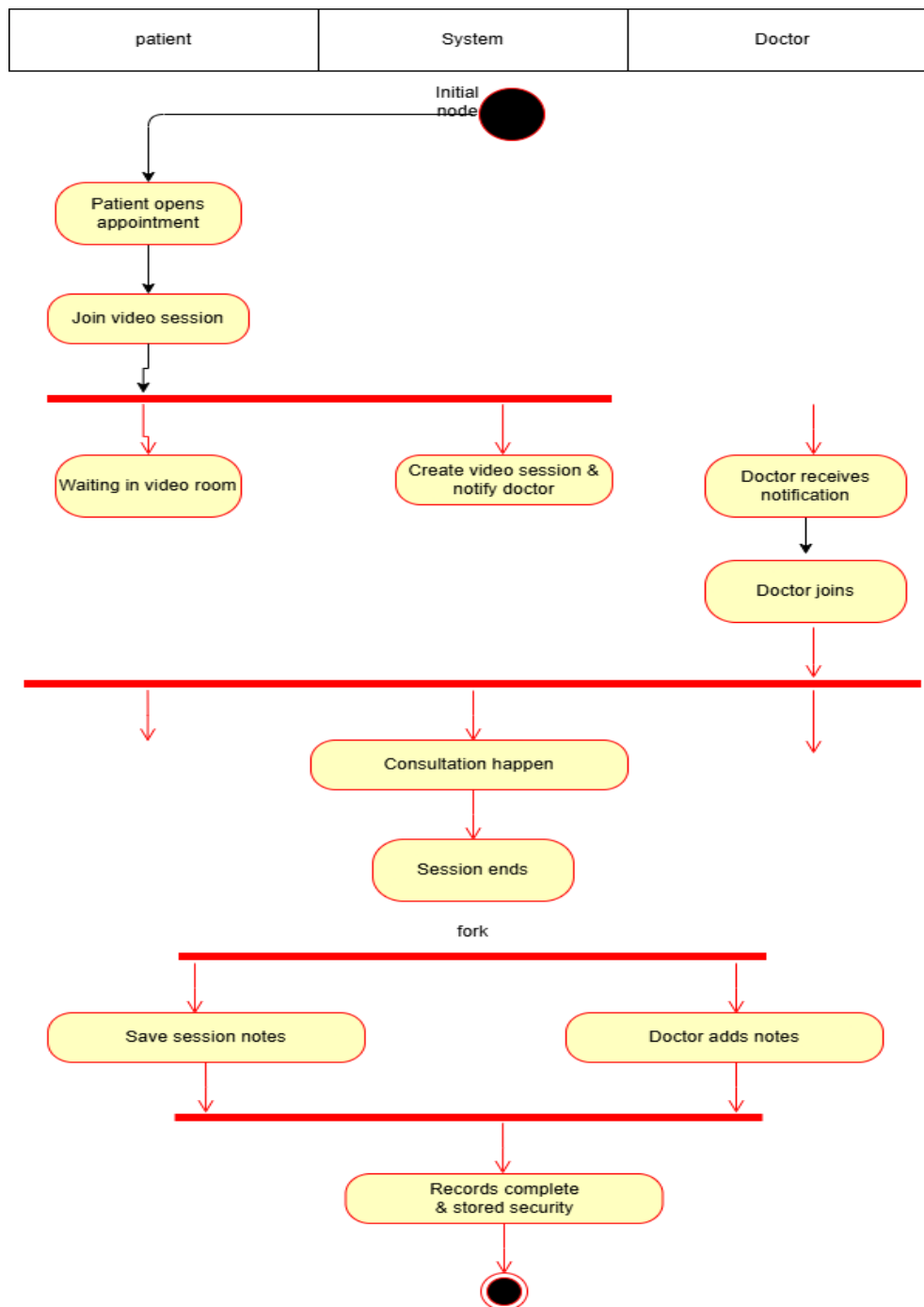


Source code used to draw diagram : [Source Code](#)



System Design Specification Document – SithCare Channeling App

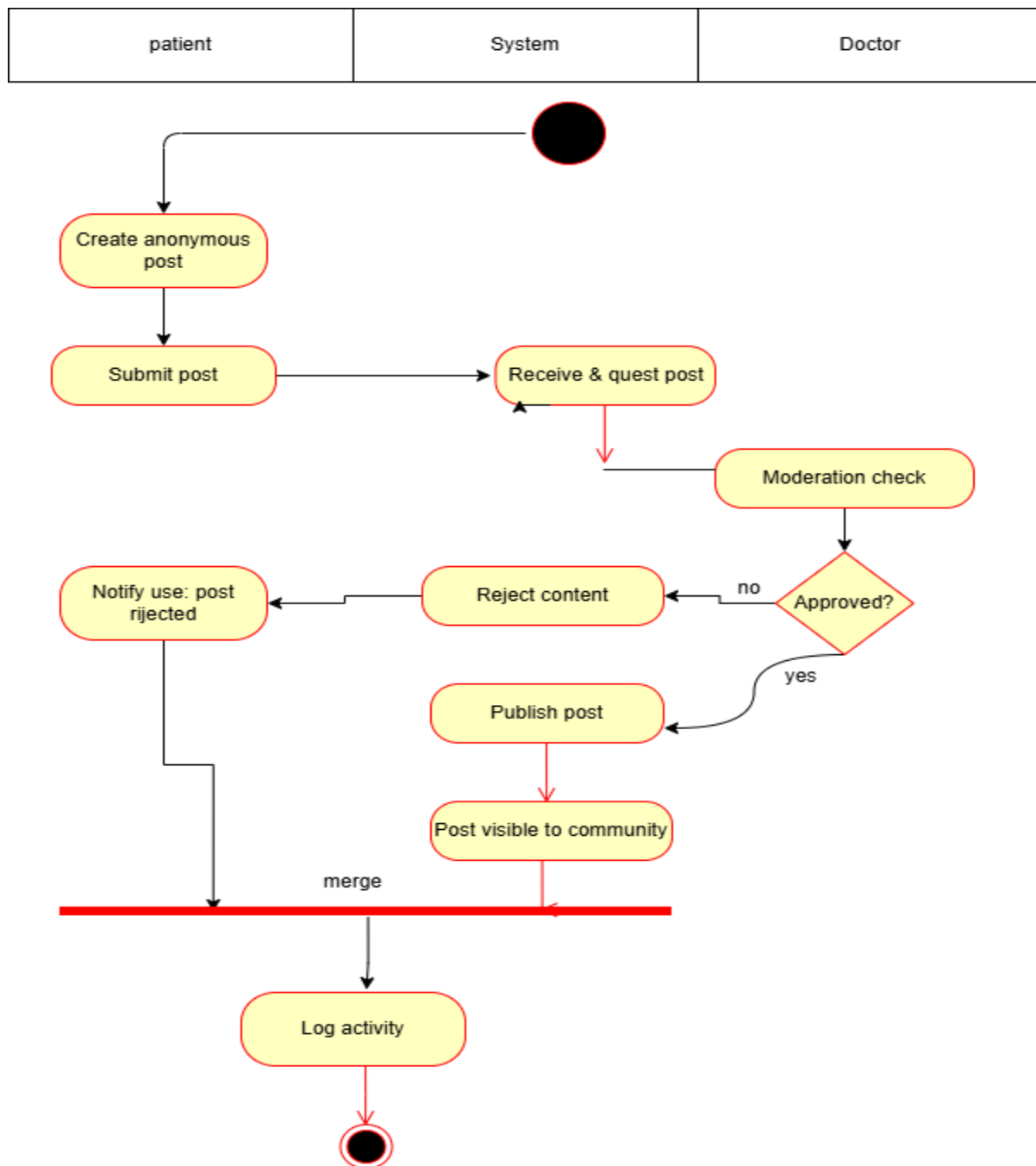
4. Tele-medicine Consultation



Source code used to draw diagram : [Source Code](#)



5. Community Posting

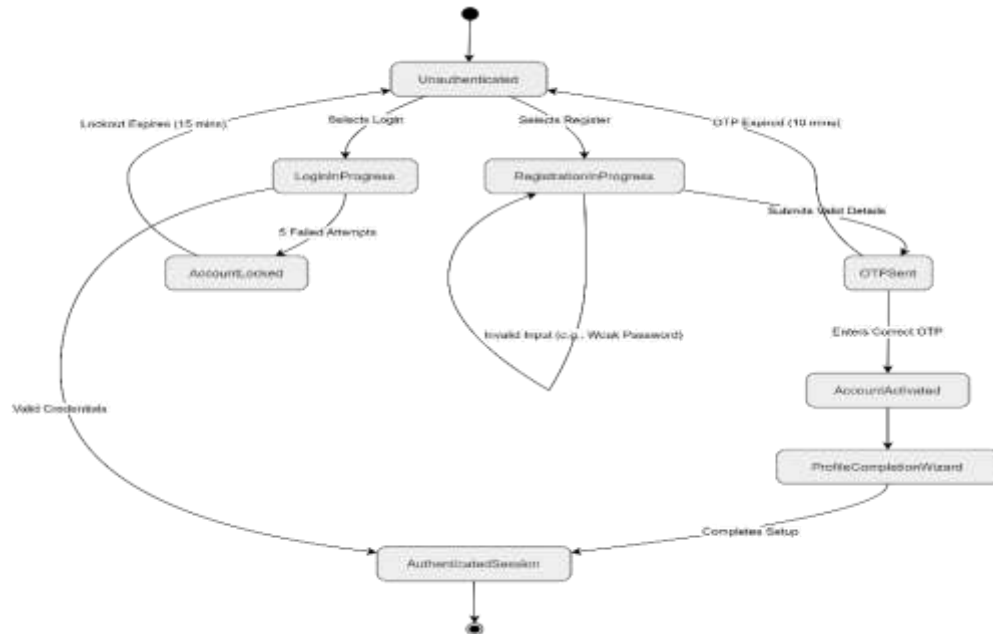


Source code used to draw diagram : [Source Code](#)

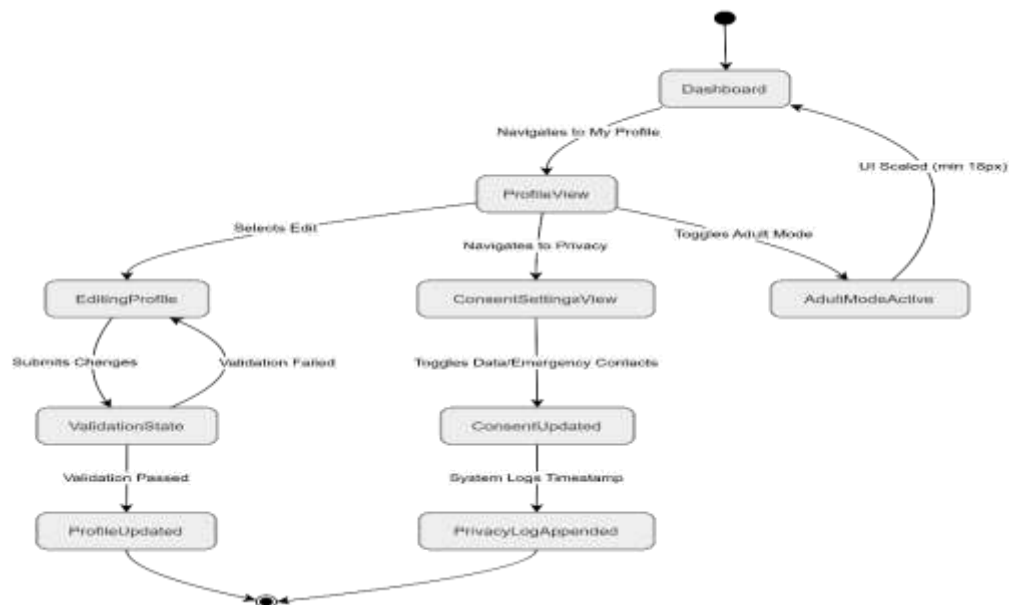


4.4. State Transition Diagrams

1. UC-01: Register and Login



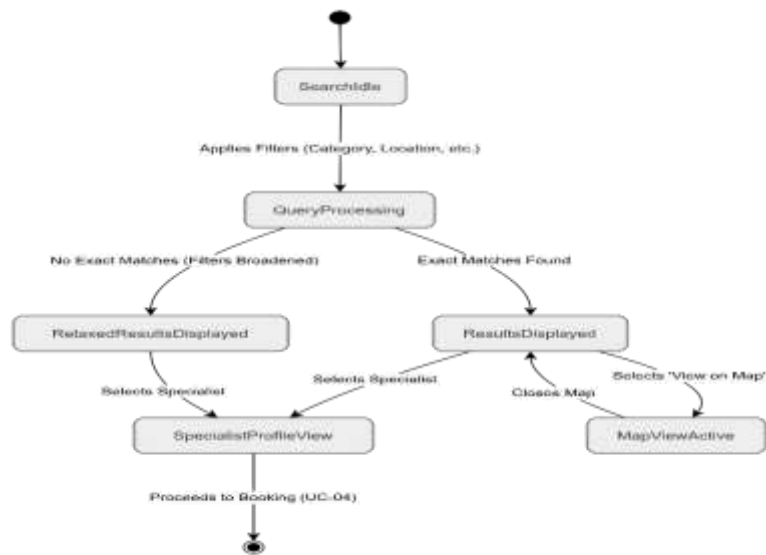
2. UC-02: Manage User Profile and Consent



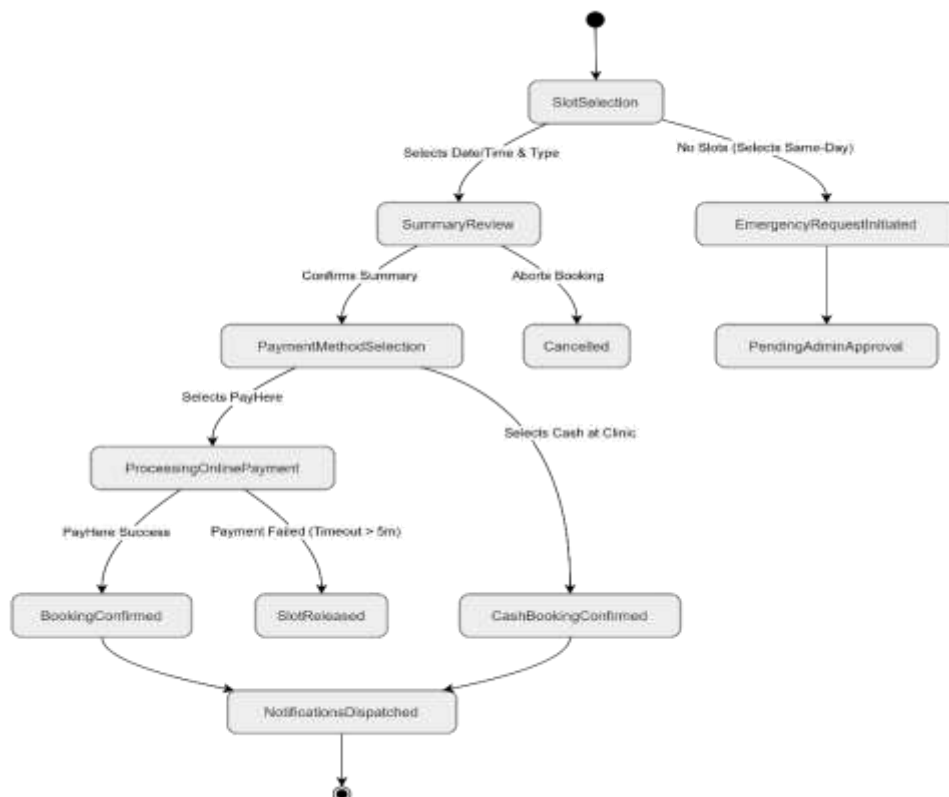
3. UC-03: Search Mental Health Specialist



System Design Specification Document – SithCare Channeling App



4. UC-04: Book Appointment

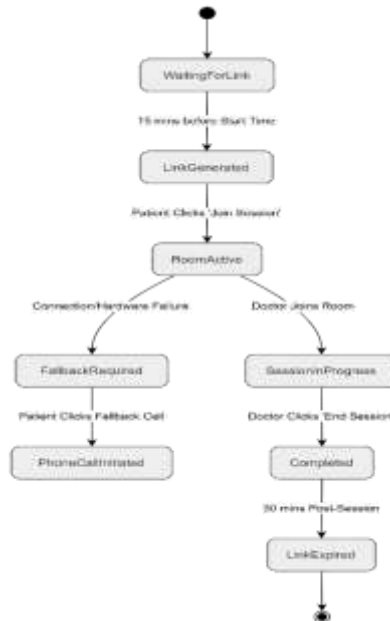


5. UC-05: Manage Doctor Availability

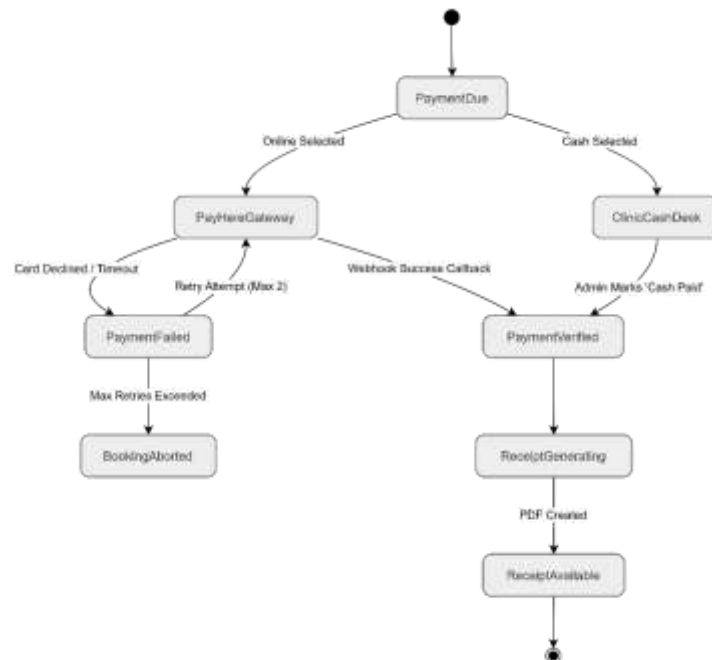




System Design Specification Document – SithCare Channeling App



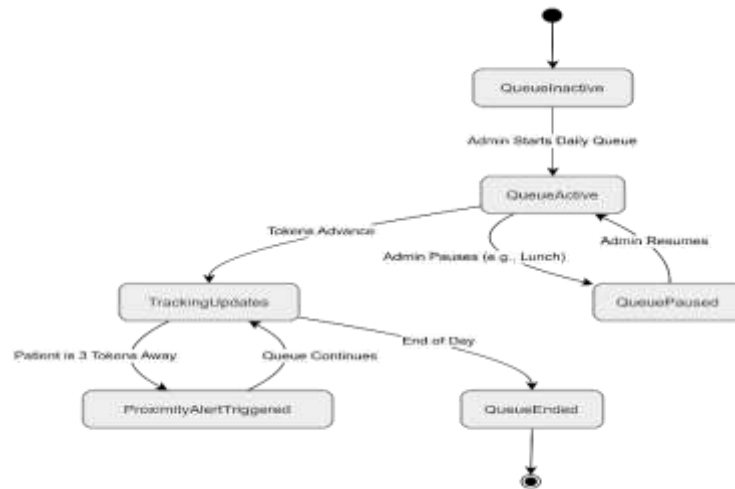
8. UC-08: Process Payment and Generate Receipt



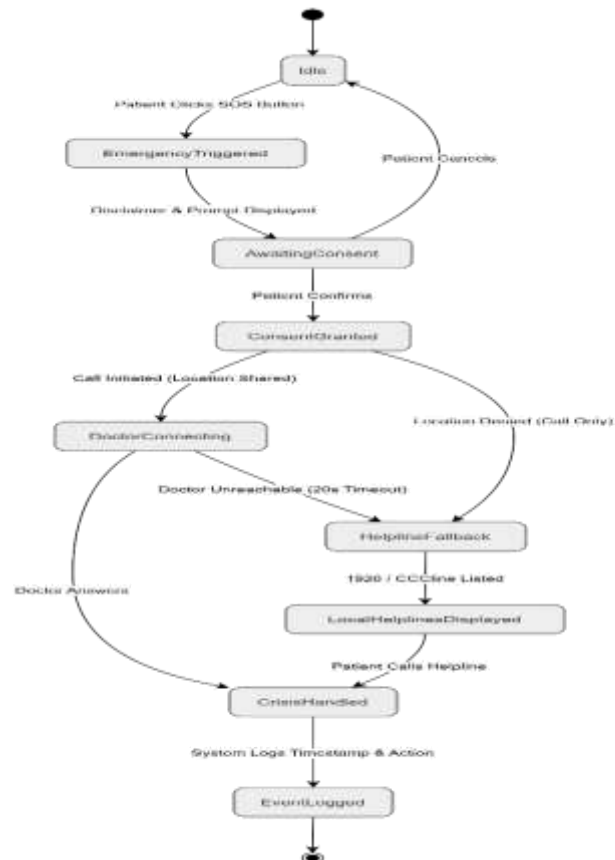
9. UC-09: Track Live Patient Queue



System Design Specification Document – SithCare Channeling App



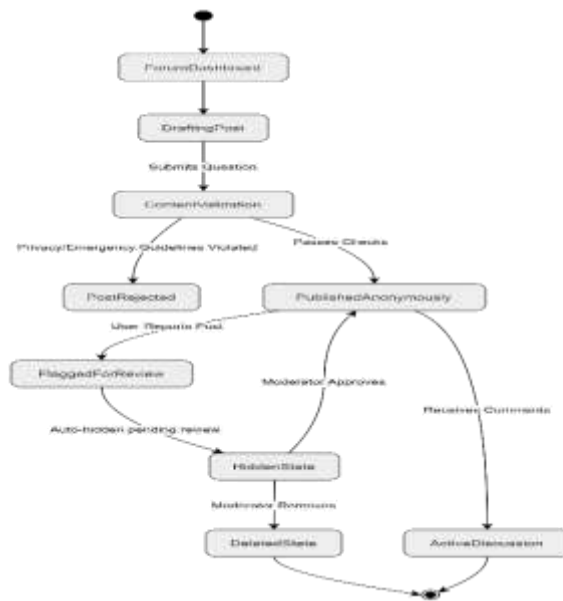
10. UC-10: Request Emergency Support



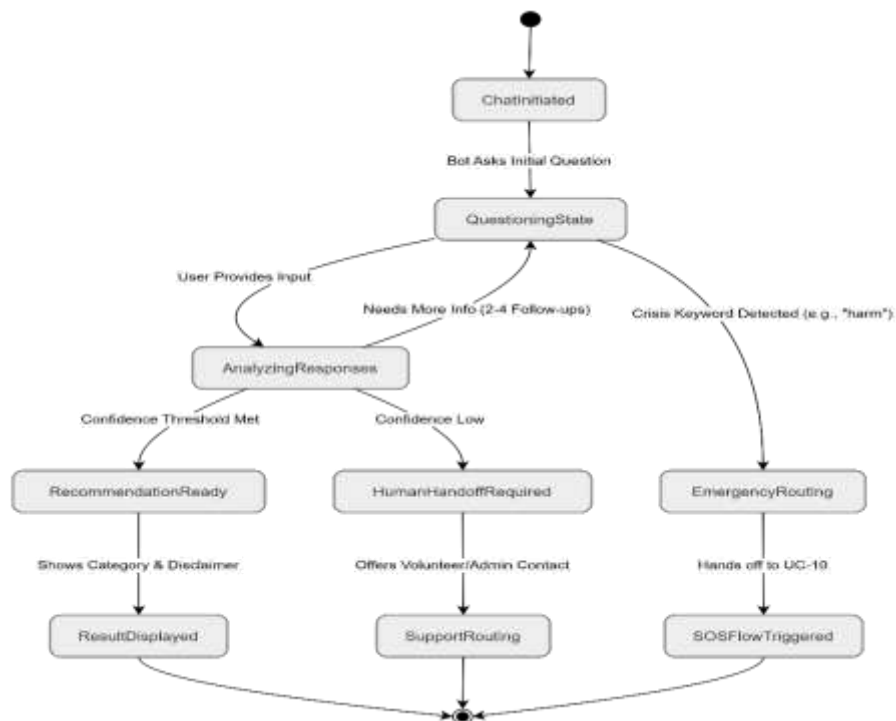
11. UC-11: Use Anonymous Community Consultation



System Design Specification Document – SithCare Channeling App



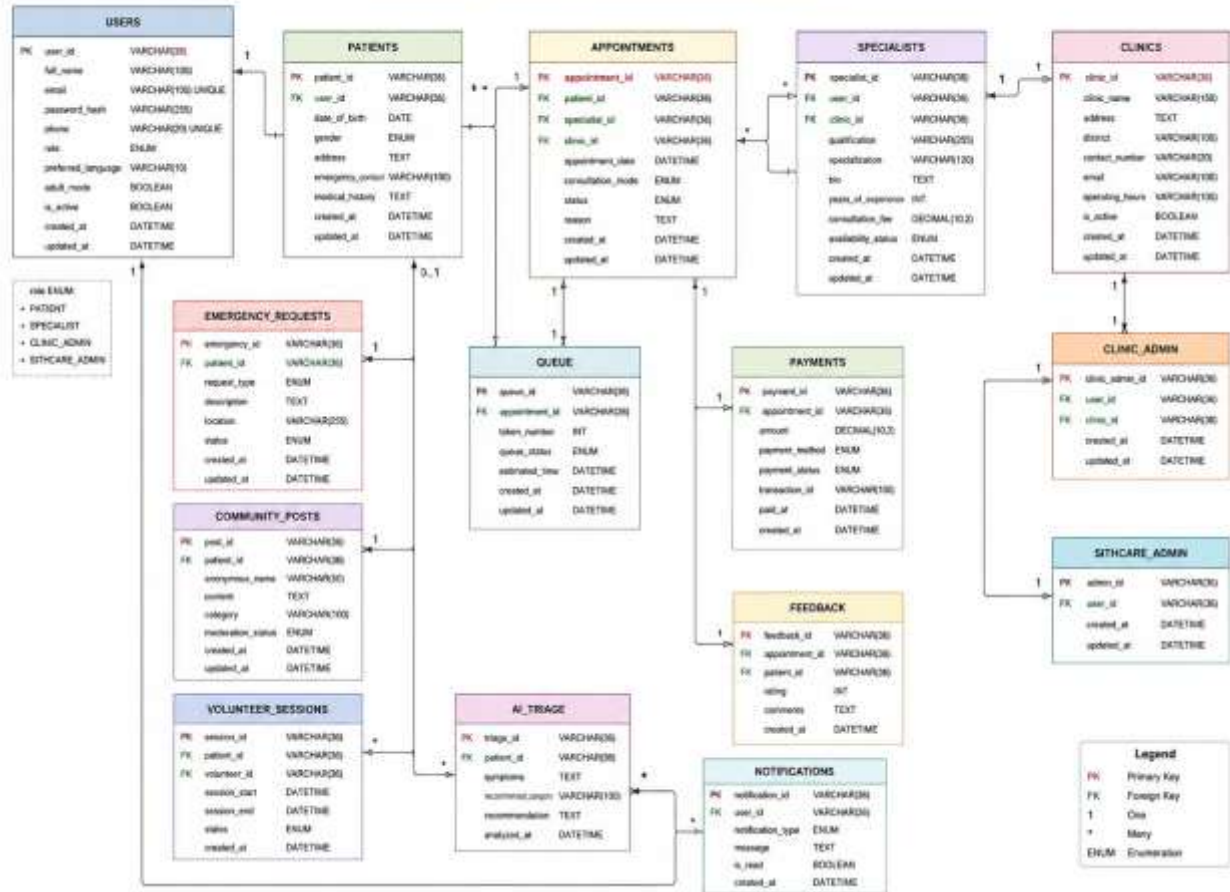
12. UC-12: Complete AI Triage Questionnaire (SithBot)





4.5. Entity-Relationship diagram

SithCare – Entity Relationship Diagram





5. System Design

This section details the frontend visual layout, core interface design, and layout architectures for the SithCare platform. To provide an optimal, role-specific user experience, the system's interface is divided into four main dashboard interfaces. Each interface is designed to meet the distinct functional and operational boundaries of its target users.

5.1. Clinic-Admin Dashboard

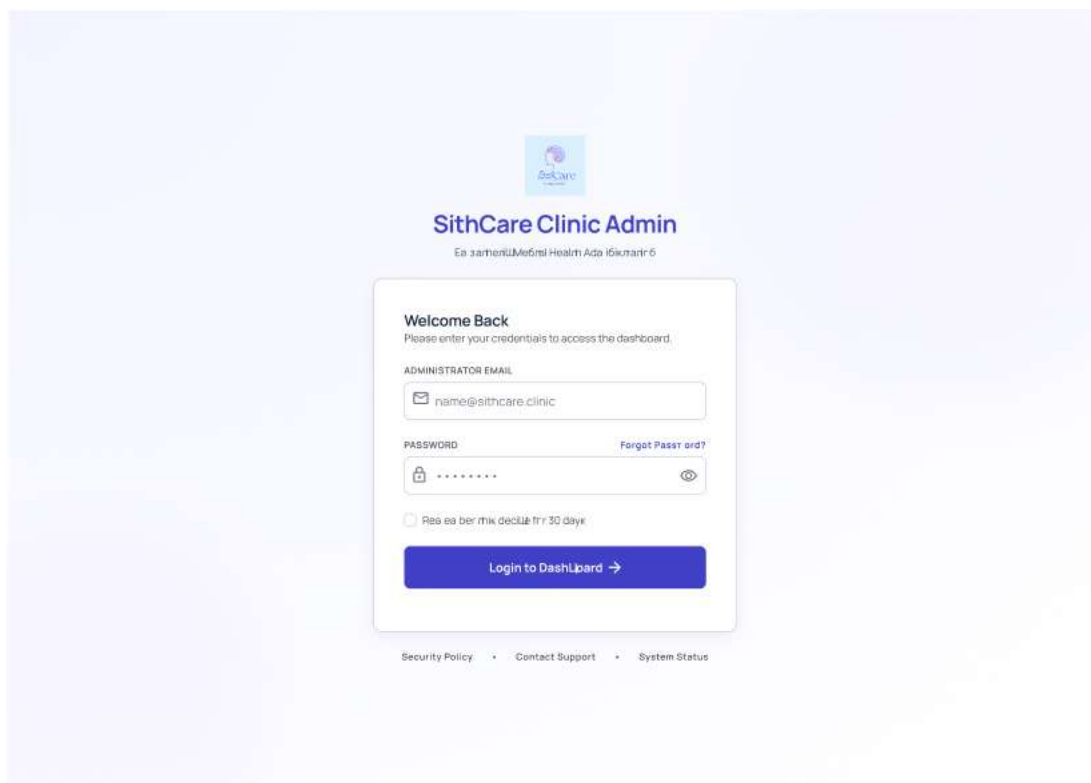


Figure 5.1.1: Login Page



System Design Specification Document – SithCare Channeling App

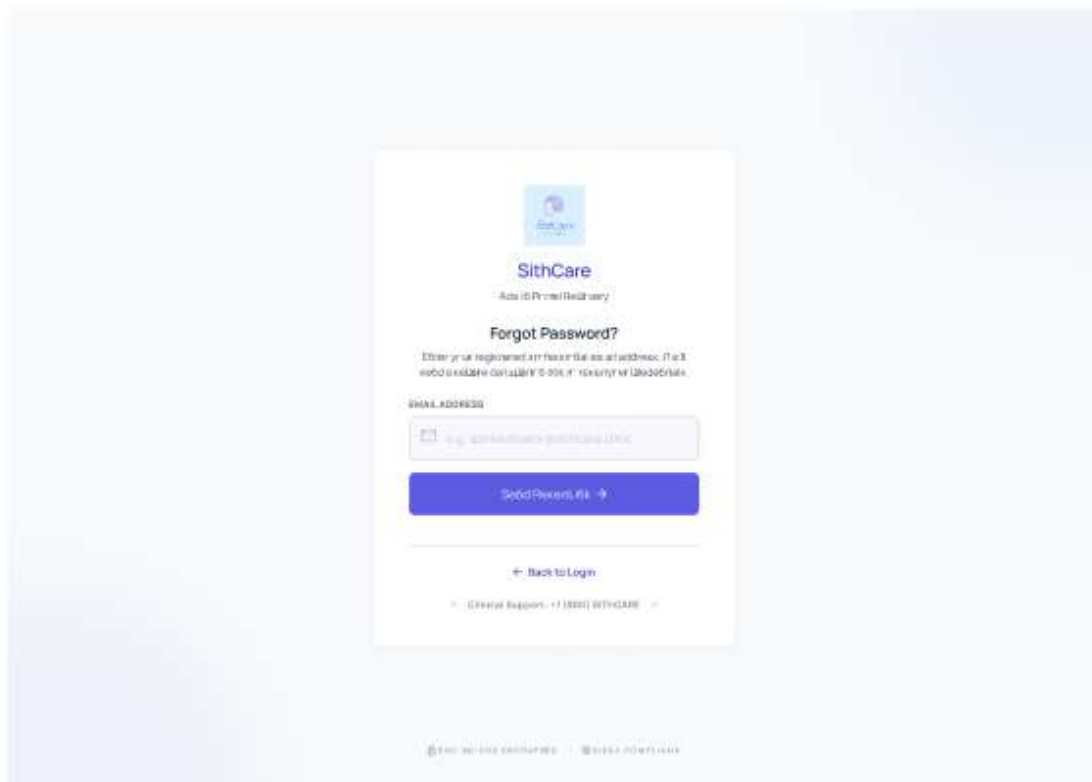


Figure 5.1.2: Forgot Password

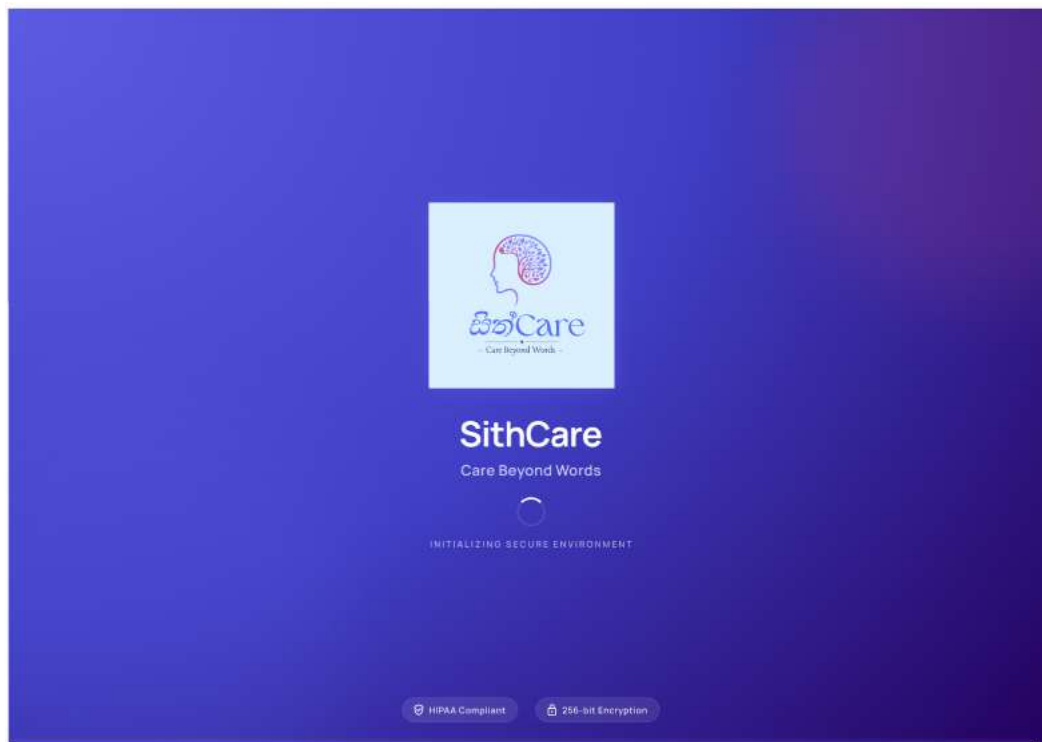


Figure 5.1.3: Splash Screen (Standardized)



System Design Specification Document – SithCare Channeling App

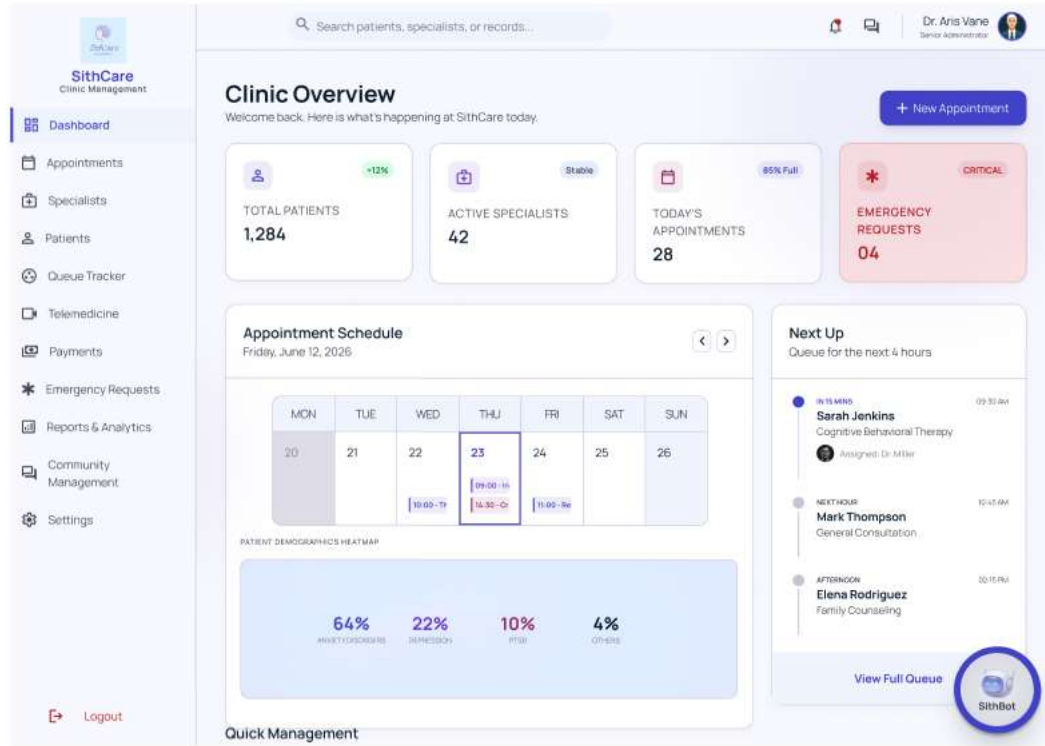


Figure 5.1.4: Dashboard Home (Scrollable)

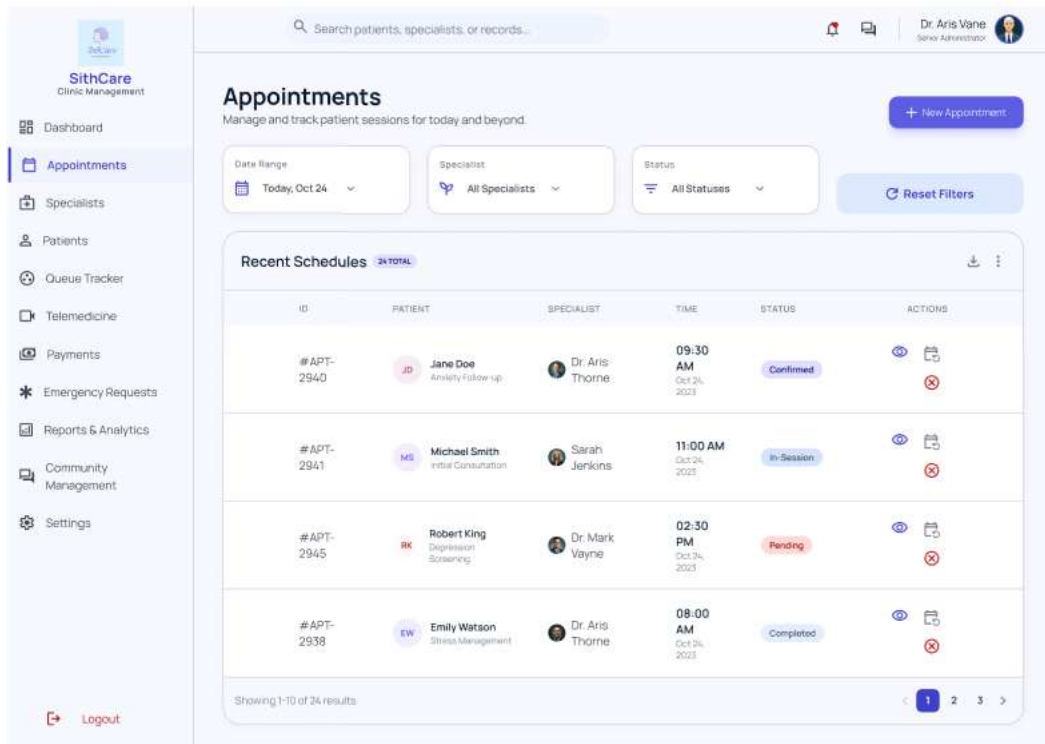


Figure 5.1.5: Appointment Management (Scrollable)



System Design Specification Document – SithCare Channeling App

SithCare
Clinic Management

Dashboard
Appointments
Specialists
Patients
Queue Tracker
Telemedicine
Payments
Emergency Requests
Reports & Analytics
Community Management
Settings

Search patients, specialists, or records...

Dr. Aris Vane
Senior Administrator

Appointments > Add New Appointment

Schedule Clinical Session

Step 1: Patient Selection & Time Allocation

STEP 01 OF 03

Select Patient [New Patient Record](#)

Search by name, patient ID, or mobile...

Arjuna Wijesekara
ID: SC-2024-8012 • Last visit: 2 weeks ago
[Selected](#) [View History](#)

Nimali Perera
ID: SC-2024-8846 • Last visit: 3 days ago

Kamal Gunawardena
ID: SC-2024-7712 • Last visit: 1 month ago

Shanika Abeywickrama
ID: SC-2024-5541 • Last visit: Never

Schedule Date < May 2024 >

S	M	T	W	T	F	S
25	26	27	28	1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24

Available Slots

09:00 AM 10:30 AM
11:00 AM 01:30 PM
03:00 PM 04:30 PM

[Cancel Scheduling](#) [Proceed to Specialist](#)

[Logout](#)

Figure 5.1.6: Add Appointment (Scrollable)

SithCare
Clinic Management

Dashboard
Appointments
Specialists
Patients
Queue Tracker
Telemedicine
Payments
Emergency Requests
Reports & Analytics
Community Management
Settings

Search patients, specialists, or records...

Dr. Aris Vane
Senior Administrator

Appointment Successfully Scheduled

Your session has been confirmed and added to the clinic's master queue. A notification has been sent to the patient.

Patient
Arjuna Wijesekara
ID: PT-8821

Specialist
Dr. Aris Thorne
Clinical Psychologist

Time
11:00 AM
Tuesday, Oct 24

[View Appointment](#) [Return to Dashboard](#)

[Need to change this? Reschedule Appointment](#)

CONFIRMATION REF: SCH-002831-X

[Logout](#)

Figure 5.1.7: Appointment Success (Standardized)



System Design Specification Document – SithCare Channeling App

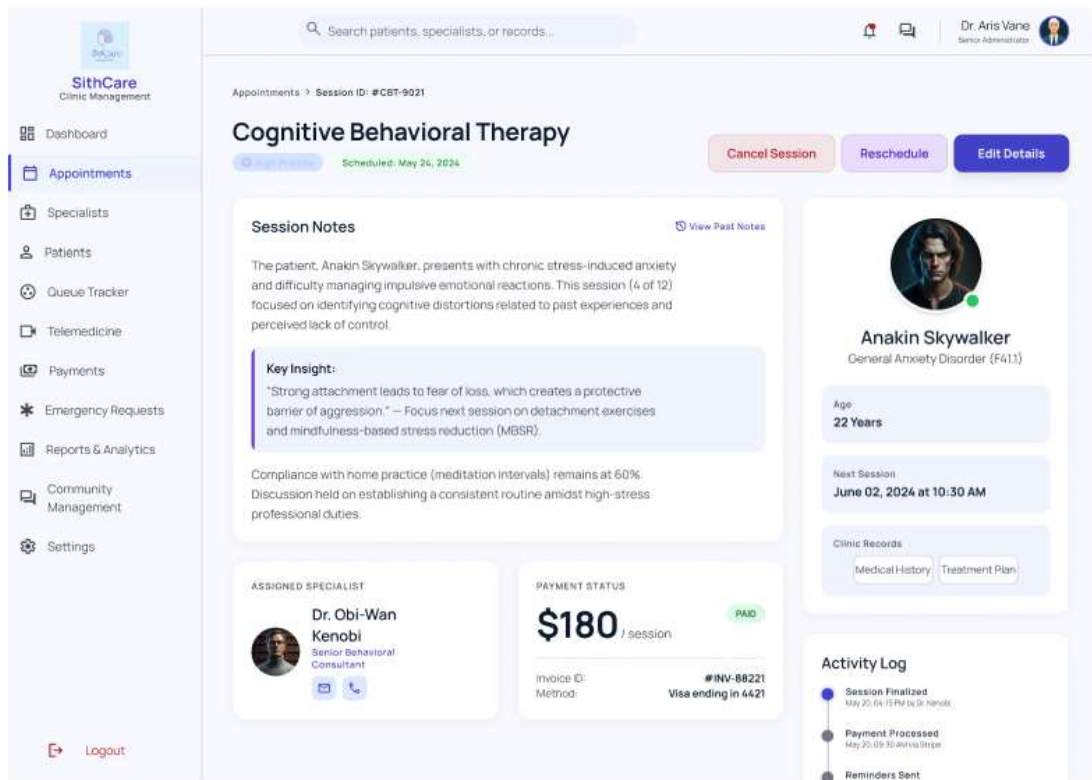


Figure 5.1.8: Appointment Details (Standardized)

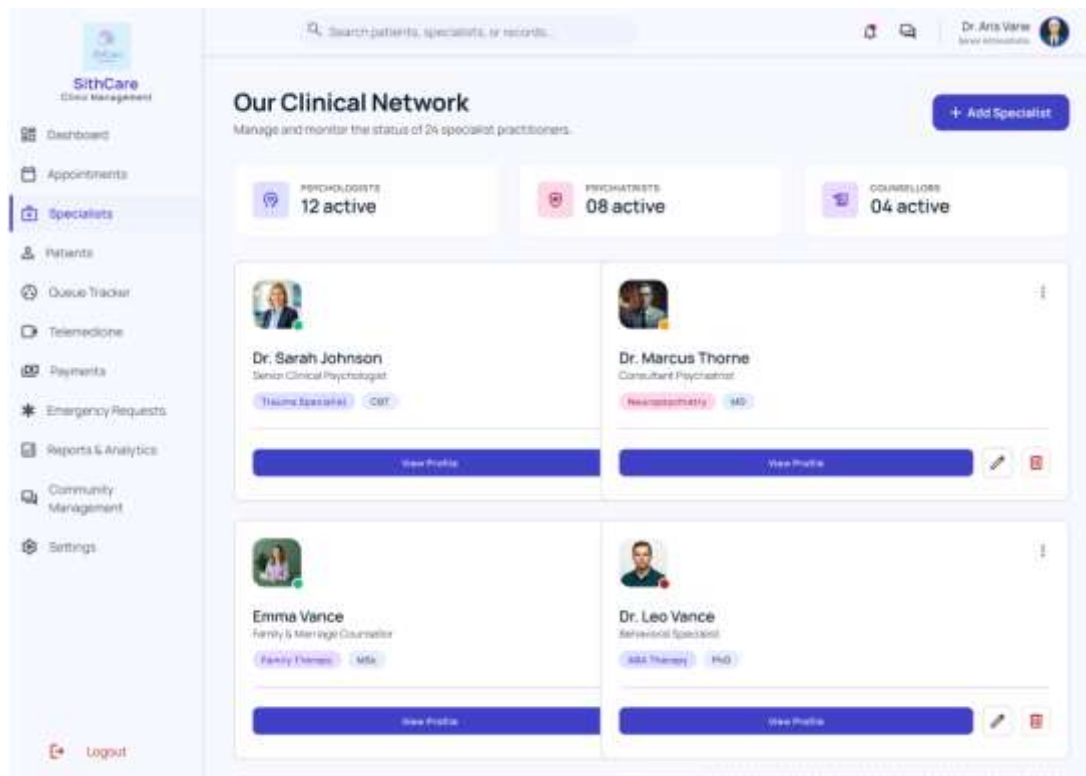


Figure 5.1.9: Specialist Management (Scrollable)



System Design Specification Document – SithCare Channeling App

SithCare
Clinic Management

Search patients, specialists, or records...

Dr. Aris Vane
Service Administrator

Specialists > Edit Profile

Edit Specialist Profile

Modify personal information, medical expertise, and clinical availability.

Cancel Save Changes

General Information

Full Name: Dr. Marcus Thorne Specialty: Clinical Psychology

Professional Biography

Dr. Marcus Thorne is a leading Clinical Psychologist with over 15 years of experience specializing in trauma-informed care and anxiety disorders. He has pioneered several mindfulness-based interventions currently utilized at SithCare. Previously, he served as the Head of Behavioral Sciences at the New Haven Institute. His approach balances clinical rigor with deep empathetic connection, ensuring every patient feels heard and supported through their recovery journey.

Expertise & Qualifications

License Number: PSY-99285-TX Years of Experience: 15

Clinical Focus Areas: Trauma Recovery X Anxiety Disorders X CBT X + Add Area

Availability

Mon: 09:00 AM - 05:00 PM
Tue: 09:00 AM - 05:00 PM
Wed: 10:00 AM - 04:00 PM
Thu: 09:00 AM - 05:00 PM
Fri: Unavailable

Logout

Figure 5.1.10: Edit Specialist (Scrollable)

SithCare
Clinic Management

Search patients, specialists, or records...

Dr. Aris Vane
Service Administrator

Specialists > Availability Management

Dr. Sarah Johnson
Clinical Psychologist • Cognitive Behavioral Therapy Specialist

Sync Calendar Save Changes

October 2023
VIEWING SCHEDULE
Week of Oct 16 - Oct 22

AVAILABLE SLOTS: 34 (+8 to last week)
BOOKED APPOINTMENTS: 22 (88% Capacity)
TOTAL OFF TIME: 12h Personal Time

Available Booked Off

Show Patient Names Bulk Edit

	MON 16	TUE 17	WED 18	THU 19	FRI 20	SAT 21	SUN 22
08:00 AM	AVAILABLE	Booked Session (Patient 1)	OFF	AVAILABLE	AVAILABLE		
09:00 AM	Booked Session (Patient 2)	AVAILABLE	AVAILABLE	Booked Session (Patient 3)	AVAILABLE		
10:00 AM	AVAILABLE	AVAILABLE	AVAILABLE	OFF	AVAILABLE		
11:00 AM	Lunch Break	Lunch Break	Lunch Break	Lunch Break	Lunch Break		
12:00 PM	AVAILABLE	Booked Time (Patient 4)	AVAILABLE	AVAILABLE	AVAILABLE		

Default Duration: 60 Minutes Session

Buffer Times: Post-session Break 15 mins

Leave Request Mark Vacation Time

Logout

Figure 5.1.11: Specialist Availability (Scrollable)



System Design Specification Document – SithCare Channeling App

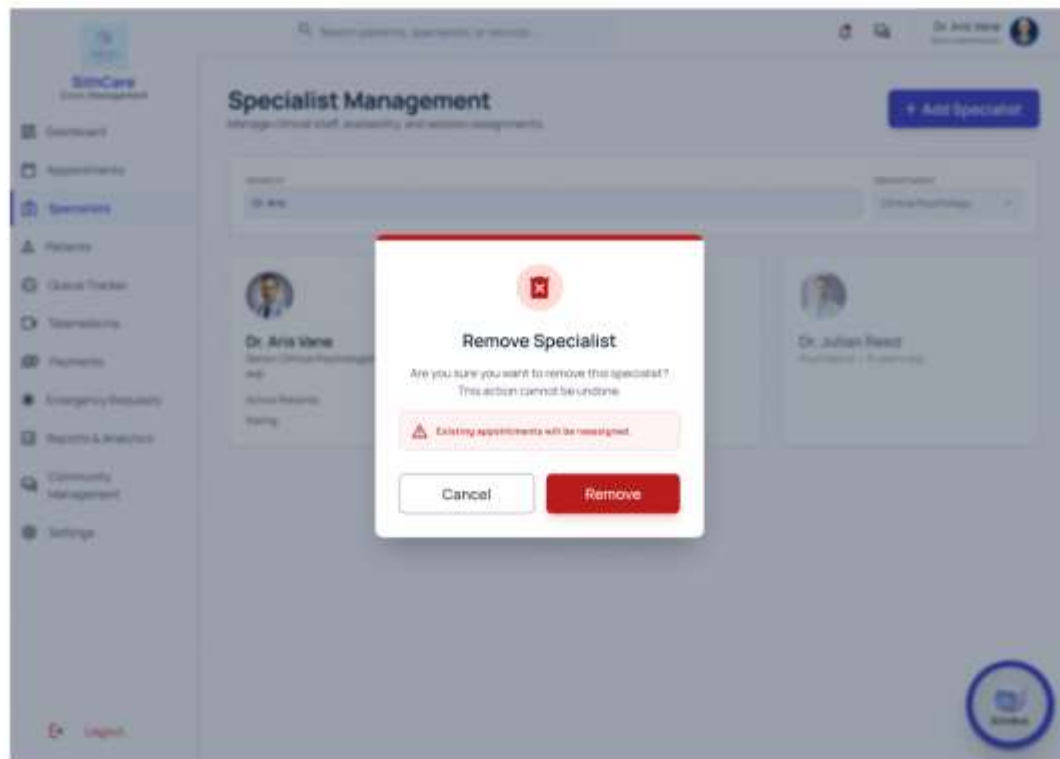


Figure 5.1.12: Remove Specialist Confirmation

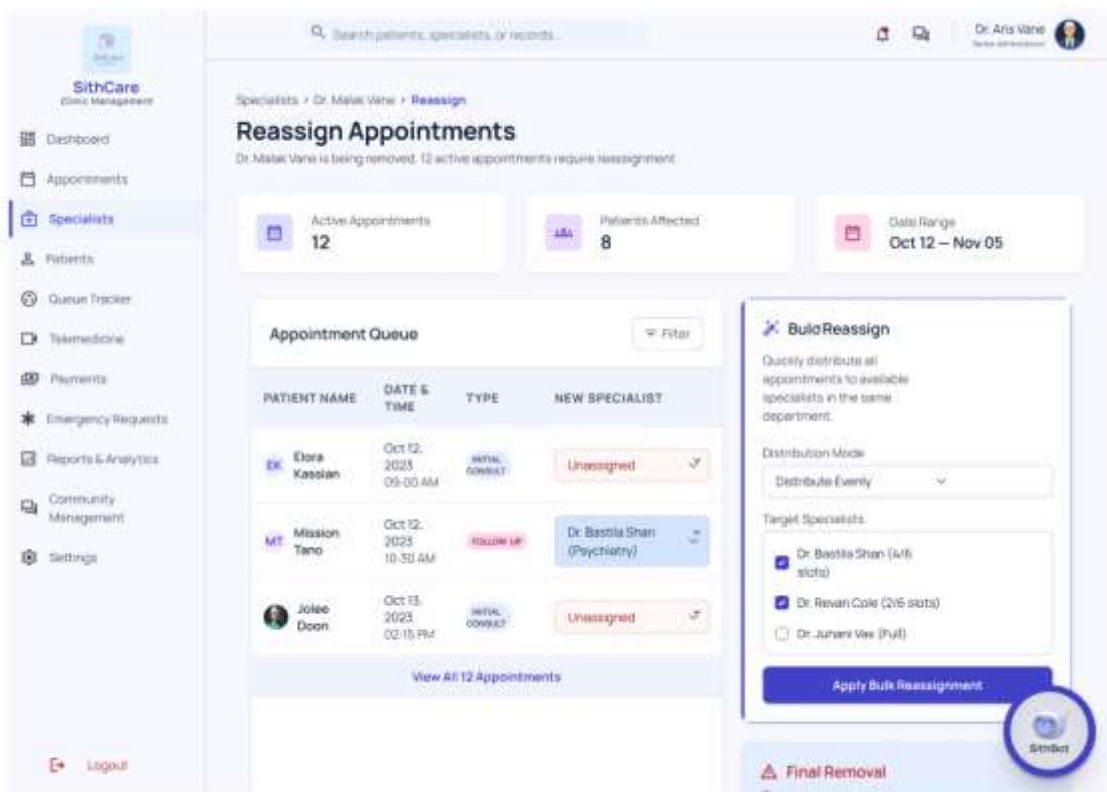


Figure 5.1.13: Specialist Appointment Reassignment



System Design Specification Document – SithCare Channeling App

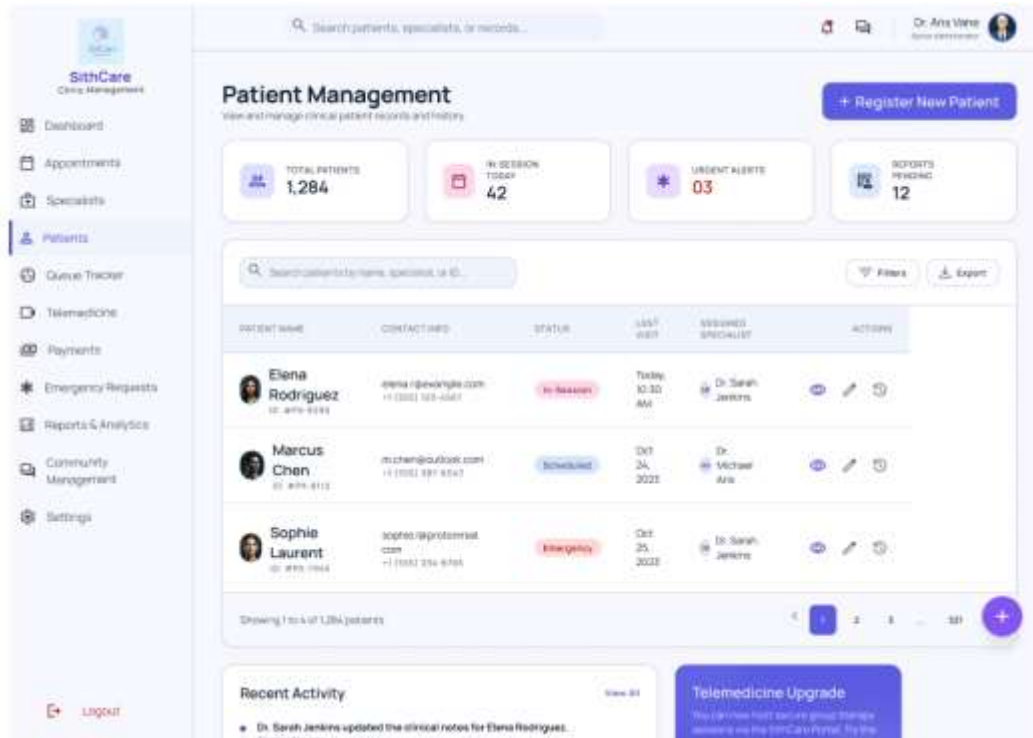


Figure 5.1.14: Patient Management (Scrollable)

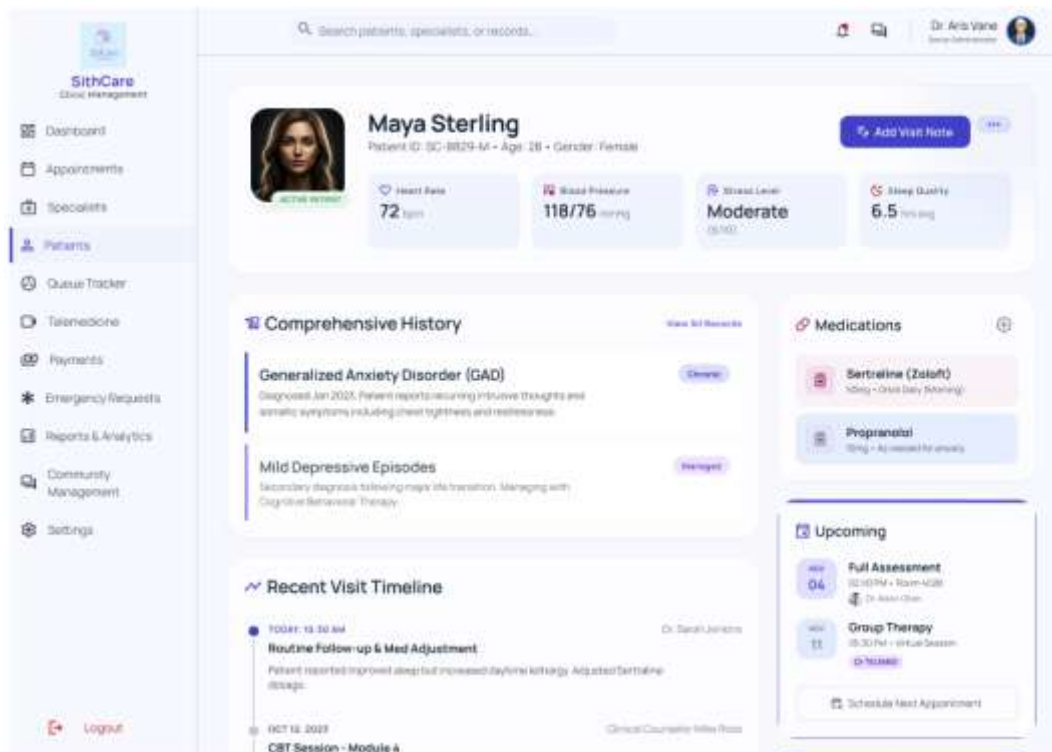


Figure 5.1.15: Patient Profile (Scrollable)



System Design Specification Document – SithCare Channeling App

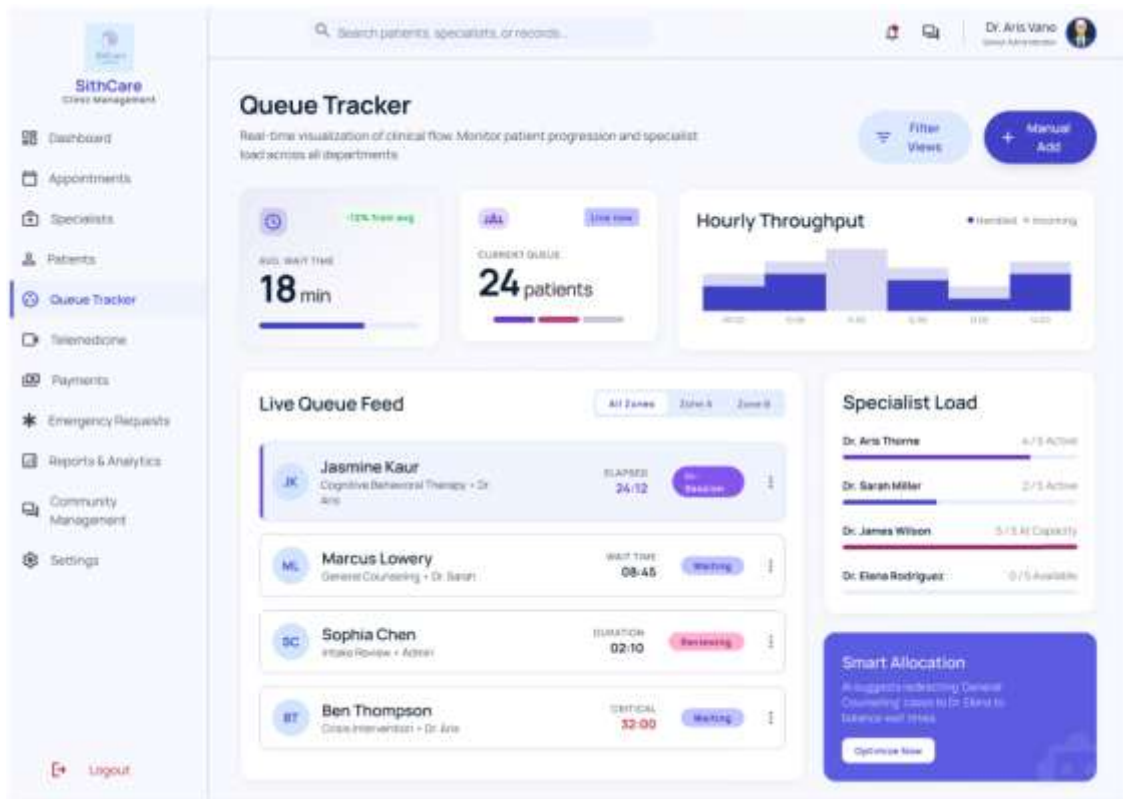


Figure 5.1.16: Queue Tracker (Scrollable)

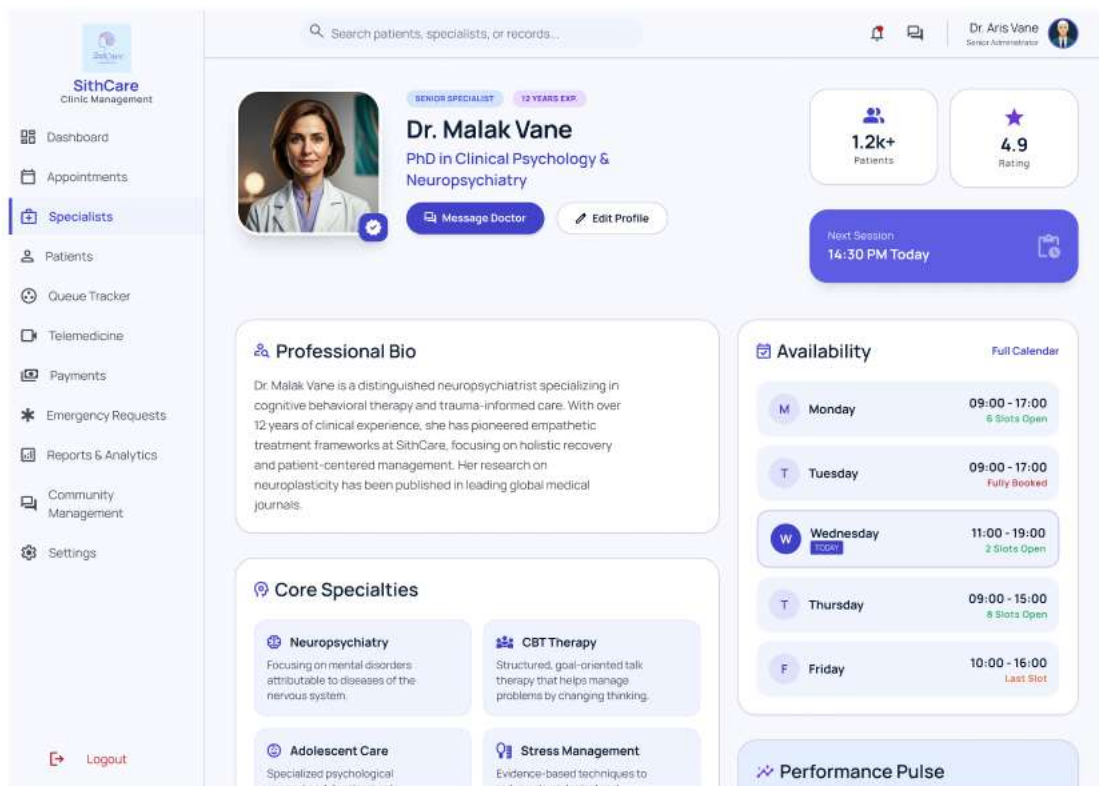


Figure 5.1.17: Specialist Profile (Scrollable)



System Design Specification Document – SithCare Channeling App

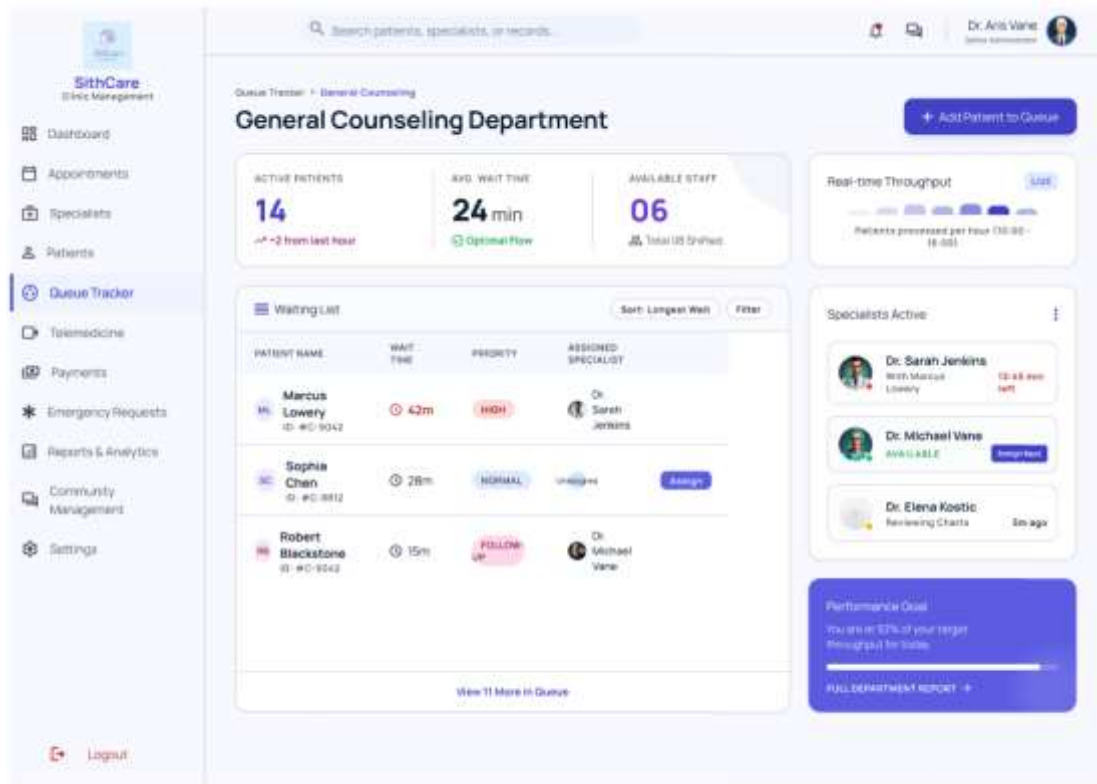


Figure 5.1.18: Queue Details (Standardized)

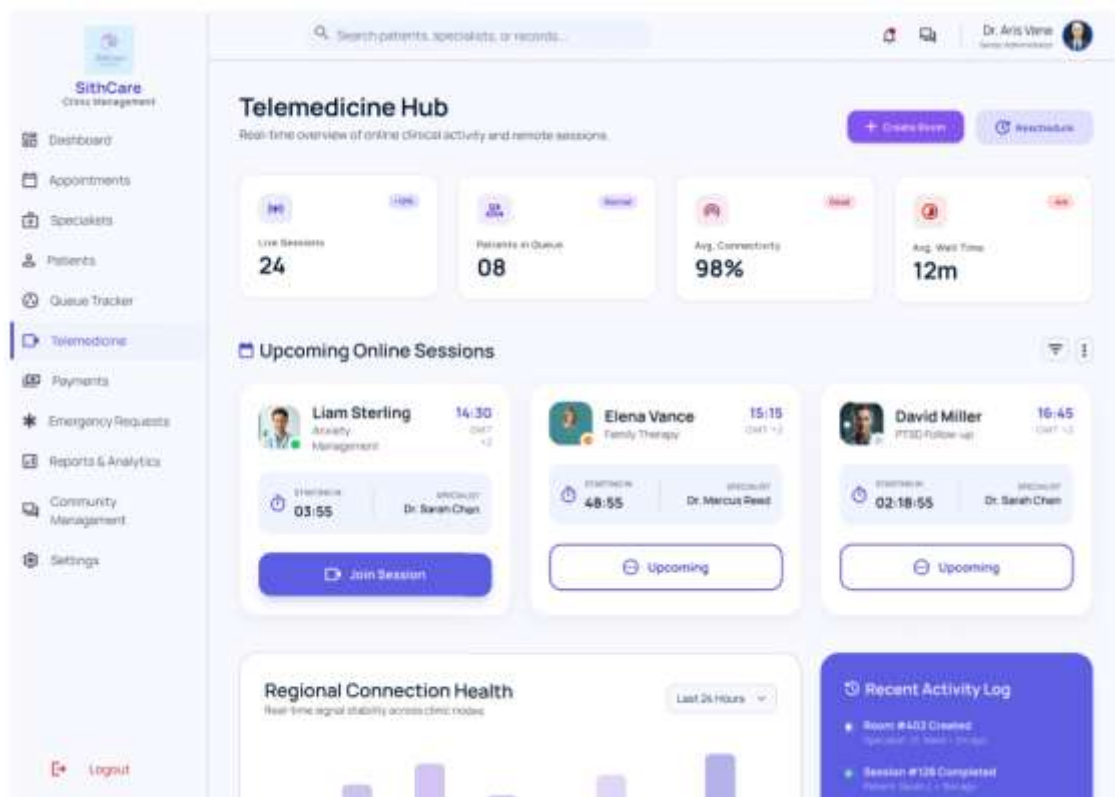


Figure 5.1.19: Telemedicine Dashboard (Scrollable)



System Design Specification Document – SithCare Channeling App



Figure 5.1.20: Active Consultation (Standardized)

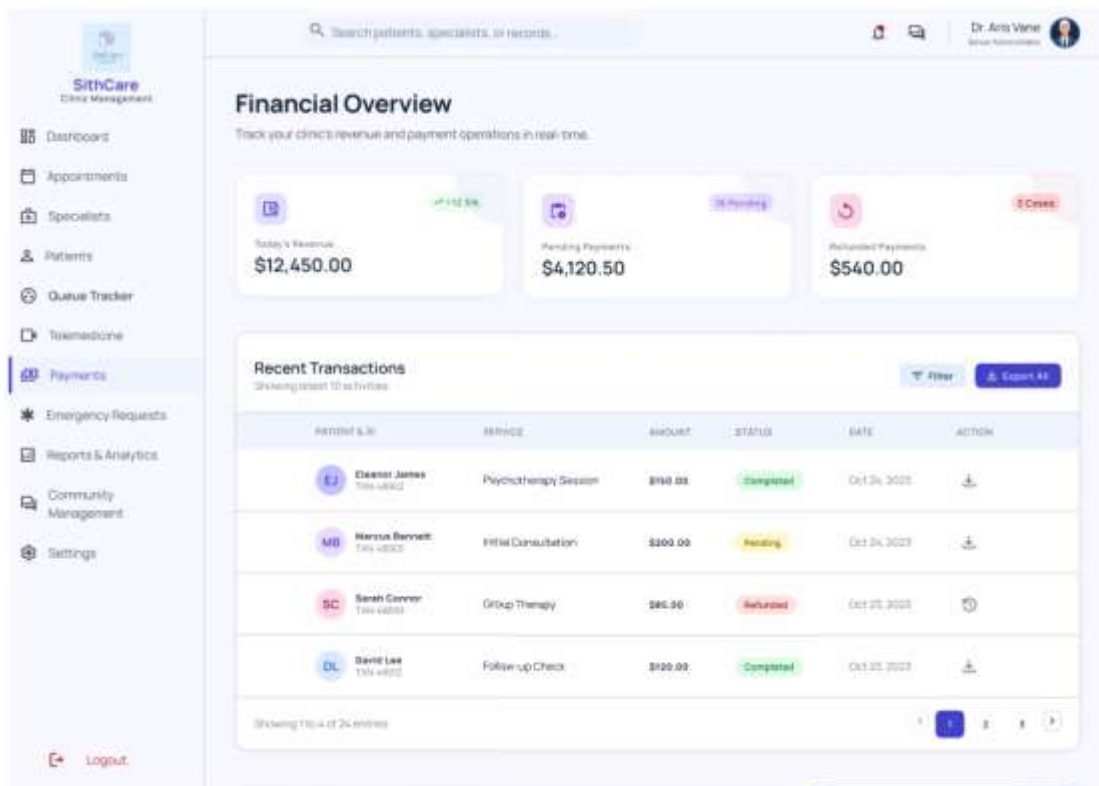


Figure 5.1.21: Payment Dashboard (Scrollable)



System Design Specification Document – SithCare Channeling App

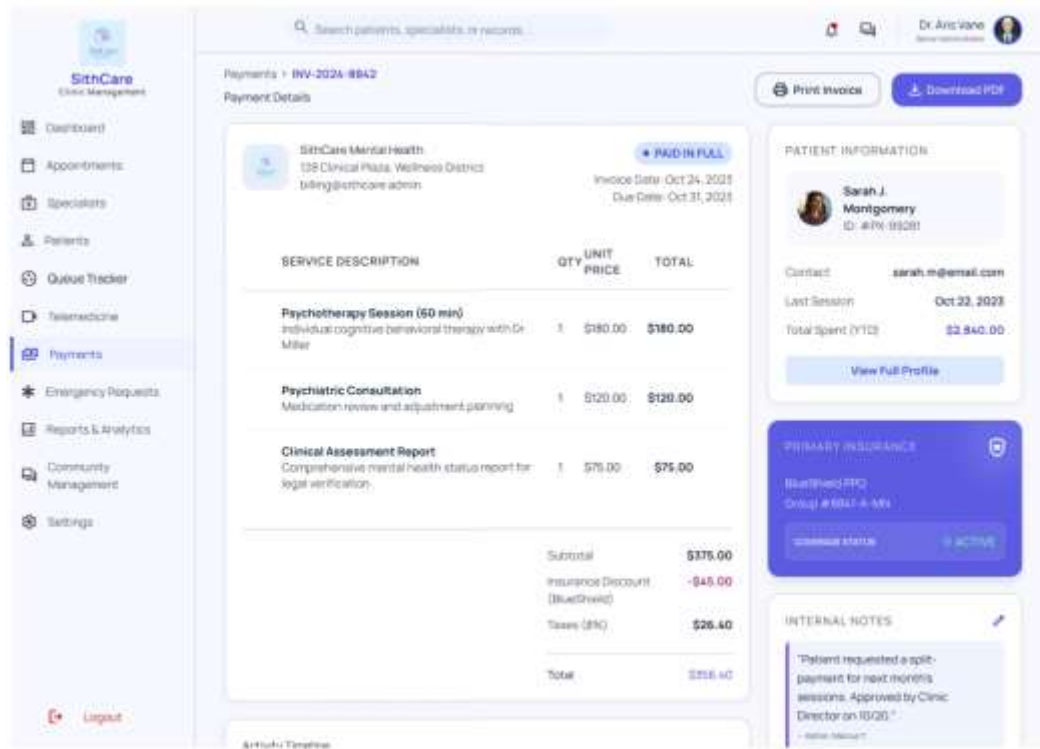


Figure 5.1.22: Payment Details (Scrollable)

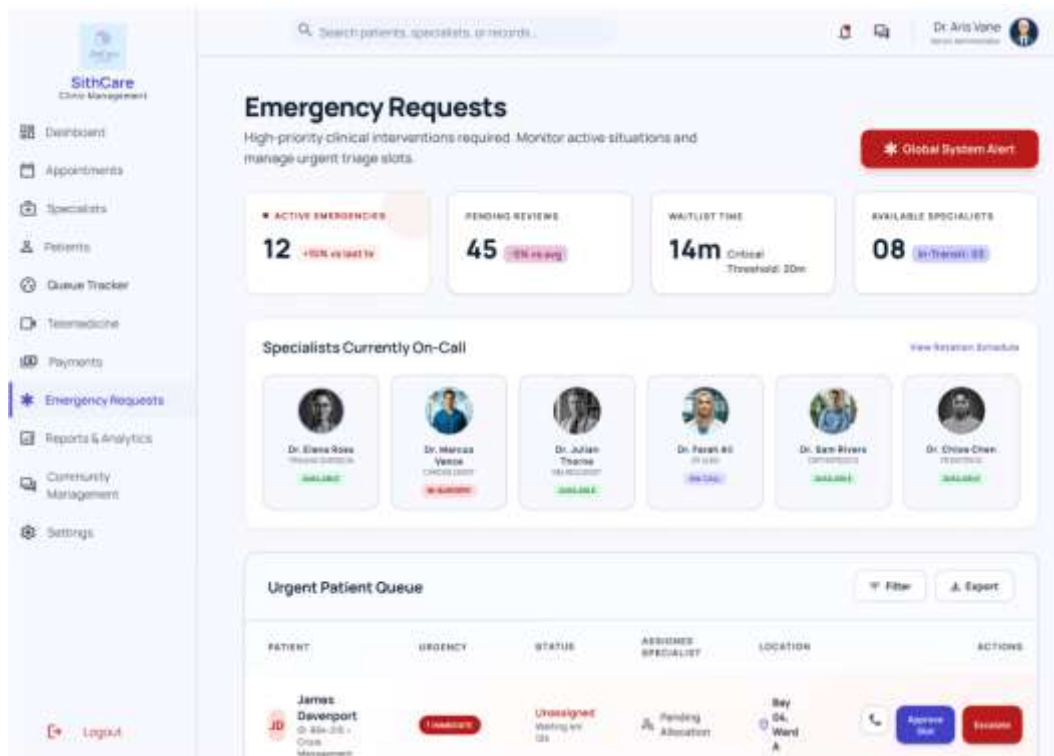


Figure 5.1.23: Emergency Requests Dashboard (Scrollable)



System Design Specification Document – SithCare Channeling App

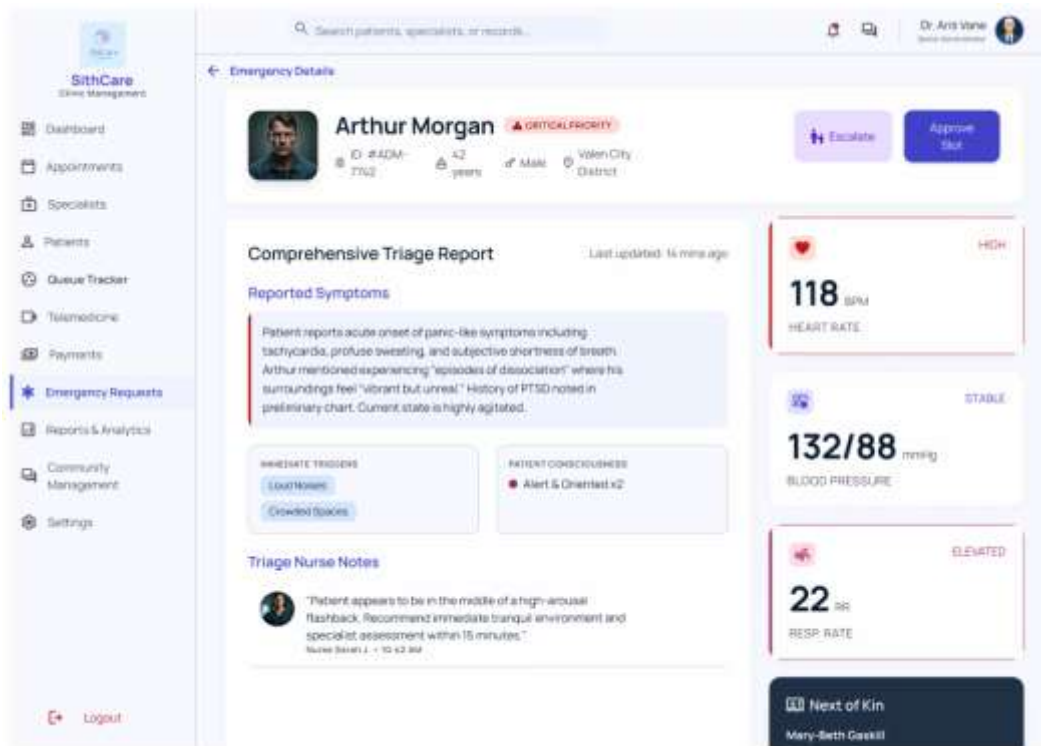


Figure 5.1.24: Emergency Request Details (Scrollable)

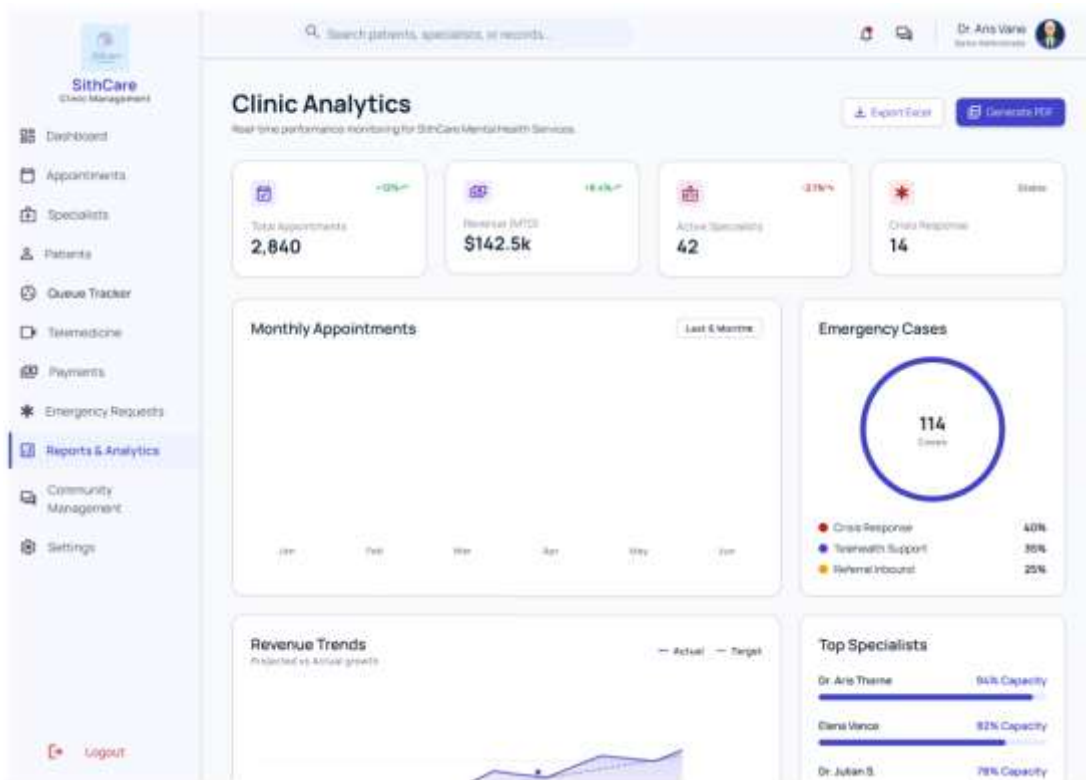


Figure 5.1.25: Reports & Analytics (Standardized)



System Design Specification Document – SithCare Channeling App

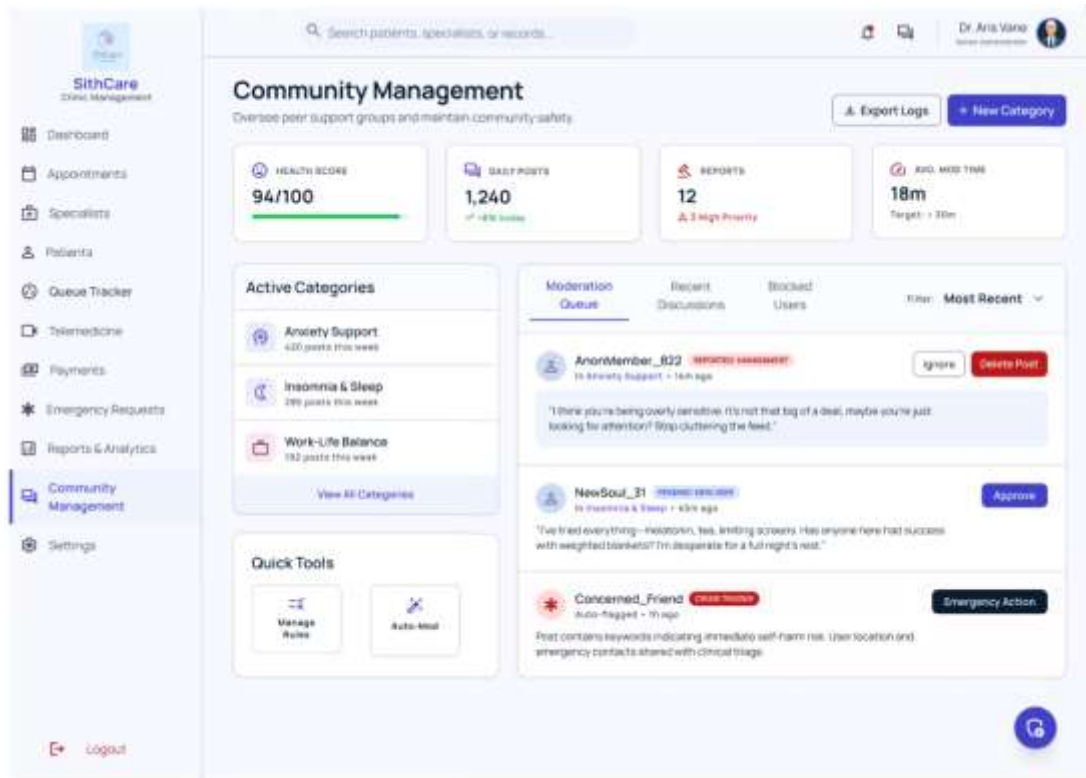


Figure 5.1.26: Community Management (Scrollable)

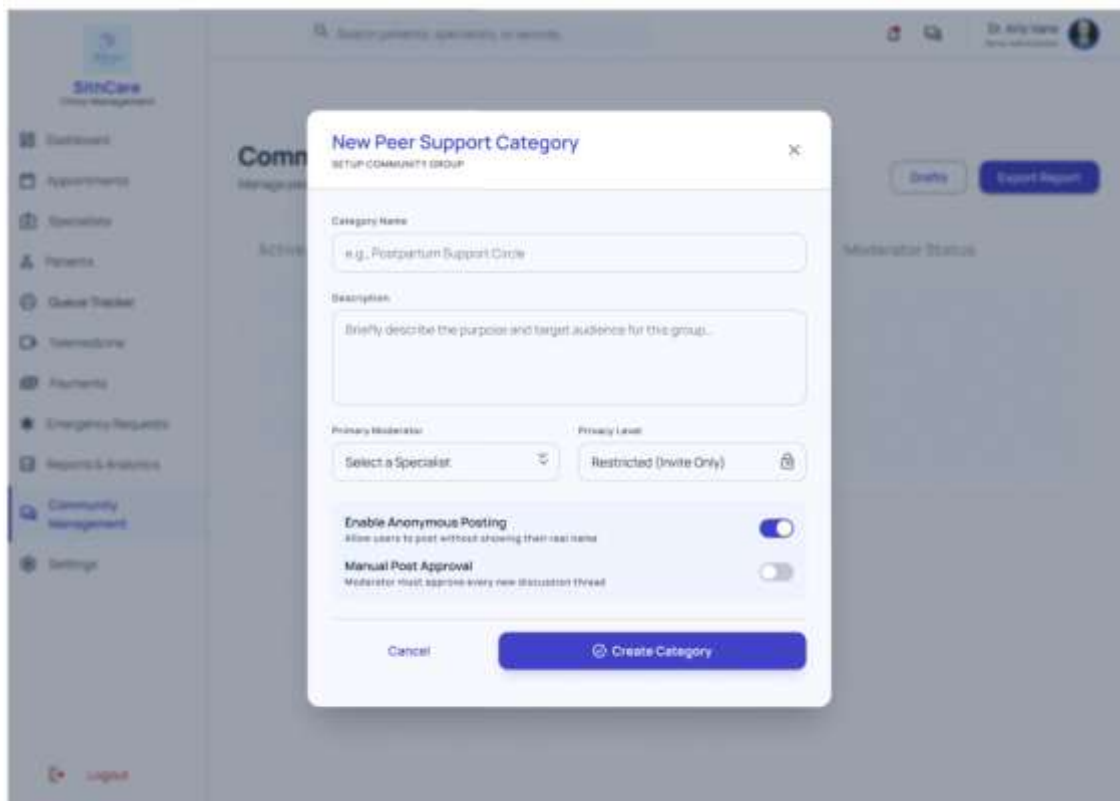


Figure 5.1.27: Community Category Creation (Standardized)



System Design Specification Document – SithCare Channeling App

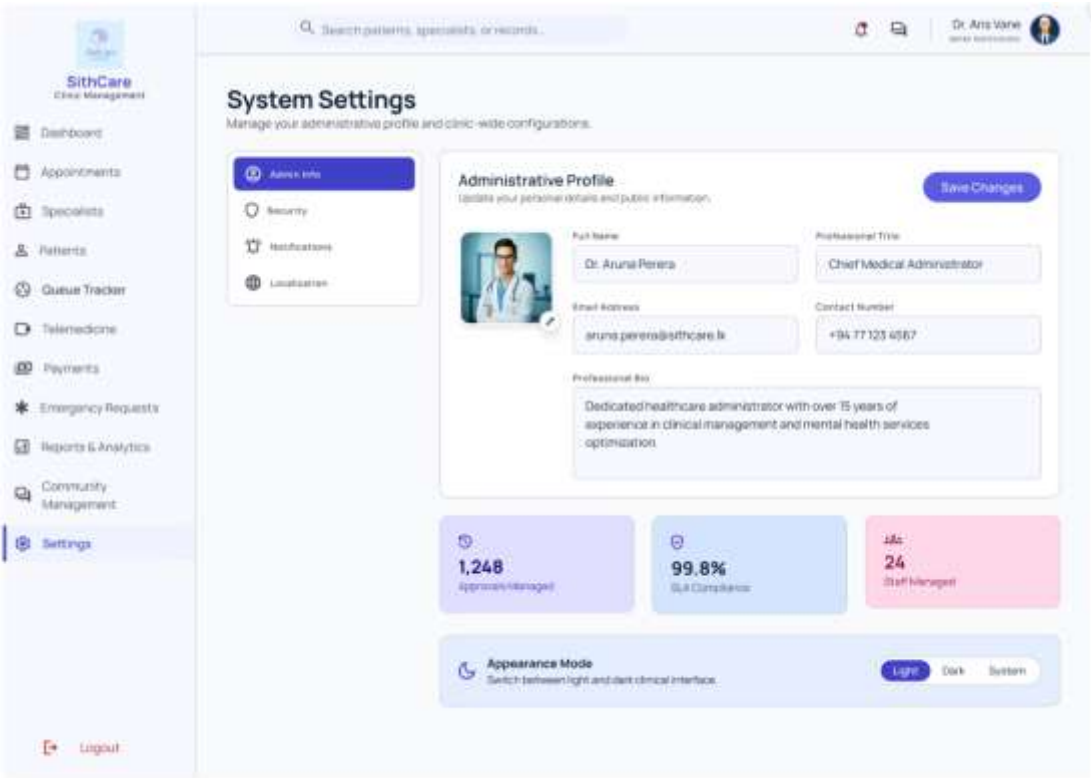


Figure 5.1.28: System Settings (Scrollable)

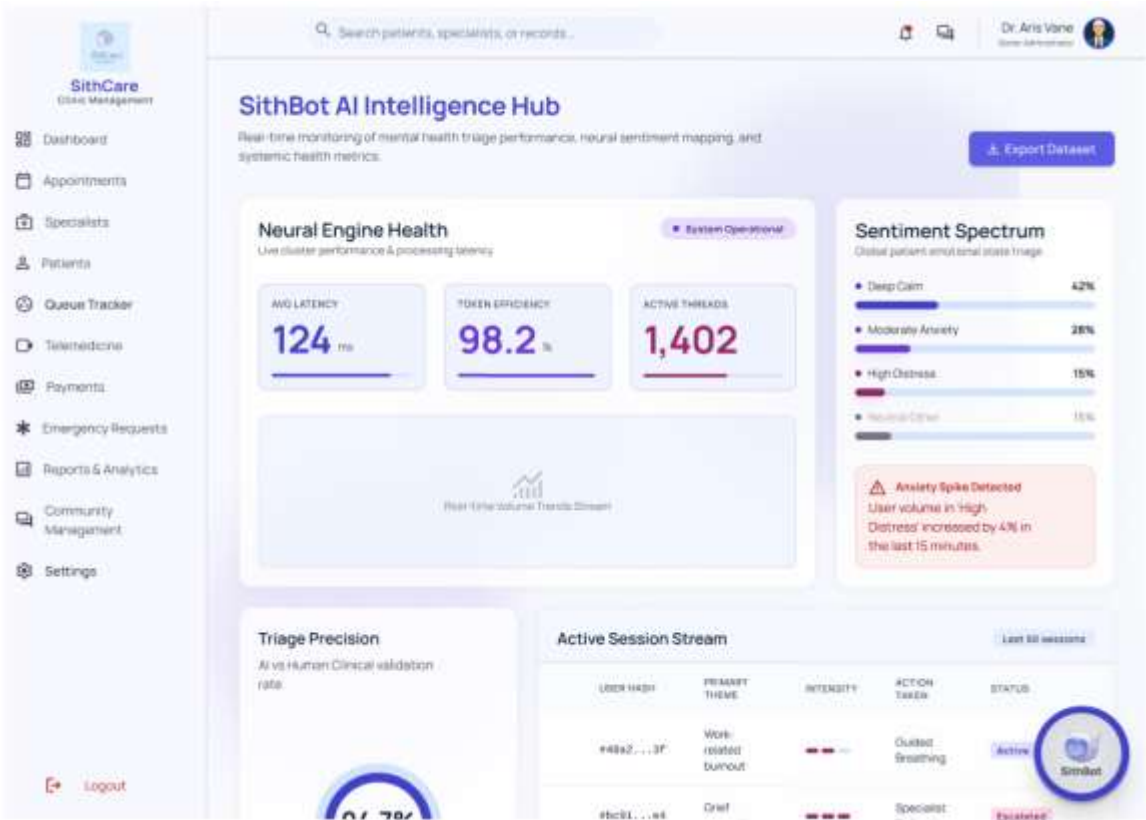


Figure 5.1.29: SithBot AI Admin Dashboard (Scrollable)



System Design Specification Document – SithCare Channeling App

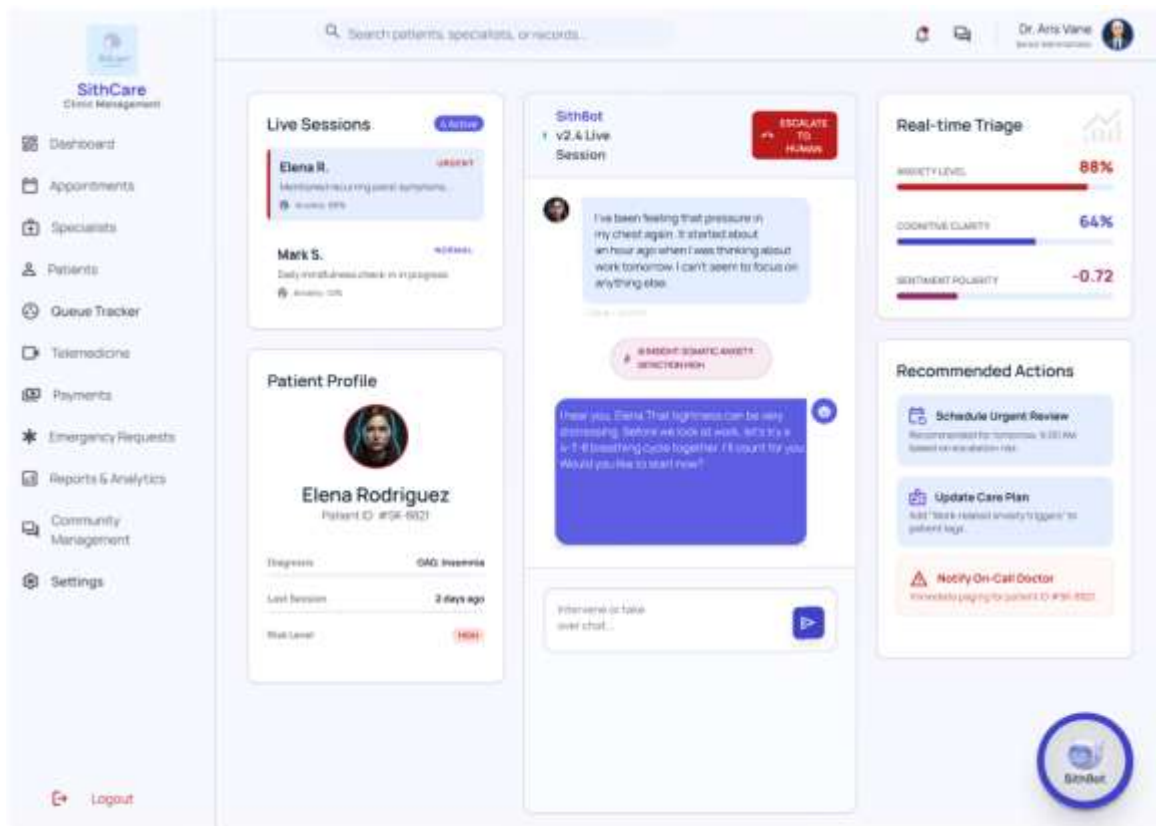


Figure 5.1.30: SithBot Chat Interface (Standardized)

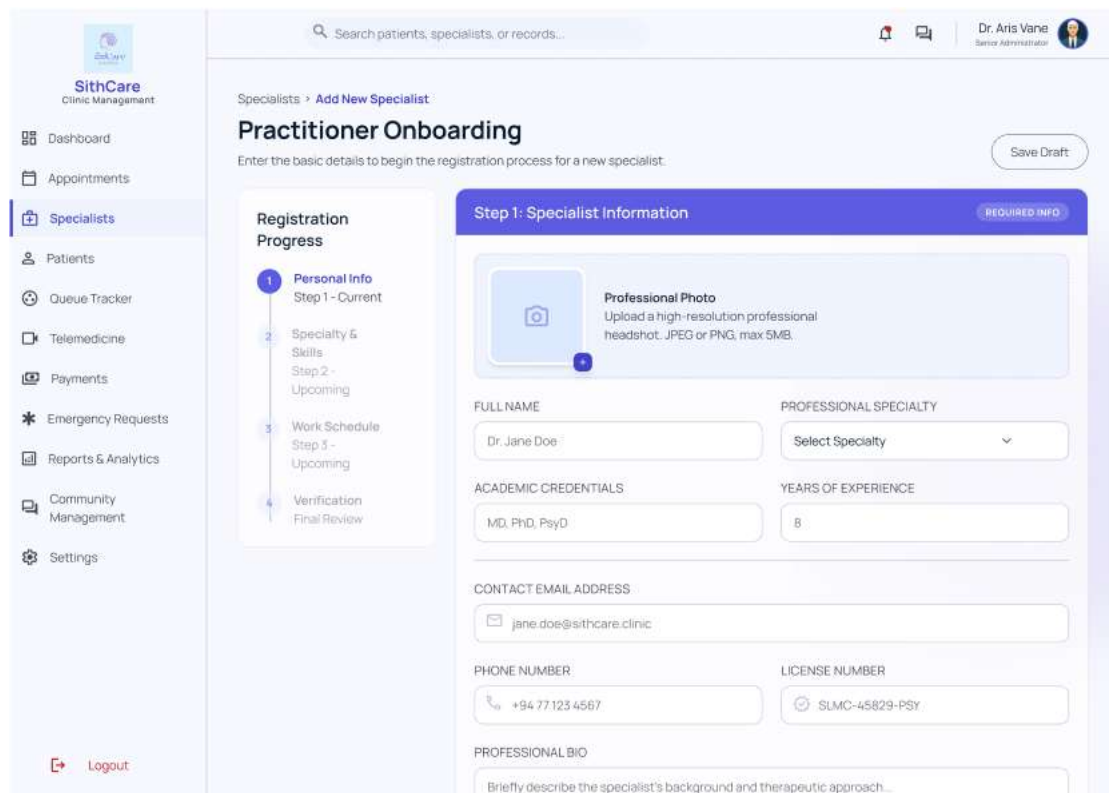


Figure 5.1.31: Add Specialist (Scrollable)



5.2. Patients Dashboard



Figure 6.2.1: Landing Page

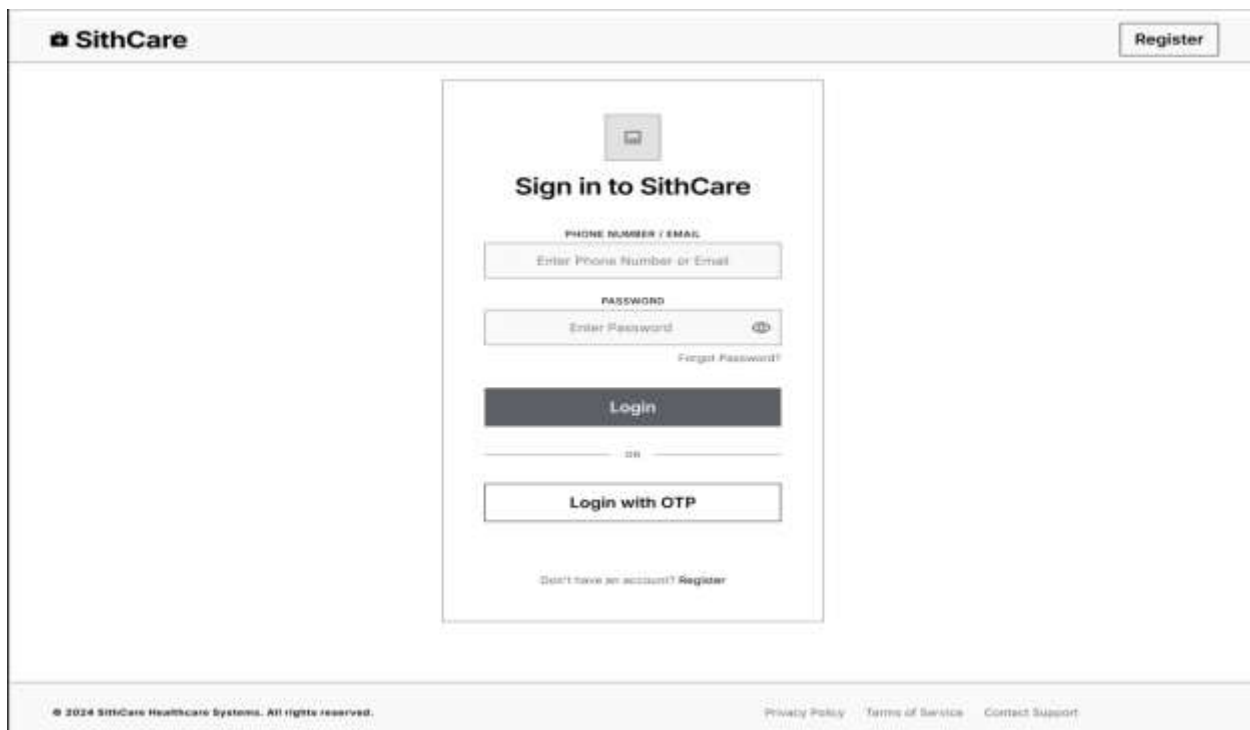


Figure 5.2.2: Login Page



System Design Specification Document – SithCare Channeling App

SithCare

LoginRegister

← Create an Account

Create a new structural identity.

FULL NAME

Enter your full name

PHONE NUMBER

Enter phone number

EMAIL ADDRESS (OPTIONAL)

Enter your email

CREATE PASSWORD

Create a password

CONFIRM PASSWORD

Confirm your password

Create Account & Send OTP

Already have an account? Login

SithCare

Privacy PolicyTerms of ServiceContact Support

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Figure 5.2.3: Register Page

←

Verify Your Number

We've sent a 6-digit code to +84 7X XXX XXXX.

Verify & Continue

Didn't receive the code? Resend in 00:28

Figure 5.2.4: OTP Verification Page



System Design Specification Document – SithCare Channeling App

SithCare

Welcome to SithCare

Let's set up your experience.

Choose your preferred language

This sets your app language and helps match you with the right therapists.

English ✓

Sithala

Tavel

Your Privacy Matters

☒ I agree to the Terms of Service and Privacy Policy. *

☐ I consent to sharing my consultation history securely with my booked specialists. *

I agree to the Community Guidelines to participate in anonymous forums.

COMPLETE SETUP

© 2024 SithCare Mental Health Portal, Clinical Wireframe V1.0

[Privacy Policy](#) [Terms of Service](#) [Accessibility](#) [Contact Support](#)

Figure 5.2.5: Complete Setup Page

Patient Portal

Medical ID: SC-7734

Dashboard

Specialists

Appointments

Records

Book New Appointment

Settings

Support

Hello, [Name]

Here is your health summary for today.

In a crisis? Get help now.

Emergency Support

Upcoming Consultation Card

Dr. Dreshel - Clinical Psychologist

Today, 10:30 AM

Join Video Call

Details

Current Queue: A-10 @Your Token: A-14

Recent Vitals

Heart Rate: 72 bpm

Weight: 165 lbs

How can we help you today?

Chat with SithBot

Find a Therapist

Anonymous Forum

Video Consultation

Recent Activity

Lab Results Available

View All

Oct 26

Figure 5.2.6: Patient Dashboard Page

60



System Design Specification Document – SithCare Channeling App

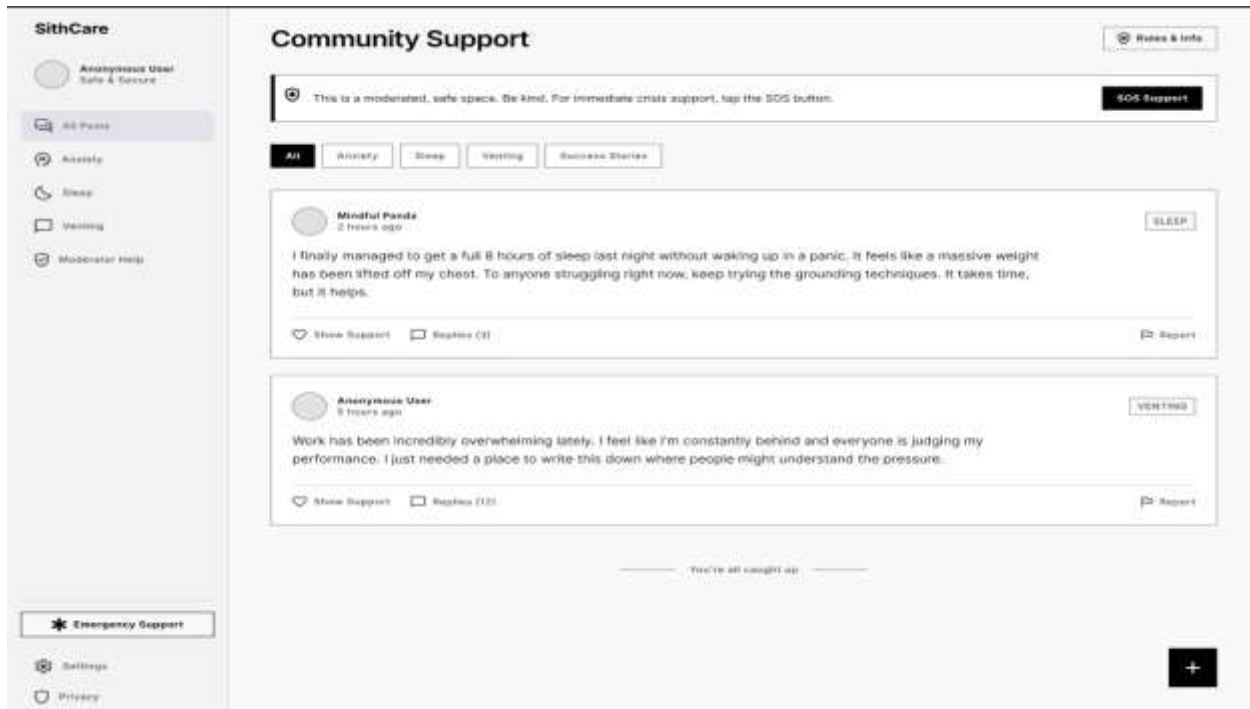


Figure 5.2.7: Anonymous Community Forum Page

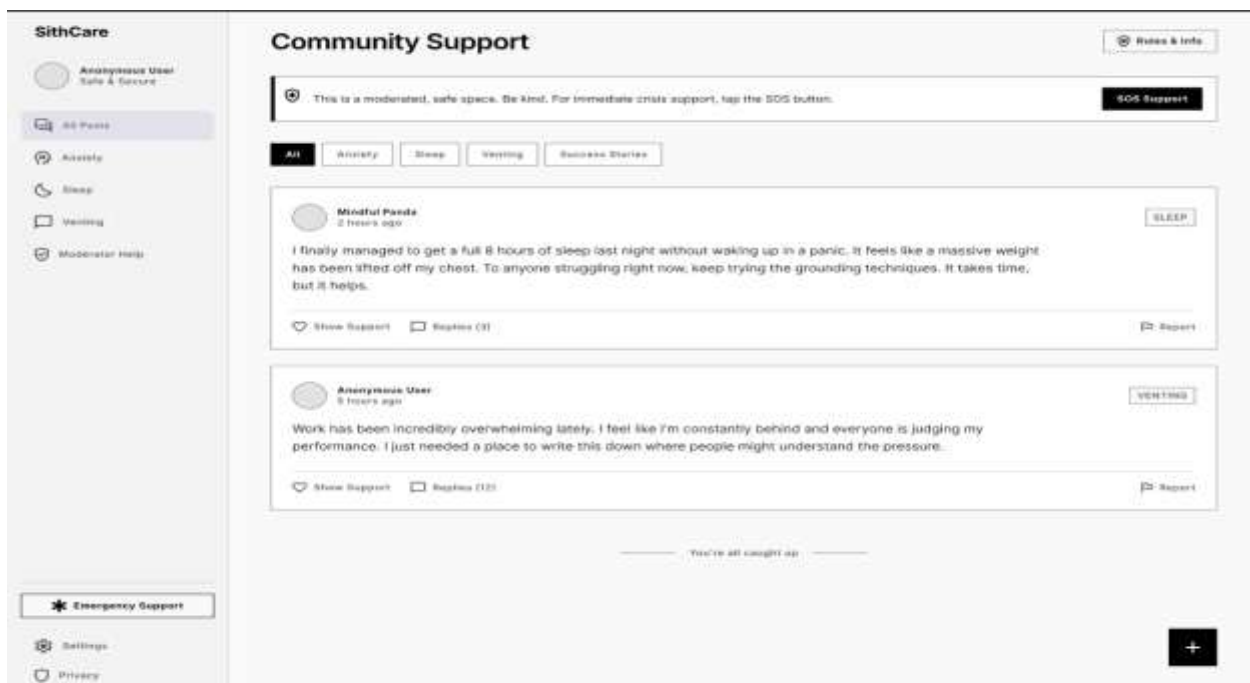


Figure 5.2.8: Create New Anonymous Post Page



System Design Specification Document – SithCare Channeling App

The screenshot shows a modal window titled "Share Anonymously" with a close button (X) in the top right corner. Inside the modal, there is a "Category" dropdown menu with the text "Select a topic". Below this is a text area labeled "Your Message" with the placeholder text "What's on your mind? Share your experience safely...". Under the text area, there is a checkbox with the text "Your real name and profile are hidden. Only moderators can see account details for safety purposes...". At the bottom left of the modal, there is a toggle switch labeled "POST ANONYMOUSLY". At the bottom right, there is a button labeled "PUBLISH POST" with a right-pointing arrow.

Figure 5.2.9: Anonymously Reply Page

The screenshot shows the SithCare app interface. At the top, there is a navigation bar with a back arrow and the text "Back to Feed" on the left, and the app name "SithCare" in the center. Below the navigation bar, there is a post by an "Anonymous User" with the topic "Anxiety". The post title is "Struggling with sleep after a long shift". The post text reads: "I've been working these 12-hour night shifts lately and it's completely wrecked my sleep schedule. Even when I'm exhausted, I lie in bed for hours just staring at the ceiling. It's starting to affect my mood and I'm feeling constantly on edge. Has anyone dealt with this kind of severe shift-work insomnia? What helped you reset?". Below the post text, there are icons for likes (12) and replies (3). Below the post, there is a section labeled "REPLIES". The first reply is from an "Anonymous Peer" and says: "I feel you. Blackout curtains and a white noise machine saved me when I was doing nights. Also, trying to stick to a consistent wake time even on days off (as painful as it is) helps anchor your circadian rhythm a bit." Below the reply, there is a like icon and the number 4. The second reply is from a "Clinical Moderator" with a verified badge and says: "Shift work sleep disorder is very common and can significantly impact mental health. While peer advice is great, if this persists for more than a few weeks, consider speaking with an occupational health specialist. In the short term, ensuring your sleep environment is completely dark, cool (around 65°F), and quiet is foundational. Avoid screens for 1 hour before your intended sleep time, even during the day." Below the reply, there is a like icon and the number 15. At the bottom of the screen, there is a text input field with the placeholder "Reply anonymously..." and a "Send" button.

Figure 5.2.10: Sithbot Page



System Design Specification Document – SithCare Channeling App

The screenshot shows the 'SithBot AI Triage Assistant' interface. At the top, a disclaimer states: 'SithBot provides guidance to help you find the right specialist. It is not a medical diagnosis and cannot handle emergencies. In a crisis? Get Emergency Support.' The main chat area displays a message from the bot: 'Hi there, I'm here to help you figure out what kind of support might be best for you today. How have you been feeling lately? 10:02 AM'. Below this, 'SUGGESTED RESPONSES' are listed: 'Feeling anxious', 'Feeling sad or low', 'Having trouble sleeping', and 'Other'. A button labeled 'Feeling anxious 10:04 AM' is shown as selected. A 'Recommendation' box follows, stating: 'Based on your answers, speaking with a Clinical Psychologist might be helpful.' and includes a button 'Find a Clinical Psychologist →'. At the bottom is a text input field 'Type your message...' and a 'Press enter to send' prompt.

Figure 5.2.11: Booking New Appointment Page

The screenshot shows the 'SithCare Patient Portal' 'New Appointment' page. The left sidebar contains navigation links: 'Dashboard', 'Specialists', 'Appointments' (highlighted), and 'Records'. The main content area is titled 'New Appointment' and features a card for 'Dr. Smith, Psychiatrist • MBBS, MD' with a fee of '₹KR 2000'. A 'CONSULTATION MODE' section has buttons for 'Video' and 'Clinic'. Below is a 'Select Date' calendar showing the week from Monday 12 to Sunday 18. Underneath, 'Available Slots' are listed: 09:00 AM, 10:30 AM, 01:00 PM (highlighted), and 02:30 PM. A 'Book New Appointment' button is at the bottom left. At the bottom right, a summary shows 'Selected Slot: Mon 12, 01:00 PM' and a 'Continue to Payment →' button.

Figure 5.2.12: Checkout & Payment Page



System Design Specification Document – SithCare Channeling App

 **Patient Portal**
Medical ID: SC-7734

Dashboard

Specialists

Appointments

Records

Book New Appointment

Settings

Support

Checkout & Payment

Search...

🔔

👤

Booking Summary

DATE & TIME

Mon 12 Nov, 10:30 AM

DOCTOR

Dr. Adithya Jayawardhana

MODE

Video Call

COST BREAKDOWN

Consultation Fee

LKR 2500

Platform Fee

LKR 200

Total Amount

LKR 2700

Choose Payment Method

☒  **Pay Online via PayHere**
Credit/Debit Card, Mobile Wallets

☐  **Pay at Clinic (Cash)**
(Unavailable for Video Consultations)

☐ I agree to the cancellation and refund policy

🔒 Confirm & Pay LKR 2700

Figure 5.2.13: Booking Confirmed Page

 **PATIENT PORTAL**
Medical ID: SC-7734

Dashboard

Specialists

Appointments

Records

Settings

Support

SithCare

🔔

👤

👤

Booking Confirmed!

Your appointment has been successfully scheduled.

DIGITAL RECEIPT

TOKEN

A-14

REFERENCE

#88291

PATIENT

Jane Doe

DATE & TIME

Oct 24, 2023 ~ 10:00 AM

DEPARTMENT

General Practice

Download PDF Receipt

Add to Calendar

Go to My Dashboard

Figure 5.2.14: Specialist Search Page



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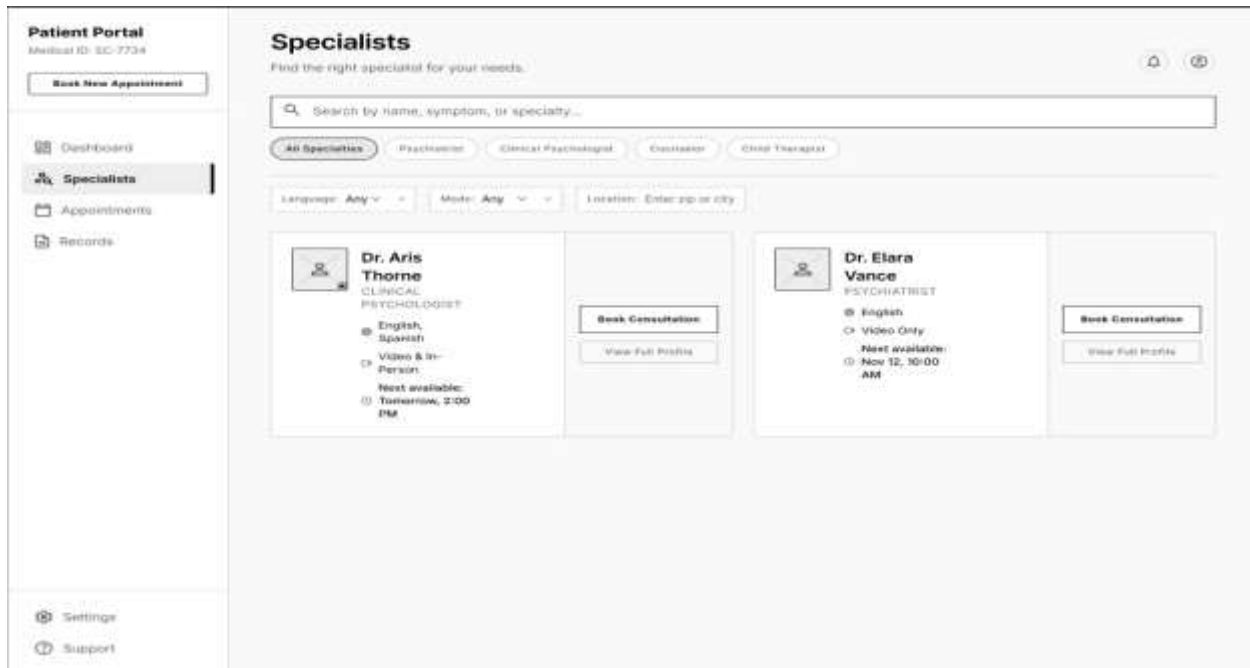


Figure 5.2.15: Specialist Empty Page

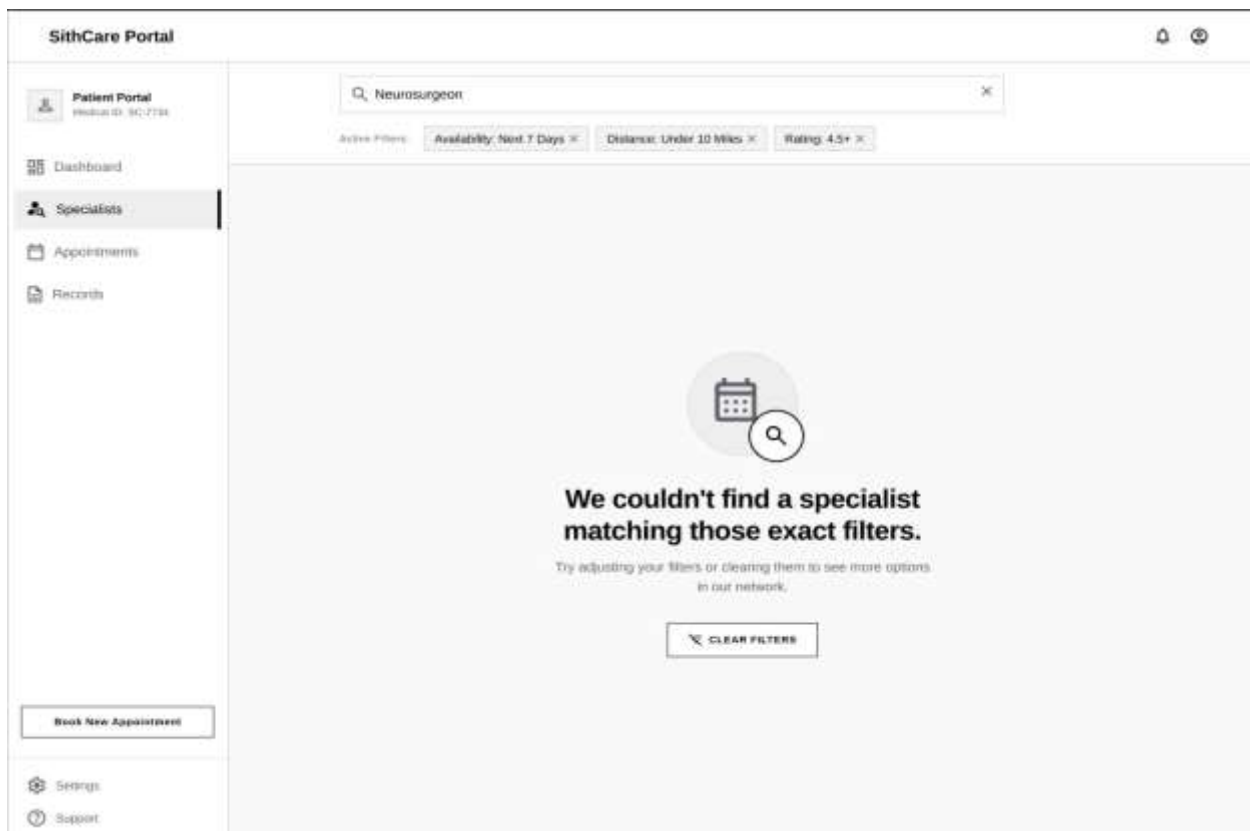


Figure 5.2.16: Video Consultation-Active page



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Figure 5.2.17: Video Consultation-Waiting Room Page

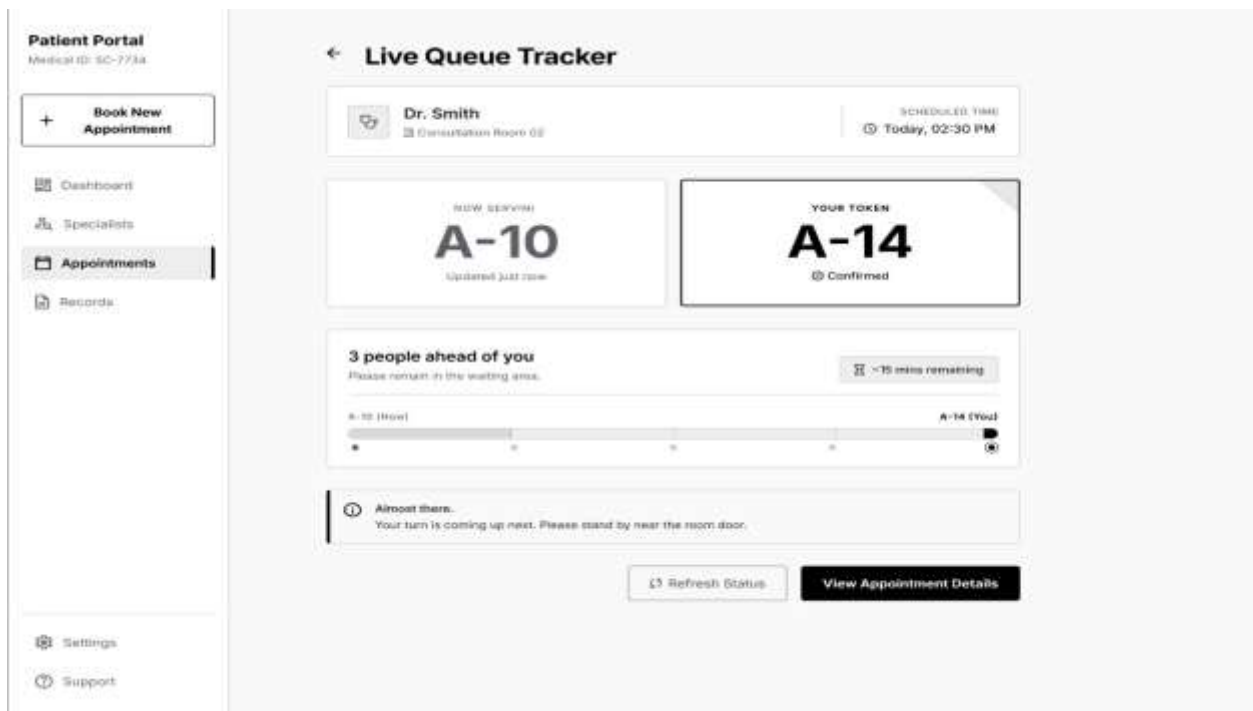


Figure 5.2.18: Live Queue Tracker Page



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Patient Portal
Medical ID: SC-7734

Dashboard

Specialists

Appointments

Records

Book New Appointment

Settings

Support

← Settings

Jane Doe | +94 77 XXX XXXX
FORUM ALIAS: MINDFULPANDA

Emergency Contacts

John Doe | Spouse
+94 71 234 5678

+ Add Trusted Contact

Privacy & Data Consent

Share consultation notes with my booked specialists
Required for doctors to view your history

Share my GPS location when I trigger an Emergency
Helps responders find you faster

Share anonymous app usage data to improve SithCare
Optional data collection for analytics

View Full Privacy Policy

Download My Data

App Preferences

App Language
English

Reminders & Alerts

Log Out

Bonnie Anonett

Figure 5.2.19: Settings Page

Patient Portal
Medical ID: SC-7734

Dashboard

Specialists

Appointments

Records

Book New Appointment

Settings

Support

Medical Records

Search doctor or date...

Upload Existing Record

All Records

Prescriptions

Consultation Notes

Lab Reports

Prescription | Oct 24, 2023
Routine Refill
Dr. Sarah Smith • Cambridge

View Online

Download PDF

Lab Report | Sep 12, 2023
Comprehensive Metabolic Panel
CorePath Laboratories

View Online

Download PDF

Consultation Notes | Aug 05, 2023
Annual Physical Summary
Dr. James Wilson • General Practice

View Online

Download PDF

Figure 5.2.20: Medical Records Page

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5.3. SithCare-Admin Dashboard



Figure 5.3.1: Admin Dashboard Page

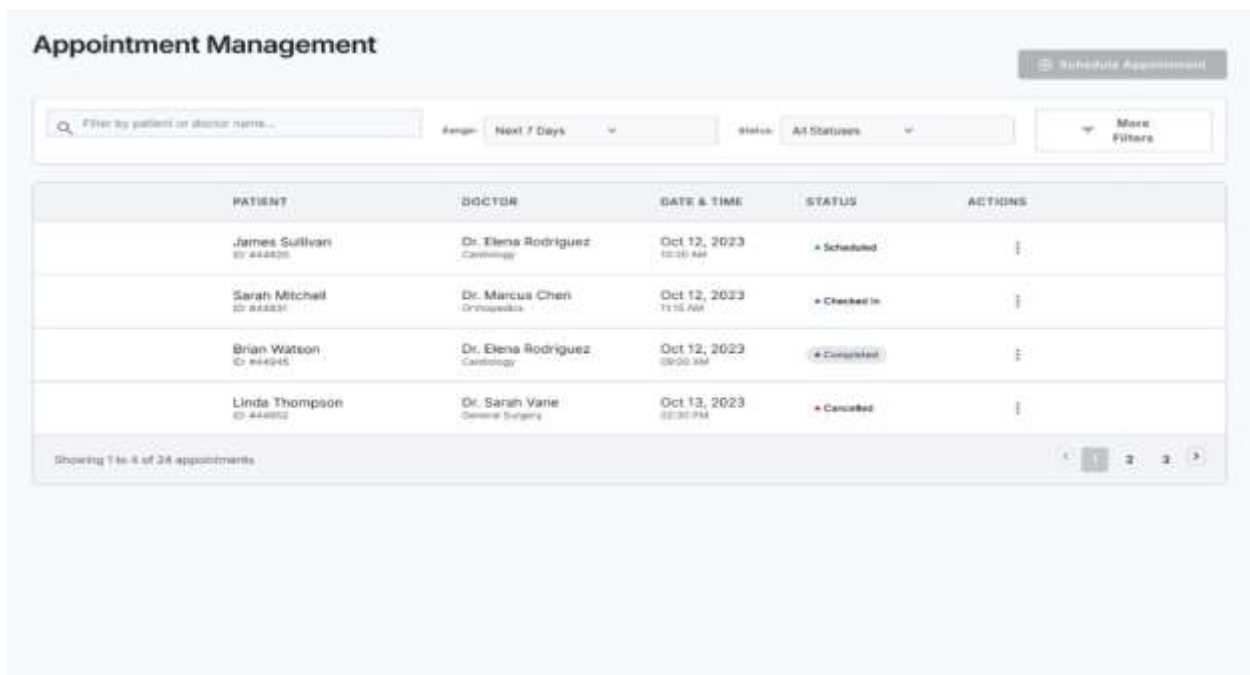


Figure 5.3.2: Appointment Management Page



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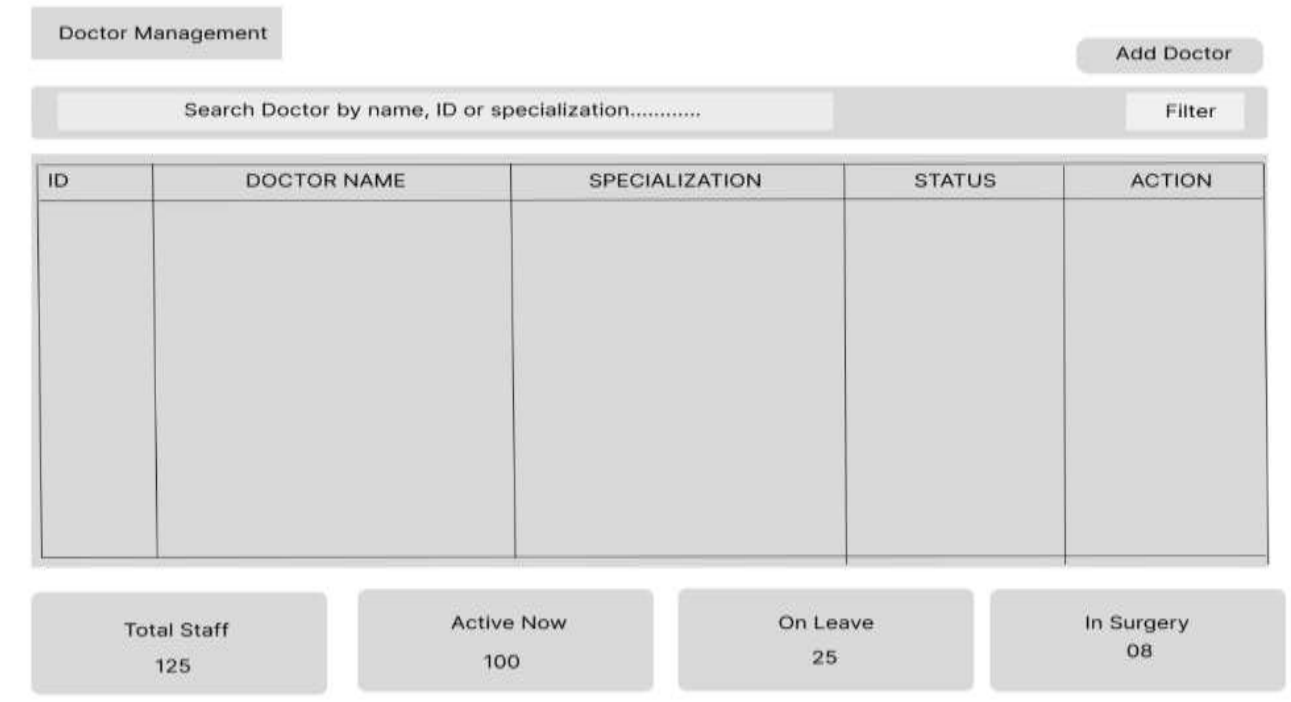


Figure 5.3.3: Doctor Management Page

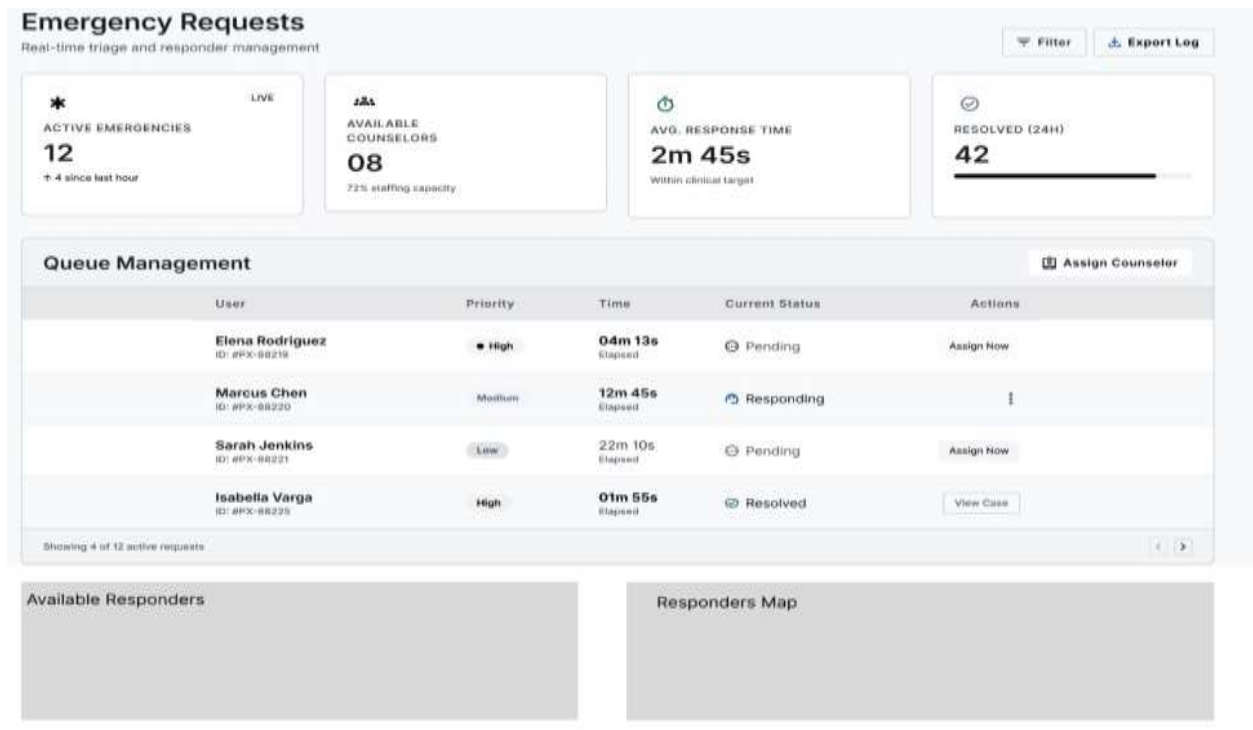


Figure 5.3.4: Emergency Requests Page



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Reports & Analytics

Comprehensive overview of facility performance and patient trends.

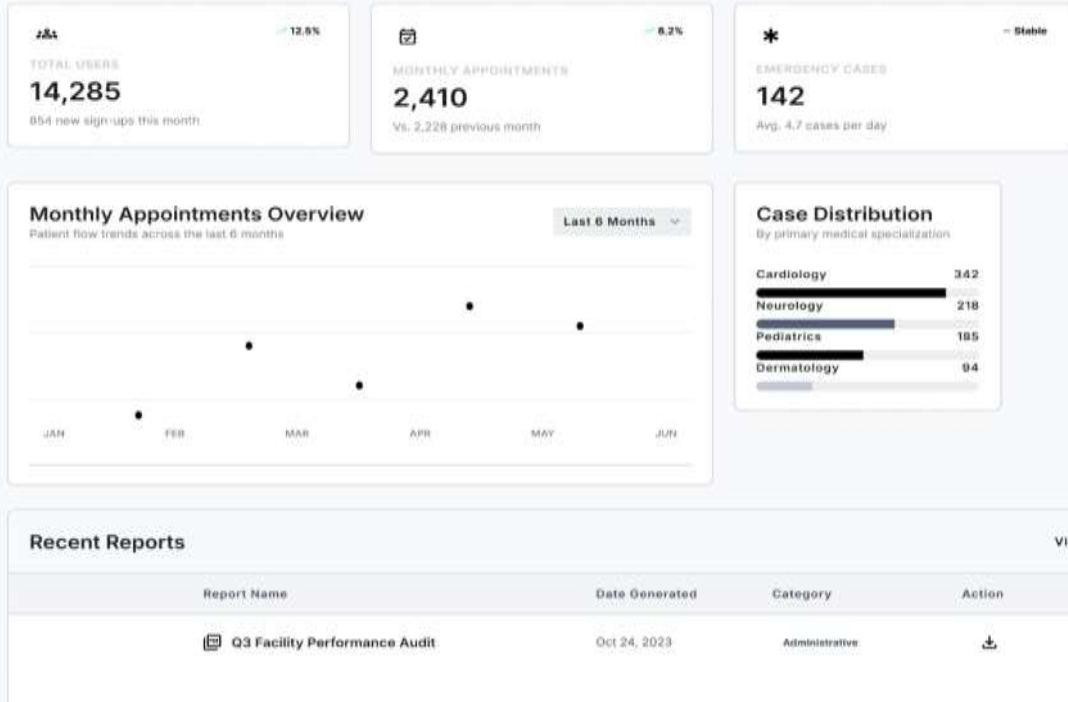


Figure 5.3.5: Reports & Analytics Page



6. Interface Design

This section defines how SithCare communicates with external services and how the internal modules of the system interact with each other. The interface design supports the role-based web application described in the SRS and follows the layered MVC architecture defined in this SDS.

6.1. External Interfaces

SithCare uses external services only where they provide clear operational value, such as payments, notifications, telemedicine, maps, and emergency helpline information. All external communication is performed through secure API calls or approved client-side SDK integrations.

External Interface	Purpose	Integration Method	Used By	Design Notes
PayHere Payment Gateway	Online appointment payments, refund handling, transaction confirmation, and LKR-based payment processing.	REST API / PayHere Checkout SDK / Webhook callbacks	Appointment Service, Payment Gateway Adapter, Patient Dashboard, Clinic Admin Panel	No card details are stored in SithCare. Payment status is verified using gateway response and webhook validation.
Jitsi Meet API / Video SDK	Telemedicine video consultation rooms for confirmed online appointments.	JavaScript SDK / secure room link generation	Telemedicine Module, Doctor Portal, Patient Dashboard	Room links are generated only for confirmed appointments and are time-limited.
Twilio SMS API / Local SMS Provider	OTP delivery, appointment reminders,	HTTPS API request from	Authentication Service, Appointment	SMS content is sent in the patient's



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	queue proximity alerts, and emergency notifications.	Notification Service	Service, Queue Manager, Emergency Handler	preferred language where possible.
Nodemailer / Email Service	Email OTPs, booking confirmations, password reset messages, appointment reminders, and payment receipts.	SMTP / email service API	Notification Service, Auth Controller, Appointment Service, Payment Service	Email templates avoid exposing sensitive health details in the subject line.
Google Maps API / OpenStreetMap	Displaying clinic locations, directions, and consent-based emergency location support.	Client-side map SDK / location service integration	Patient Dashboard, Emergency Support Module	Location is used only after user permission and explicit consent.
Emergency Helpline Data	Display of Sri Lankan crisis support and emergency contact numbers.	Static configuration maintained by Platform Admin	Emergency Support Module, SithBot	Helpline information is visible even if other non-critical modules fail.

6.2. User Interface Interfaces

The frontend is a React Single Page Application. Users interact with SithCare through role-based dashboards and selected modules according to their permissions.

- **Patient Interface:** registration, profile and consent management, specialist search, appointment booking, telemedicine access, payments, queue tracking, community features, SithBot, and emergency support.



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- **Doctor / Specialist Interface:** availability management, appointment list, telemedicine session start, patient consultation status updates, and limited access to assigned patient records.
- **Clinic Administrator Interface:** clinic profile management, doctor allocation, appointment approval or rescheduling, queue control, cash payment tracking, and daily appointment monitoring.
- **Platform Administrator Interface:** user and role management, clinic verification, specialist verification, system reports, crisis log review, moderation oversight, and configuration management.
- **Volunteer and Moderator Interfaces:** volunteer listener access, community post review, report handling, and escalation to platform administrators when required.

6.3. Internal Module Interfaces

Internal communication follows a controller-service-repository structure. The frontend communicates with backend controllers through REST endpoints. Controllers validate requests and call service modules. Service modules apply business rules and use repositories or Mongoose models to access MongoDB Atlas.

Internal Module	Communicates With	Main Interface Methods	Responsibility
Auth Controller	User Controller, Notification Service, User Repository	register(), login(), verifyOTP(), refreshToken(), resetPassword()	Creates secure sessions, verifies OTPs, manages JWT tokens, and routes users to role-based dashboards.
User/Profile Service	Auth Controller, Consent Repository, Emergency	getProfile(), updateProfile(), updateConsent(), updateEmergencyContact()	Maintains profile, preferred language, consent status,



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	Contact Repository		and emergency contacts.
Specialist Search Service	Specialist Repository, Clinic Repository, Availability Service	searchSpecialists(), filterByLocation(), getSpecialistProfile()	Returns verified specialists based on category, language, location, consultation type, and availability.
Appointment Service	Availability Service, Payment Service, Notification Service, Queue Manager	createAppointment(), cancelAppointment(), rescheduleAppointment(), updateStatus()	Controls booking, confirmation, cancellation, rescheduling, and appointment state changes.
Availability Service	Doctor Portal, Clinic Admin Panel, Appointment Repository	createSlot(), blockSlot(), updateSchedule(), validateConflict()	Prevents schedule conflicts and exposes available slots to patients.
Payment Service	PayHere Adapter, Appointment Service, Receipt Generator	initiatePayment(), verifyWebhook(), recordCashPayment(), generateReceipt()	Tracks online and cash payments and generates PDF receipts.
Queue Manager	Clinic Admin Panel, Appointment Service, Notification Service	startQueue(), advanceQueue(), pauseQueue(), resetQueue()	Maintains queue tokens and notifies patients when they are close to their turn.



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Telemedicine Service	Jitsi Adapter, Appointment Service	generateRoomLink(), validateJoinAccess(), expireRoomLink()	Creates secure video consultation access only for valid online appointments.
Emergency Handler	Notification Service, Location Service, Crisis Log Repository, SithBot	requestEmergencySupport(), confirmConsent(), logCrisisEvent(), showHelplines()	Handles emergency flow, consent-based location sharing, fallback helpline display, and crisis logging.
Community Service	Moderator Panel, User Repository, Content Filter	createPost(), hidePost(), reportPost(), moderatePost()	Supports pseudonymous posting and moderation without exposing real identities.
SithBot Service	Recommendation Engine, Emergency Handler, Specialist Search Service	startTriage(), detectCrisisKeywords(), recommendCategory()	Provides triage guidance without diagnosis and redirects high-risk input to emergency support.

6.4. API Design Principles

The backend exposes RESTful JSON APIs over HTTPS. API endpoints are grouped by functional domain to improve maintainability and testing.

- All protected API requests must include a valid JWT access token in the Authorization header.
- Request and response payloads use JSON format with consistent field names and error structures.



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- Input validation is performed at controller level before business logic execution.
- Sensitive operations such as payments, emergency actions, role changes, and consent updates are logged for audit purposes.
- API responses must not expose stack traces, database errors, passwords, tokens, or unnecessary patient health data.

API Group	Main Purpose
/api/auth	Registration, login, OTP verification, password reset, token refresh
/api/users	Profile update, consent management, emergency contact management, preferred language update
/api/specialists	Specialist search, specialist profile view, availability summary
/api/appointments	Appointment creation, cancellation, rescheduling, approval, status tracking
/api/payments	Payment initiation, webhook verification, cash payment record, receipt generation
/api/queue	Daily queue start, queue advance, pause, reset, patient queue status
/api/telemedicine	Video room link generation, consultation join validation, room expiry
/api/emergency	Emergency confirmation, location consent, helpline display, crisis event logging
/api/community	Anonymous post creation, reply, report, moderation actions
/api/sithbot	Triage conversation, crisis keyword detection, specialist category recommendation



6.5 Hardware and Device Interfaces

SithCare does not require dedicated medical hardware for the MVP. It uses standard browser-accessible device features when required.

- Camera and microphone are accessed only during telemedicine sessions after browser permission is granted.
- Device location is requested only during emergency support and only after explicit user confirmation.
- Mobile and desktop browsers are supported through responsive React and Tailwind CSS layouts.
- If camera, microphone, or location permissions are denied, the system provides fallback instructions or alternative contact options.



7. Security Design

SithCare handles sensitive mental health-related information. Therefore, the security design focuses on confidentiality, integrity, availability, accountability, and user consent. The design also supports the security requirements defined in the SRS, including TLS 1.3, AES-256 encryption, JWT expiry rules, RBAC, pseudonymous community identities, and audit logging.

7.1. Authentication Strategy

Authentication verifies the identity of users before they access protected parts of the system. SithCare uses a token-based authentication mechanism suitable for a web-based MERN application.

- Users register using email or phone number and verify the account using OTP.
- Passwords are stored only as salted hashes using bcrypt. Plain-text passwords are never stored or logged.
- JWT access tokens expire after 15 minutes. Refresh tokens expire after 7 days and are used to request new access tokens securely.
- The system automatically logs users out after 30 minutes of inactivity where applicable.
- Clinic administrators and platform administrators may use optional multi-factor authentication for stronger protection.
- After repeated failed login attempts, the account is temporarily locked to reduce brute-force attacks.

7.2. Authorization Strategy

Authorisation controls what each authenticated user can access or perform. SithCare implements Role-Based Access Control (RBAC) with clear separation between patient, doctor, clinic administrator, platform administrator, volunteer, and moderator functions.

Role	Allowed Access
Patient	Own profile, consent settings, own appointments, own payment receipts, own queue status, community posts under pseudonym, SithBot, emergency support.



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Doctor / Specialist	Own availability, assigned appointments, assigned patient consultation information, telemedicine session controls.
Clinic Administrator	Clinic branch details, doctor schedules, appointment approval, queue control, payment tracking, clinic reports.
Platform Administrator	User and role management, clinic onboarding, specialist verification, crisis log review, compliance reports, system settings.
Volunteer Listener	Volunteer session dashboard and assigned listener sessions only. No clinical records or diagnosis features.
Community Moderator	Reported community content, moderation actions, escalation tools. No access to private patient medical details.

- Every protected endpoint checks the user role before executing business logic.
- Patients can access only their own records. Doctors can access only records related to assigned appointments.
- Administrative actions such as role changes, specialist verification, clinic changes, and crisis log review are audit logged.
- Community posts display pseudonymous identities and never publicly expose real user names.

7.3. Data Protection Design

The system protects data throughout its lifecycle: collection, transmission, storage, processing, access, and deletion/minimisation.

- Data in transit is protected using HTTPS with TLS 1.3 between browser, frontend hosting, backend APIs, and external services where supported.



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- Patient health-related data at rest is protected using AES-256 encryption through database/storage configuration and encryption controls.
- Payment card data is not stored in SithCare. Card handling is delegated to PayHere.
- Consent records are stored with timestamp, user identifier, action, and version of consent text where applicable.
- Emergency location sharing is session-based and requires explicit confirmation each time.
- Backup data is protected using the same access control and encryption policies as production data.
- Logs must avoid storing passwords, OTP values, full JWT tokens, card information, and unnecessary health details.

7.4. Input Validation and Output Encoding Strategy

Input validation is applied before any data is processed or stored. Validation occurs both in the frontend for user experience and in the backend for security enforcement.

- Email, phone number, date of birth, password, appointment time, payment amount, and file uploads are validated against strict rules.
- User-generated community content is checked for harmful content, personal identifying information, and crisis keywords before publishing.
- File uploads such as profile photos are limited by file type and size. Only JPG and PNG formats are accepted for profile images.
- MongoDB queries use controlled model methods and do not directly accept untrusted query objects from users.
- All displayed user-generated content is encoded to prevent script injection in the browser.



7.5. OWASP Top 10 Considerations

OWASP Risk Area	SithCare Design Response
Broken Access Control	RBAC middleware, ownership checks, route protection, admin action logging, and test cases for each user role.
Cryptographic Failures	TLS 1.3 for transit, AES-256 for stored sensitive data, bcrypt for passwords, and no storage of payment card data.
Injection	Backend validation, Mongoose schema validation, parameterised model access, sanitisation of search filters and community content.
Insecure Design	Security considered in architecture through controller validation, service isolation, audit logging, and emergency fallback design.
Security Misconfiguration	Environment variables for secrets, secure CORS configuration, disabled debug output in production, and secure HTTP headers.
Vulnerable and Outdated Components	Dependency review, package updates, GitHub security alerts, and controlled use of third-party SDKs.
Identification and Authentication Failures	OTP verification, strong password rules, token expiry, refresh token control, MFA for privileged roles, and account logout.
Software and Data Integrity Failures	GitHub pull requests, code review, CI checks, controlled deployment pipeline, and verification of PayHere webhook responses.
Security Logging and Monitoring Failures	Logging for authentication events, role changes, payment status, emergency



	actions, failed access attempts, and moderation actions.
Server-Side Request Forgery	Backend does not fetch arbitrary user-provided URLs; external service endpoints are allow-listed and configured through environment variables.

7.6. Security Logging and Monitoring

Security-related events are logged to support auditing, incident review, and troubleshooting.

- Authentication events: login, logout, failed login, OTP verification, password reset, and account lockout.
- Authorisation events: denied access attempts, role changes, admin actions, and specialist verification changes.
- Healthcare-sensitive events: consent update, emergency support activation, crisis event logging, and location-sharing confirmation.
- Payment events: payment initiation, webhook confirmation, payment failure, cash payment record, and refund status.
- Community events: post creation, report, hide, delete, escalation, and moderation decision.



8. Non-Functional Design Decisions

This section explains how the SithCare design addresses the non-functional requirements defined in the SRS. Each design decision is linked to the relevant quality attribute so that implementation and testing can be aligned with the original requirements.

8.1. Performance Design Decisions

NFR ID	Requirement	Design Decision
NFR-P-01	Page load time $\leq$ 3 seconds for 95% of requests on 4G LTE.	React with Vite is used for fast frontend builds. Static assets are hosted through Vercel/Netlify CDN. Tailwind CSS keeps production CSS lightweight. Images should be compressed before deployment.
NFR-P-02	Appointment booking transaction $\leq$ 5 seconds excluding PayHere redirect.	Booking logic is handled through a dedicated Appointment Service with simple database operations, slot reservation, and asynchronous notification triggering.
NFR-P-03	Support at least 500 concurrent users.	Node.js non-blocking I/O, stateless REST APIs, and cloud hosting allow horizontal scaling of the backend when traffic increases.
NFR-P-04	Queue updates visible within 60 seconds.	Queue Manager stores the current queue state centrally and patient views refresh automatically within



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		the defined interval. SMS/push proximity notifications are triggered when the patient is near their turn.
NFR-P-05	SithBot response time ≤ 2 seconds for 90% of queries.	SithBot uses a guided triage flow with limited question sets and lightweight recommendation logic for MVP performance.

8.2. Security Design Decisions

NFR ID	Requirement	Design Decision
NFR-S-01	Encrypt data in transit using TLS 1.3.	All frontend-backend and external API communication uses HTTPS. Hosting platforms must provide valid SSL certificates.
NFR-S-02	Encrypt patient health data at rest using AES-256.	MongoDB Atlas/database storage encryption is used for stored sensitive data and backups.
NFR-S-03	JWT expiry: access token 15 minutes, refresh token 7 days.	Auth Controller issues short-lived access tokens and validates refresh tokens before issuing new sessions.
NFR-S-04	Implement OWASP Top 10 mitigations.	Validation, sanitisation, RBAC middleware, rate limiting, secure headers, dependency review, and



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		audit logging are included in the backend design.
NFR-S-05	Patient health/session records restricted to authoring doctor and Platform Administrator.	Ownership checks and role checks are applied before medical/session information is returned by the API.
NFR-S-06	Community posts use pseudonymous IDs.	Community Service stores and displays pseudonymous identifiers instead of real names for forum interactions.
NFR-S-07	Payment card data not stored by SithCare.	PayHere handles card entry and processing. SithCare stores only transaction reference, amount, status, and receipt metadata.
NFR-S-08	Authentication events logged with timestamp and IP address.	Auth events are written to the audit log for security review and incident analysis.

8.3. Usability Design Decisions

NFR ID	Requirement	Design Decision
NFR-U-01	Achieve SUS score ≥ 70 .	The interface uses simple role-based dashboards, clear navigation, plain-language labels, and guided workflows for booking, payment, emergency support, and SithBot.



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NFR-U-02	New users complete register, find doctor, and book appointment within 5 minutes.	The patient flow is designed as a step-by-step process: register/login, search specialist, select slot, choose payment, and confirm booking.
NFR-U-03	Adult Mode with minimum 18px font and single-level navigation.	Adult Mode provides simplified navigation, larger text, reduced feature set, and direct access to booking and emergency support.
NFR-U-04	WCAG 2.1 Level AA accessibility.	UI design should include sufficient colour contrast, keyboard navigation, semantic labels, screen-reader-friendly forms, and accessibility testing using tools such as axe or WAVE.
NFR-U-05	Plain-language error messages in selected language.	Frontend validation and backend error responses are mapped to user-friendly messages in English, Sinhala, and Tamil where supported.

8.4. Reliability Design Decisions

NFR ID	Requirement	Design Decision
NFR-R-01	System availability $\geq 99.5\%$ excluding scheduled maintenance.	Frontend and backend are deployed on cloud hosting platforms with uptime monitoring. Critical errors are logged and monitored.



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NFR-R-02	Maintenance not during 08:00–22:00 SLST.	Deployment and maintenance planning avoids peak patient and clinic operating hours.
NFR-R-03	Automated database backups every 6 hours; RPO = 6 hours.	MongoDB Atlas backup scheduling or equivalent backup strategy is used. Restore testing is planned during system testing.
NFR-R-04	Emergency support available even if non-critical modules fail.	Emergency helpline numbers are stored as static configuration and shown as fallback. Emergency UI is kept independent from optional modules such as community and ratings.

8.5. Scalability Design Decisions

NFR ID	Requirement	Design Decision
NFR-SC-01	Application layer supports horizontal scaling without code changes.	The backend is designed as stateless REST APIs. Session state is maintained using JWT rather than server memory.
NFR-SC-02	Support up to 200 clinics without schema migration.	MongoDB document-based modelling supports flexible clinic, doctor, availability, and appointment records. Clinic identifiers separate tenant-related data.



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NFR-SC-03	Handle 5× traffic spike during public health campaigns.	Frontend CDN delivery, Node.js backend scaling, database indexing, and modular service boundaries support increased traffic.
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8.6. Maintainability Design Decisions

NFR ID	Requirement	Design Decision
NFR-M-01	Consistent coding standards documented in CONTRIBUTING.md.	The project uses GitHub, pull requests, code reviews, linting, and shared coding guidelines.
NFR-M-02	Minimum 70% backend unit test coverage.	Backend services are designed as separate modules, making them easier to test independently.
NFR-M-03	All API endpoints documented using OpenAPI 3.0.	REST endpoints are grouped by domain and documented with request/response schemas for frontend-backend coordination.
NFR-M-04	No hardcoded secrets in source code.	Environment variables are used for JWT secrets, database URLs, PayHere keys, SMS/email credentials, and map API keys.
NFR-M-05	All code changes through pull requests with peer review.	GitHub branch protection and PR workflow ensure review, traceability, and controlled merging.



8.7. Summary

The selected design provides a practical and secure foundation for SithCare. The layered MVC architecture, MERN stack, role-based access control, encrypted communication, modular service boundaries, and cloud deployment strategy collectively support the SRS requirements for performance, security, usability, reliability, scalability, and maintainability. These decisions also keep the system feasible for the academic project timeline while allowing future expansion into more advanced healthcare features.